

Some Basic Concept of Chemistry

1. 16 gm sulphur is burnt to form SO_2 which is oxidised by Cl_2 water. The resulting is treated with BaCl_2 solution. Specify the amount of BaSO_4 precipitate & amount of Cl_2 used in oxidation of SO_2 respectively :

[Atomic weight of Ba = 137]

- (A) 233 gm, 71 gm
- (B) 58.25 gm, 17.75 gm
- (C) 116.5 gm, 35.5 gm
- (D) 58.25 gm, 35.5 gm

Ans. (C)

2. 50 ml, 0.04 solution of sodium sesquicarbonate dihydrate $[\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 \cdot 2\text{H}_2\text{O}]$ is titrated against 0.05 M HCl solution. X ml of acid is required to reach the phenolphthalein end point while Y ml of same acid was required when methyl orange is used as indication for end point in separate titration.

Which of the following statements are correct ?

- (A) $X + Y = 160$ ml
- (B) $X + Y = 120$ ml
- (C) $Y - X = 80$ ml
- (D) If the titration is started with phenolphthalein indicator and methyl orange is added at the end point, 2x ml HCl would be required further to reach end point

Ans. (A, C, D)

3. Mass of a molecule of a compound is 7.348×10^{-23} gm. Each molecule of compound contains two oxygen atom than correct statements are about compound :

- (A) 22 gm of gaseous compound occupy 11.2 litre NTP
- (B) 5.5 gm of compound contains $\frac{1}{8} N_A$ oxygen atoms
- (C) 0.4 gm molecule of compound contains 12.8 gm oxygen
- (D) Mass of 0.2 mole of compound is 8.8 gm

Ans. (A, C, D)

4. 'X' and 'Y' are two elements which form X_2Y_3 and XY_2 molecules. If 0.15 mole of X_2Y_3 and XY_2 weighs 15.9 gm & 9.3 gm respectively. If oxidation state of 'y' in both compounds is +2 than correct options are :

- (A) Atomic weight of X = 26 & Y = 18.
- (B) Equivalent weight of $\text{X}_2\text{Y}_3 = 17.6$ & $\text{XY}_2 = 15.5$
- (C) Atomic weight of X = 26 & Y = 36.
- (D) Equivalent weight of $\text{X}_2\text{Y}_3 = 26.66$ & $\text{XY}_2 = 24.5$

Ans. (A, B)

5. 20 ml mixture of CH_4 , C_2H_4 & C_3H_8 in the ratio 1 : 1.5 : 2.5 respectively is burnt in excess of air. If all measurement are made at same temperature & pressure than specify :
- (A) Volume of O_2 needed = 38 ml
 (B) Volume of O_2 needed = 76 ml
 (C) Volume of CO_2 formed = 46 ml
 (D) Volume of CO_2 formed = 23 ml

Ans. (BC)

6. Match the following.

Column-I

Column-II

- | | |
|--|------------------------|
| (A) 120 gm CH_3COOH in 1 litre solution
($d_{\text{Solution}} = 1.2 \text{ gm/ml}$) | (P) $M = 2$ |
| (B) 120 gm glucose dissolved in 1 litre solution
($d_{\text{Solution}} = 1.2 \text{ gm/ml}$) | (Q) 10% w/w solution |
| (C) $X_{\text{NH}_2\text{CONH}_2} = \frac{1}{31}$ (aq. solution) | (R) 12% (w/v) solution |
| (D) 19.6 (w/v) H_2SO_4 solution
($d_{\text{Solution}} = 1.2 \text{ gm/ml}$) | (S) $m = 1.85$ |

Ans. [(A) → (P, Q, R, S); (B) → (Q, R); (C) → (Q, S); (D) → (P)]

7. Number of correct statements are :

- (i) In 400 amu ammonium nitrate number of H-atoms are $20 N_A$.
 (ii) Mass of 0.6 gm atom of sodium is 13.8 gm.
 (iii) Number of moles in 5.6 litre of $\text{N}_2\text{O(g)}$ at 1 atm & 273 K temperature is 0.25.
 (iv) 42 gm N_2 contains same number of molecules as number of atoms present in 11.2 litre $\text{NO}_2\text{(g)}$ at NTP.
 (v) Mass of a molecule of oxalic acid is $1.5 \times 10^{-26} \text{ kg}$. [$N_A = 6 \times 10^{23}$]

Ans. (3)

8. 100 ml of a sample of hard water requires 20 ml of 0.03 N H_2SO_4 for complete reaction. The hardness of water (density = 1 gm/ml) is :

[Give your answer as sum of digits and fill OMR sheet]

Ans. (3)

9. Two acids A and B are titrated against 10 ml of 2.5 N Na_2CO_3 solution separately which requires 10 ml of 'A' and 40 ml of 'B' for complete neutralisation. x litre of 'A' and y litre of 'B' are required to mix to produce one litre of 1N acid solution ? Specify $y \times 10$.

Ans. (8)

10. Volume of 1 mole of water at 1 atm & 273 K is (in ml) :

(Density of $\text{H}_2\text{O} = 1 \text{ gm/ml}$)

[Give your answer as sum of digits and fill OMR sheet]

Ans. (9)

Atomic Structure

1. In a sample of hydrogen atom in ground state electrons make transition from ground state to a particular excited state where path length is five times de-broglie wavelength, electrons make back transition to the ground state producing all possible photons. If photon having 2nd highest energy of this sample can be used to excite the electron in a particular excited state of Li^{2+} atom then find the final state of Li^{2+} atom.

(A) 6

(B) 12

(C) 9

(D) 15

Ans. (B)

2. Which of the following statements are True (T) and False (F) ?

(i) An orbital with $l = 0$ is symmetrical about the nucleus

(ii) Radial function $[R(r)]$ a part of wave function is dependent on quantum number n only.

(iii) Angular function depends only on the direction, and is independent to the distance from the nucleus.

(iv) $\psi^2(r, \theta, \phi)$ is the probability density of finding the electron at a particular point in space.

(v) Radial distribution function $(4\pi r^2 R^2)$ gives the probability of the electron being present at a distance r from the nucleus.

(A) T F T T T

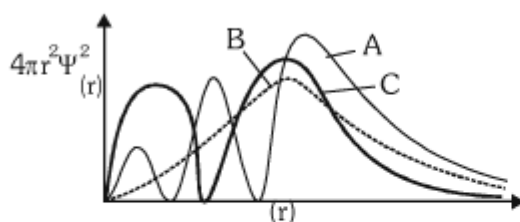
(B) T T T T F

(D) T F T F T

(D) F F T T T

Ans. (A)

3. From the graph given below, select the option which correctly matches the orbitals for which the radial probability distribution is given :



- (A) A - 3p, B - 3d, C - 3s
 (B) A - 3s, B - 3d, C - 4p
 (C) A - 4p, B - 3d, C - 5f
 (D) A - 3p, B - 4d, C - 4p

Ans. (C)

4. Which of the following is/are true.

- (A) Multiplicity in Fe^{3+} is greater than that in Co^{3+}
 (B) Ti^{3+} , Cr^{+} , Sc^{2+} ions are diamagnetic.
 (C) If we consider the energy of orbital $\propto n$ then the value of $(n + l + m)$ for last electron of K can be 4.
 (D) The value of Z_{eff} for 3s electron of Cl is 10.9.

Ans. (AC)

5. A gas of identical H-like atom has some atoms in the lowest (ground) energy level A and some atoms in a particular upper (excited) energy level B and there are no atoms in any other energy level. The atoms of the gas make transition to a higher energy level by absorbing monochromatic light of photon energy 2.7 eV. Subsequently, the atoms emit radiation of only six different photons energies. Some of the emitted photons have energy 2.7 eV. Some have more and some have less than 2.7 eV.

- (A) The principal quantum number of initially excited level B = 2.
 (B) The ionisation energy for the gas atoms is 28.8 eV.
 (C) The maximum and the minimum energies of the emitted photons will be 13.5 eV and 0.699 eV respectively.
 (D) Nucleus is stationary & e^{-} will revolve around the nucleus in circular path.

Ans. (AC)

6. Which of the following statements is/are incorrect ?

- (A) Number of electrons having ' $l = 0$ ' is 10 in Pd.
 (B) The value of Z_{eff} for 3d electron of Cr & 3d electron of Mn is same as number of electron in 'd' subshell of Cr & Mn are same.
 (C) Multiplicity of Fe is equal to Ni^{2+} .

- (D) Value of $\left(\frac{\ell}{n}\right)$ for last electron of element having atomic number 57 is 0.4.

Ans. (ABC)

7. Which of the following options represent correct statements regarding quantum numbers.
- (A) A hydrogen like one electron wave function obeying Pauli's Exclusion Principle will have unique set of values of all the 4 quantum numbers.
 - (B) Shape and size of hydrogen like atomic orbitals depends only on principal and azimuthal quantum number.
 - (C) Except spin quantum number rest all three quantum numbers affect orbital angular momentum of the electron in a hydrogen like atomic orbital.
 - (D) Probability density of electron at the nucleus in hydrogen like atom is independent of all quantum numbers.

Ans. (ABC)

8. When electron jumps from 3rd excited state to 1st excited state of He^+ , then correct statements regarding above are : ($R_H = 109600 \text{ cm}^{-1}$)
- (A) The emitted energy in above transition is 10.2 eV
 - (B) This transition is correspond to $n = 2$ to $n = 1$ in H-atom
 - (C) The radiation emitting out will fall in UV region
 - (D) Wave number for this transition is 82200 cm^{-1}

Ans. (ABCD)

9. Which is / are correct statements ?

- (A) Electronic configuration $\boxed{\uparrow\downarrow} \boxed{\uparrow\uparrow\downarrow} \boxed{\uparrow}$ violates Hund's rule.
- (B) Cu^{+2} is paramagnetic species.
- (C) Emission spectrum of $\text{H}_2(\text{g})$ is Line emission spectrum
- (D) De-Broglie wave length associated with electron moving in Hydrogen atom having 3.4eV energy of separation is equal to $2.116\pi \text{ \AA}$

Ans. (BD)

10. $\psi = \frac{2(Z/a_0)^{3/2}}{8\sqrt{3}} \left(27 - 18\frac{Zr}{a_0} + \frac{2Z^2r^2}{a_0^2} \right) e^{\frac{-Zr}{3a_0}}$; This is the equation representing an

orbital. Which of the following is/are correct ?

- (A) This equation is of 3s orbital
- (B) Radial probability curve touches r-axis at three finite points
- (C) Above equation can be used to deduce n & ℓ quantum numbers.
- (D) Probability density curve shows four maximas

Ans. (ABC)

Paragraph-1

Consider a hypothetical atom where p_x , p_y , p_z , d_{xy} , d_{xz} , d_{yz} and $d_{x^2-y^2}$ orbital are present for principal quantum number $n = 3$.

11. Find the number of other orbital which lobes are fully present in the nodal plane of p_x orbital :
- (A) 2
(B) 3
(C) 4
(D) 5

Ans. (B)

12. Which of the following orbitals lobe is not present at all either in the nodal plane of p_x orbital or in the nodal plane of p_y orbital.
- (A) d_{xy}
(B) $d_{x^2-y^2}$
(C) d_{yz}
(D) p_z

Ans. (A)

13. Out of following which orbital will never be used in the formation of π -bond.
- (A) $d_{x^2-y^2}$
(B) d_{z^2}
(C) d_{xy}
(D) d_{yz}

Ans. (A)

14. Match the following.

Column-I

- (A) Energy of ground state of Li^{2+}
(B) Potential energy of 1st orbit of H-atom
(C) K.E. of 2nd excited state of He^+
(D) Ionisation potential of He^+

Column-II

- (P) 6.04 eV
(Q) 54.4 eV
(R) -122.4 eV
(S) -27.2 eV

Ans. [(A) $\rightarrow$ (R); (B) $\rightarrow$ (S); (C) $\rightarrow$ (P); (D) $\rightarrow$ (Q)]

15. If wave function of an orbital, for H-atom is represented by

$$\Psi_{n,l,m_l}(r, \theta, \phi) = \frac{\sqrt{2}}{81\sqrt{\pi} a_0^{3/2}} \left[6 - \frac{r}{a_0} \right] \frac{r}{a_0} e^{-r/3a_0} \cos \theta, \text{ where } a_0 \text{ is the Bohr's radius}$$

then match the following.

List-I

- (P) No. of angular nodes
(Q) Total no. of nodes
(R) Value of n
(S) Value of (n + l)

List-II

- (1) 2
(2) 3
(3) 4
(4) 1

Codes :	P	Q	R	S
(A)	4	1	2	3
(B)	2	1	3	4
(C)	3	2	1	3
(D)	4	2	1	3

Ans. (A)

16. Knowing the fact that CIA agents are poor in chemistry. Russian's started using code names based on chemistry. Please help CIA officers in decoding the given code with the help of information given.

CODE NAME : ROCK

First letter (R) : It represent the ratio of principal quantum no. and azimuthal quantum no. of the orbital having 4 radial nodes and xz plane as only angular node.

Second letter (O) : Ratio of potential energy to total energy for the electron in second orbit of Li^{+2} ions

Third letter (C) : Represents the wavelength fo photoelectrons in "nm", emitted by a metal plate having work function 40 meV when exposed to radiation of wavelength $12.4 \mu\text{m}$.

(use : $\lambda = \sqrt{\frac{150}{V(\text{volt})}} \text{ \AA}$; $hc = 1240 \text{ nm, eV}$)

Fourth letter (K) : Represents the number of possible spectral lines when electrons jump from $n = 4$ to $n = 2$ in H-atom species.

[Give your answer as sum of digits when you get a single digit answer and fill OMR sheet]

Ans. (7)

Periodic Table

1. Correct order for solubility is :

- (A) $\text{CuS} > \text{ZnS} > \text{Na}_2\text{S}$
- (B) $\text{ZnS} > \text{Na}_2\text{S} > \text{CuS}$
- (C) $\text{Na}_2\text{S} > \text{CuS} > \text{ZnS}$
- (D) $\text{Na}_2\text{S} > \text{ZnS} > \text{CuS}$

Ans. (D)

2. Which of the following statement is not correct for given electronic configuration
 $[\text{Xe}]4f^{14}5d^{10}6s^2$

- (A) It belongs to II-B group and 6th period
- (B) It is liquid at room temperature
- (C) It is a transition element
- (D) It is diamagnetic in +1 oxidation state

Ans. (C)

3. One mole of 'F' atoms are ionised to F^- the energy released is 'X' Joules. Then :

- (A) X Joules is sufficient to ionise 1 mole of gaseous Cl into Cl^+ .
- (B) X Joules is sufficient to ionise 1 mole of gaseous F into F^+ .
- (C) X Joules is sufficient to ionise 1 mole of gaseous F into F^+ as well as 1 mole of gaseous Cl into Cl^+ .
- (D) Less than 1 mole of gaseous F or Cl will be ionised to F^+ or Cl^+ .

Ans. (D)

4. Consider the successive ionisation energy for an element 'A' are 100 eV, 150 eV, 181 eV, 2000 eV, 2200 eV. Select correct statements for element 'A'.

- (A) Element 'A' should have valence e^- configuration ns^2np^2
- (B) 'A' may form chloride of ACl_3 type
- (C) Oxide of 'A' may be amphoteric
- (D) Metal 'A' may form a protective layer over it in presence of an oxidising agent

Ans. (BCD)

5. X, Y and Z are three consecutive elements of second period whose I.P. data (in eV) are available. By using these data specify incorrect statements regarding X, Y and Z.

Element	IP ₁	IP ₂	IP ₃	IP ₄
X	5.39	75.6	122.4	Not possible
Y	8.2	25.1	37.9	259.2
Z	9.3	18.2	217.6	363.7

- (A) $X_{(g)} + e^- \longrightarrow A_{(g)}^-$ $\Delta H = \text{positive}$
 (B) $Z_Y > Z_Z$ (Z is atomic number)
 (C) Oxide of 'Y' is amphoteric
 (D) X^{+1} is stable than X^{2+}

Ans. (AC)

6. How many of following statements are not correct ?

- (i) I.P. of $F^- > Cl^-$
 (ii) $S_{(g)} + e^- \longrightarrow S_{(g)}^-$ is endothermic process
 (iii) I.P. of Y < Zn
 (iv) Size of $Ga^{3+} > Cu^+$

Ans. (3)

7. Write the number of pairs in which size of first element or ion is higher as compare to IInd out of following eight pairs.

$(Ba^{+2}_{aq}, Be^{+2}_{aq})$, (Kr, Ne), (Na, Na^+), (Cl, Cl^-), (I^- , Cl^-), (Li_{Aq}^+ , Na_{Aq}^+),
 (Li, Na), (Li^+ , Na^+)

Ans. (4)

8. How many of these have more electron affinity than oxygen ?

B, C, N, F, S, Se, P, Cl, Li, Na

Ans. (4)

9. Amongst the following, the total number of orders which are correct with respect to the property indicated against each is:

(i) $\text{Mg} > \text{Al} > \text{Si} > \text{P}$;	Covalent radius
(ii) $\text{Na}^+ < \text{O}^{2-} < \text{F}^- < \text{N}^{3-}$;	Ionic size
(iii) $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Li}^+ < \text{K}^+$;	Ionic size
(iv) $\text{C} < \text{Si} > \text{P} > \text{N}$;	Electron affinity value
(v) $\text{N} < \text{C} < \text{O} < \text{F}$;	Electron affinity value
(vi) $\text{F} > \text{Cl} > \text{Br} > \text{I}$;	Electron affinity value
(vii) $\text{Si} > \text{Mg} > \text{Al} > \text{Na}$;	First ionisation energy
(viii) $\text{O} > \text{F} > \text{N} > \text{C}$;	Second ionisation energy
(ix) $\text{N} > \text{P} > \text{Sb} > \text{As}$;	Third ionisation energy

Ans. (6)

10. How many of the following have correct comparison of ionisation energy in isolated gaseous state ?

(i) $\text{P} > \text{S}$	(ii) $\text{F} > \text{O}$
(iii) $\text{Sn} > \text{Pb}$	(iv) $\text{Ca}^{2+} > \text{Ar}$
(v) $\text{Ne} > \text{F}^-$	(vi) $\text{O}^+ > \text{F}^+$
(vii) $\text{Cl}^- > \text{F}^-$	(viii) $\text{Al} > \text{Ga}$
(ix) $\text{Al}^{3+} > \text{Al}^{2+}$	(x) $\text{He} > \text{He}^+$

Ans. (7)

Chemical Bonding

1. If s-p mixing does not occur in B_2 molecule, then which of the following must be **FALSE** for this molecule? (Assume Z-axis as internuclear axis)

- (A) Paramagnetic
 (B) Bond order = 1
 (C) HOMO is σ_{2p_z}
 (D) LUMO is π_{2p_x} or π_{2p_y}

Ans. (A)

2. The correct order of strength of H-bonding is
- (A) $AH \dots\dots A < A_1H \dots\dots A_1 < BH \dots\dots B < B_1H \dots\dots B_1$
 (B) $A_1H \dots\dots A_1 > AH \dots\dots A < B_1H \dots\dots B_1 < BH \dots\dots B$
 (C) $AH \dots\dots A > BH \dots\dots B > A_1H \dots\dots A_1 > B_1H \dots\dots B_1$
 (D) $A_1H \dots\dots A_1 > B_1H \dots\dots B_1 > BH \dots\dots B > AH \dots\dots A$

Where B = First element of group 15

B_1 = First element of group 16

A_1 = First element of group 17

A = Second element of group 17

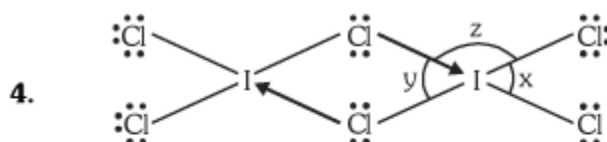
Ans. (D)

3. Which of the following statements are correct ?

- (I) Vapour pressure of H_2O is more than H_2Te at given temperature.
 (II) In both N_2O_5 and N_2O_4 all N–O bond lengths are equivalent.
 (III) In both crystalline $NaHCO_3$ and $KHCO_3$, HCO_3^- forms different type of polymeric anion through hydrogen bond.
 (IV) Amongst B_2 , C_2 , N_2^- and O_2 , N_2^- and O_2 on further ionization (losing single electron) forms thermodynamically more stable species.

- (A) I and II
 (B) III and IV
 (C) II and III
 (D) I and IV

Ans. (B)



Compare x, y, z bond angle from the above given molecule :

- (A) $x > y > z$
 (B) $x > z > y$
 (C) $y > x > z$
 (D) $z > y > x$

Ans. (B)

5. Select the incorrect order :

- (A) Thermal stability : $LiNO_3 < NaNO_3 < KNO_3$
 (B) Solubility : $LiNO_3 < NaNO_3 < KNO_3$
 (C) Thermal stability : $Be(OH)_2 < Ca(OH)_2 < Sr(OH)_2$
 (D) Solubility : $Be(OH)_2 < Ca(OH)_2 < Sr(OH)_2$

Ans. (B)

6. How many of the following molecules have two π -bonds ?

$B_2, C_2, N_2, Li_2, C_2H_2$

(A) 1

(B) 2

(C) 3

(D) 4

Ans. (C)

7. In which of the following N-N bond length is the shortest ?

(A) N_2O

(B) N_3^-

(C) N_2O_4

(D) N_2O_3 (unsymmetrical)

Ans. (A)

8. Which of the set of isomers of $C_6H_4Cl_2$ is having equal dipole moment with C_6H_5Cl and C_6H_6 respectively

(A) ortho and meta

(B) meta and para

(C) ortho and para

(D) para and ortho

Ans. (B)

9. Select the correct statement.

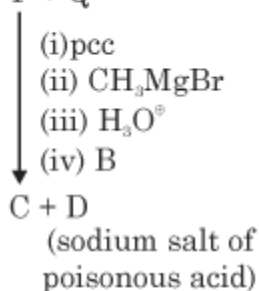
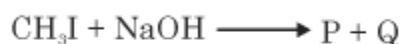
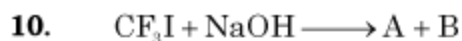
(A) If molecule has any polar bond then it is always polar

(B) Solubility of noble gases increases in water down the group when their size increases because London dispersion force increases

(C) First ionisation energy of Al is greater than the gallium.

(D) XeF_5^+ has square pyramid geometry.

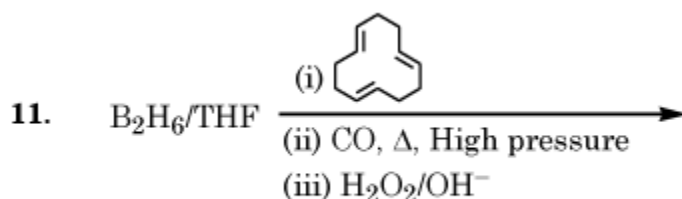
Ans. (D)



Find the correct statement.

- (A) $\text{A} > \text{C}$ (Boiling point)
- (B) $\text{A} < \text{C}$ (Vapour pressure)
- (C) $\text{A} < \text{C}$ (Water solubility)
- (D) $\text{A} > \text{C}$ (Dipole moment)

Ans. (D)



Hybridisation of Boron and Carbon are :

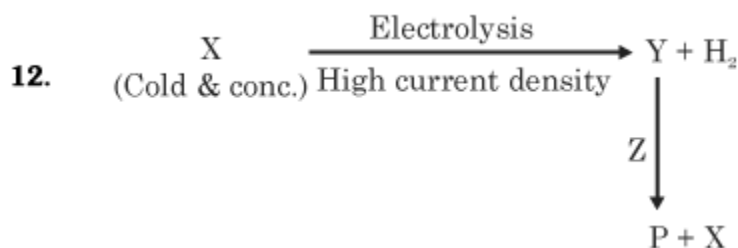
Boron

- (A) $\text{sp}^3, \text{sp}^2, \text{sp}$
- (B) sp^3, sp^2 only
- (C) sp^3, sp only
- (D) sp^3, sp^2 only

Carbon

- $\text{sp}^3, \text{sp}^2, \text{sp}$
- $\text{sp}^3, \text{sp}^2, \text{sp}$
- sp^3, sp^2 only
- sp^3, sp

Ans. (A)



'Z' is used in preparation of bakelite then :

- (A) Dipole moment of Z is more than dipole moment of P
- (B) Boiling point of P is more than Boiling point of Z
- (C) P is less water soluble than Z
- (D) P is non polar molecule

Ans. (B)

- 18.** Select the correct statement(s) :
- (A) Due to the absence of vacant orbital of N-atom, NCl_3 does not undergo hydrolysis.
 - (B) Order of rate of hydrolysis in SF_6 , SeF_6 and TeF_6 , is $\text{SF}_6 < \text{SeF}_6 < \text{TeF}_6$.
 - (C) Only proton donor acids are formed as final hydrolyzed product of SF_6 .
 - (D) Hybridization of central atom in transition state changes during hydrolysis of BCl_3 .

Ans. (BCD)

- 19.** For reaction : $\text{NaF} + \text{XeF}_4 \rightarrow \text{P}$, the correct statement(s) for anionic part of compound 'P' is/are
- (A) Six atoms lie in a plane
 - (B) Non polar
 - (C) Non axial d-orbital used in hybridization of xenon has two xz and yz nodal planes.
 - (D) All bonds are equivalent.

Ans. (ABCD)

- 20.** Which of the following are ionic compounds :

- (A) $\text{PCl}_5(\text{s})$
- (B) $\text{CaC}_2(\text{s})$
- (C) $\text{PBr}_5(\text{s})$
- (D) $\text{CO}_2(\text{s})$

Ans. (ABC)

- 21.** Which is/are correct for B_2H_6 structure ?

- (A) It has 4 B—H terminal bonds and two 3C—2e bonds.
- (B) It has six B—H terminal bonds and one 3C—2e bond.
- (C) It has four B—H terminal bonds two 3C—2e bonds and one B—B bond.
- (D) It has ionic interaction between $[\text{BH}_2]^+$ and $[\text{BH}_4]^-$ in solid state

Ans. (AD)

- 22.** The **incorrect** order of bond angle is/are :

- (A) $\text{H}_2\text{O} > \text{OF}_2 > \text{SF}_2 > \text{H}_2\text{S}$
- (B) $\text{H}_2\text{O} > \text{SF}_2 > \text{OF}_2 > \text{H}_2\text{S}$
- (C) $\text{H}_2\text{O} > \text{OF}_2 > \text{H}_2\text{S} > \text{SF}_2$
- (D) $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{OF}_2 > \text{SF}_2$

Ans. (BCD)

- 23.** What is/are true for SF_4 ?
- (A) Hybridisation of 'S' involves d_{z^2} orbital.
 - (B) Angle distortion between two axial bonds is less than that between two equatorial bonds.
 - (C) None of the bond angle is 90° .
 - (D) SF_4 on hydrolysis gives H_2SO_3 and HF.

Ans. (ABCD)

- 24.** The species/molecule is having same shape but different hybridisation.

- (A) XeF_2 , CO_2
- (B) I_3^- , HgCl_2
- (C) CO_3^{2-} , NO_3^-
- (D) SO_2 , OCl_2

Ans. (ABD)

- 25.** The **correct** order of Lewis basic strength is/are :

- (A) $\bar{\text{C}}\text{Cl}_3 < \bar{\text{C}}\text{F}_3$
- (B) $(\text{CH}_3)_2\text{O} < \text{O}(\text{SiH}_3)_2$
- (C) $(\text{CH}_3)_3\text{N} > \text{N}(\text{SiH}_3)_3$
- (D) $\text{NH}_3 > \text{PH}_3$

Ans. (ACD)

- 26.** The dipole moment of AX_3 , BX_3 and CY_3 are 4.97×10^{-30} , 0.60×10^{-30} and 0.00 C.M. respectively, then the shape of molecules may be

- (A) Pyramidal, T-shape, Trigonal planar
- (B) Pyramidal, pyramidal, T-shape
- (C) Pyramidal, pyramidal, Trigonal planar
- (D) Pyramidal, pyramidal, linear

Ans. (AC)

- 27.** Incorrect statement(s) is/are :

- (A) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$: Lewis basic character
- (B) CH_4 can not act as lewis base but can act as lewis acid
- (C) Lewis acidic strength of a species does not depend on charge density of central atom
- (D) SO_2 can acts as lewis acid as well as base

Ans. (BC)

28. What are the hydrolysed products of BeCl_2 formed during hydrolysis

- (A) $[\text{Be}(\text{OH})_4]^{2-}$
- (B) $\text{Be}(\text{OH})_2$
- (C) HCl
- (D) BeH_2

Ans. (BC)

29. Product formed on hydrolysis of BCl_3

- (A) $[\text{B}(\text{OH})_4]^-$
- (B) HCl
- (C) $\text{B}(\text{OH})_3$
- (D) HOCl

Ans. (BC)

30. CCl_4 is inert towards hydrolysis under ordinary condition because

- (A) No vacant orbital on attacking site of C-atom
- (B) H_2O molecule can not approach to the anti-bonding M.O. of C-Cl bond due to steric crowding
- (C) Bond dissociation energy of C-Cl bond is very high
- (D) CCl_4 is non polar and does not react with polar H_2O molecule

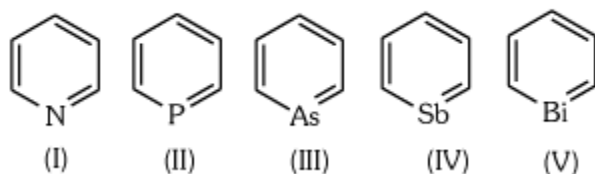
Ans. (AB)

31. Which of the following molecule/molecules have $3\text{C} - 4\text{e}^-$ bond ?

- (A) Si_2Cl_6
- (B) Al_2Cl_6
- (C) I_2Cl_6
- (D) Al_2Br_6

Ans. (B, C, D)

32. Consider the following compounds :



Which of the following is/are correct ?

- (A) $\text{I} > \text{II} > \text{III} > \text{IV} > \text{V}$ (Bond angle at group XV element)
- (B) $\text{I} < \text{II} < \text{III} < \text{IV} < \text{V}$ (C-A bond length where A is group XV element)
- (C) $\text{I} > \text{II} > \text{III} > \text{IV} > \text{V}$ (Stability)
- (D) Lone pair present on the central group XV atom is delocalised in the ring.

Ans. (ABC)

33. All atoms are present in a plane in the molecular structure of -

- (A) $C_2(CN)_4$
- (B) N_3H
- (C) $NH_2(SiH_3)$
- (D) $CO(CN)_2$

Ans. (ABD)

34. The compounds having Keesom, and London forces are

- (A) H_2O_2
- (B) H_2O
- (C) Xe
- (D) ICl

Ans. (ABD)

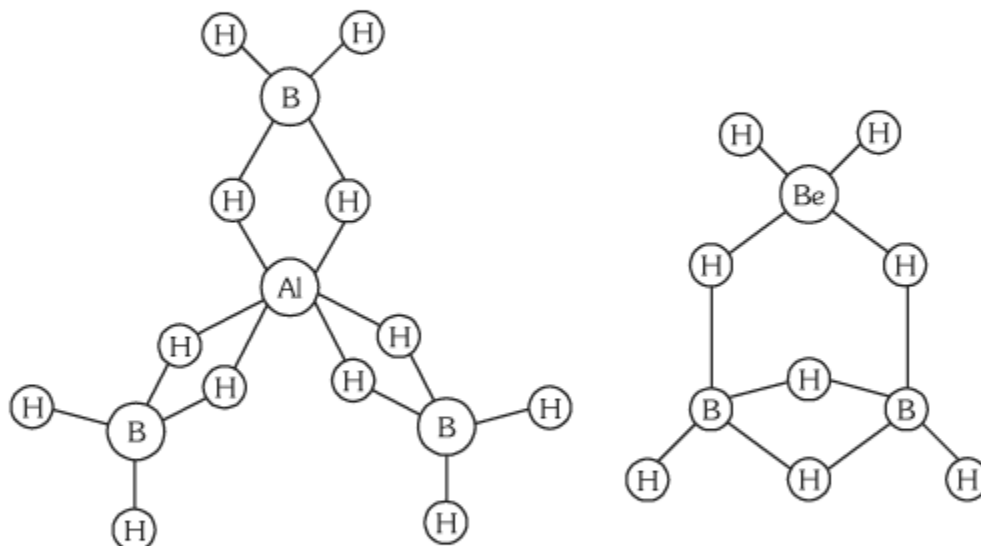
35. Which of the following species(s) having bond order equal or less than one.

- (A) H_2^+
- (B) C_2
- (C) O_2^+
- (D) B_2

Ans. (AD)

Paragraph-1

NaBH_4 is ionic compound contain BH_4^- tetrahydridoborate ion and in solid state NaBH_4 has sodium chloride structure. Not all tetrahydridoborates are ionic. The beryllium, aluminium and transition metal borohydrides become increasingly covalent and volatile. In these type of tetrahydrido borate, the BH_4^- form covalent bond with metal ion. One or more H atoms in a BH_4^- act as a bridging bond to metal, forming a three centre bond with two electrons shared by three atoms. The BH_4^- is usually in that it may form one two or three such three centre bonds to the metal ion when forms covalent bond. $\text{Be}(\text{BH}_4)_2$ and $\text{Al}(\text{BH}_4)_3$ are covalent and structures are given below :



36. Select incorrect statement about $\text{Li}[\text{AlH}_4]$.
- (A) Hybridisation of Al is same as B in $\text{Na}[\text{BH}_4]$
 - (B) Geometry around Al is same as AlCl_4^-
 - (C) AlH_4^- , BH_4^- , AlCl_4^- are iso-structural
 - (D) AlH_4^- , AlCl_4^- , BH_4^- are iso-electronic.

Ans. (D)

37. Select correct about $\text{Al}(\text{BH}_4)_3$.
- (A) Each tetrahydride borate form two hydrogen bridges.
 - (B) Two BH_4 form 2 hydrogen bridges and one BH_4 form one hydrogen bridge.
 - (C) One BH_4 form 2 hydrogen bridge and two BH_4 form one hydrogen bridge.
 - (D) B form only 2c-2e bond.

Ans. (A)

38. Total 2c-2e and 3c-2e bonds in $\text{Be}(\text{BH}_4)_2$ are respectively :

- (A) 6, 4
- (B) 4, 6
- (C) 4, 4
- (D) 4, 8

Ans. (C)

Paragraph-2

Hybridisation is the chemists attempt to explain the observed molecular shape by constructing hybridised atomic orbitals with the appropriate inter orbital angles. The molecule for which deviation from normal bond angle is observed, VSEPR theory suggest electron pair repulsive interaction ($\ell\text{p} - \ell\text{p} > \ell\text{p} - \text{bp} > \text{bp} - \text{bp}$). While from hybridisation point of view that is departure from normal hybridisation because the angle between any equivalent hybrid orbitals determine the fraction of s and p character of the hybrid and vice - versa.

39. An element 'A' has outer shell configuration of $5s^25p^6$. If A forms covalent compound AO_2F_2 with fluorine and oxygen. The orbitals used by 'A' for bonding are

- (A) d-orbitals
- (B) p-orbitals
- (C) sp-hybridised orbital
- (D) sp^3d hybrid orbital

Ans. (D)

40. In which species number of lone pair on iodine and number of d - orbitals used in hybridisation by iodine are same

- (A) ICl_2^+
- (B) ICl_2^-
- (C) IF_7
- (D) ICl_4^-

Ans. (D)

41. Which line of properties A, B, C or D correctly listed, properties of the given three compounds.

XeF_4	CCl_4	Zn^{2+}
(A) Square planar	Non-planar	$\mu = \sqrt{8}$ BM
(B) $\mu = 0$	Tetrahedral	Diamagnetic
(C) Non-planar	B.A. = $109^\circ 28'$	$\mu = \text{Zero}$
(D) Symmetrical molecule	Square planar	Total number of 12 electrons having m (Magnetic quantum number) = Zero

Ans. (B)

Paragraph-3

When two compounds ACl_3 and DCl_3 of two elements A and D are mixed together a compound ADCl_6 is formed. Structural analysis showed that ADCl_6 is an ionic compound. Given that DCl_3 is trigonal planar and ACl_3 is trigonal pyramidal.

- 42.** If anion has See - Saw shape then shape of cation formed is
(A) Linear
(B) Bent
(C) Pentagonal bipyramidal
(D) Trigonal planar

Ans. (A)

- 43.** If shape of anion is tetrahedral then shape of cation formed is
(A) Tetrahedral
(B) Bent
(C) Linear
(D) Trigonal planar

Ans. (B)

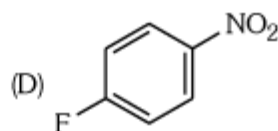
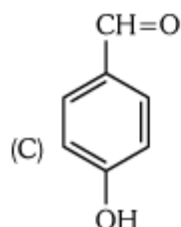
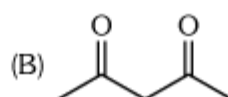
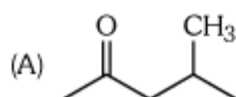
- 44.** Shape of DCl_4^- is
(A) See - Saw
(B) Perfect tetrahedral
(C) Square planar
(D) none of these.

Ans. (B)

Paragraph-4

One hydrogen bonding represented as $\text{X} - \text{H} \cdots \text{D}$ denotes the interaction between the donor species D and the acceptor species $\text{X} - \text{H}$ through the hydrogen end. H atom should remain between the atoms (i.e. X and D) of more electronegative non metallic elements.

45. In which of the following can show intramolecular hydrogen bonding in solution ?



Ans. (B)

46. In which of the following compound does not have hydrogen bonding.

- (A) K_2HPO_4
 (B) K_2HPO_3
 (C) Chloralhydrate
 (D) H_2O

Ans. (B)

47. The strength of H-bonding order is

- (A) $H_2O > H_2O_2 > HF$
 (B) $H_2O_2 > H_2O > HF$
 (C) $HF > H_2O > H_2O_2$
 (D) $HF = H_2O = H_2O_2$

Ans. (C)

Paragraph-5

Covalent radius for Homodiatomic molecule A_2 is $r_A = \frac{d_{A-A}}{2}$. For Hetrodiatomic

molecular bond length between A and B in A-B molecule is given by Stevenson schomaker formula

$$d_{A-B} = r_A + r_B - 0.09 (\chi_A - \chi_B)$$

Vader waal radius is larger than metallic as well as covalent radius.

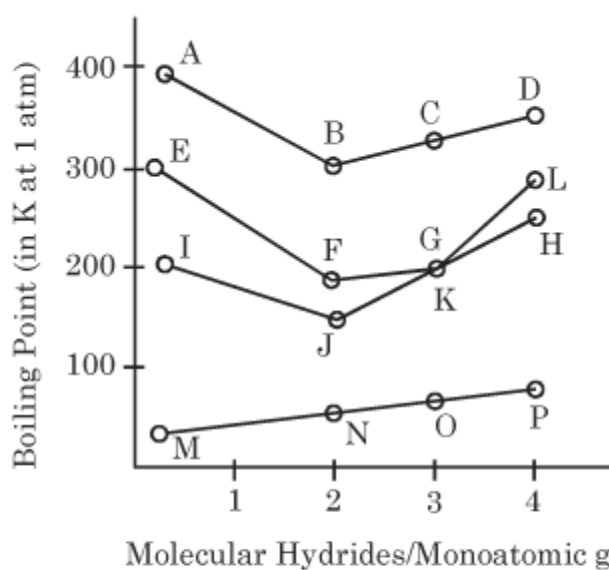
Boiling point of compound of depends on their intermolecular force of attraction.

48. The bond length of Cl-Cl is $1 \text{ \AA}$ and C-C is $1.54 \text{ \AA}$. If electronegativity of C and Cl are 2.5 and 3 respectively. Find C-Cl bond length.

- (A) $1.28 \text{ \AA}$
 (B) $1.225 \text{ \AA}$
 (C) $1.54 \text{ \AA}$
 (D) $1.08 \text{ \AA}$

Ans. (B)

49. Consider the boiling point of some molecular hydrides and monoatomic gases



- According to the given information the incorrect match is :
- (A) A to D compounds are H_2O , H_2S , H_2Se , H_2Te respectively
- (B) E to H compounds are HF , HCl , HBr , HI respectively
- (C) I to L compounds are CH_4 , SiH_4 , GeH_4 , SnH_4 respectively
- (D) M to P are Ne , Ar , Kr , Xe respectively

Ans. (C)

Paragraph-6

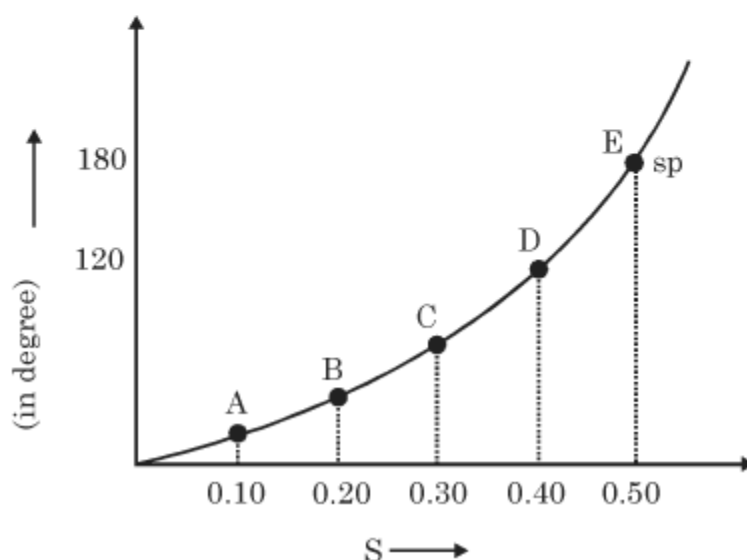
Compound, MX_n type ($n = 2$ or 3 or 4)	Value of $\cos \theta$ ($\theta =$ bond angle between equivalent hybrid orbitals)
P	-0.241
Q	-0.292
R	-0.5
S	-0.325
T	-0.469

In all expected compounds each central atom only uses its s and p-orbitals in hybridisation. The relationship between bond angle ' θ ' and decimal fraction of s and p character present in the equivalent hybrid orbitals is given by.

$$\cos \theta = \frac{S}{S-1} = \frac{P-1}{P},$$

S = decimal fraction of s-character in the equivalent hybrid orbital

P = decimal fraction of p-character in the equivalent hybrid orbital



50. The correct order of % p-character in bond pairs of central atoms in the following $\oplus$ compounds is

- (A) $P > T > S > Q > R$
- (B) $S > R > T > P > Q$
- (C) $P > Q > S > R > T$
- (D) $P > Q > S > T > R$

Ans. (D)

51. The incorrect statement is

- ⊕ (A) The ratio of % p-character to % s-character is less than four, for the bond pair of central atom of compound S.
 (B) Central atom uses three hybrid orbitals to form compound R.
 (C) Central atom uses four hybrid orbitals to form compound S.
 (D) There are three compounds present between point C to E, according to % s-character in bond pair of central atom.

Ans. (D)

52. Match the column

Column-I
(Molecules/Species)

Column-II
(Characteristics which is/are not correct for the given molecule/species in column-I)

- | | |
|-----------------------------------|--|
| (A) CFCIBrI | (P) All possible bond angles are identical |
| (B) XeF ₄ | (Q) Specie/ molecule is planar |
| (C) SO ₃ ²⁻ | (R) Specie/ molecule is non-polar |
| (D) BFCI ₂ | (S) Specie/ molecule is non-planar |

Ans. [(A) → (P, Q, R); (B) → (P, S); (C) → (Q, R); (D) → (P, R, S)]

53. **Column-I**
(Compound)

Column-II
(Properties)

- | | |
|----------------------------------|--|
| (A) IF ₇ | (P) Non-planar with different M-X bond length. |
| (B) BrF ₃ | (Q) Polar with different M-X bond length. |
| (C) NO ₃ ⁻ | (R) Planar with same M-X bond length. |
| (D) SF ₄ | (S) Non-polar with different M-X bond length. |

Ans. [(A) → (P, S); (B) → (Q); (C) → (R); (D) → (P, Q)]

54. Match the column

Column-I (molecules)

Column-II (correct about molecule)

- | | |
|--|---|
| (A) PCl ₅ (θ = 120°, (θ' = 90°) | (P) Number of the given bond angle (θ or θ') is = One |
| (B) NH ₃ (θ = 107°) | (Q) Number of the given bond angle (θ or θ') is = Three |
| (C) SiH ₄ (θ = 109° 28') | (R) Number of the given bond angle (θ or θ') is = Six |
| (D) SO ₂ (θ = 116°) | (S) d-orbital(s) is/are involved in hybridisation of central atom |

Ans. [(A) → (Q, R, S); (B) → (Q); (C) → (R); (D) → (P)]

55. Match the compounds with the correct statement regarding their hydrolysis reaction.

Column-I

- (A) PCl_3
(B) SO_2Cl_2
(C) NCl_3
(D) H_2SO_5

Column-II

- (P) At least one of the hydrolysis product undergoes tautomerism
(Q) Produces at least one acid having basicity two on hydrolysis
(R) The nucleophilic attack takes place on the surrounding atom during hydrolysis
(S) The nucleophilic attack takes place at the central atoms during hydrolysis

Ans. [(A) → (P, Q, S); (B) → (Q, S); (C) → (R); (D) → (Q, S)]

56. Match the Column.

Column-I

- (A) Dimer of BeCl_2
(B) Dimer of AlBr_3
(C) Solid BeCl_2
(D) XeF_5^-

Column-II

- (P) All atoms are sp^3 hybridised
(Q) Non-planar
(R) Central atoms are electron deficient
(S) All bond angles are of 72°

Ans. [(A) → (R); (B) → (P, Q); (C) → (P, Q); (D) → (S)]

57. Consider the following compounds.

- (i) IF_5
(iii) XeO_2F_2
(v) BCl_3
(vii) AsCl_5
(ix) NO_2^-

- (ii) ClF_4^-
(iv) NH_2^-
(vi) BeCl_2
(viii) BO_3^{3-}

List-I

(Species in which central atom uses valency shell orbital)

- (A) All three p-orbitals are involved in hybridisation of central atom
(B) Only two p-orbitals are involved in hybridisation of central atom
(C) Only one p-orbital is involved in hybridisation of central atom
(D) At least one d-orbital is involved in hybridisation of central atom

List-II

(Number of species out of given above)

- (P) One
(Q) Five
(R) Three
(S) Four

Ans. [(A) → (Q); (B) → (R); (C) → (P); (D) → (S)]

58. Out of following how many species can not exist.

- | | |
|-------------------------------|-----------------------|
| (i) $\text{Sn}(\text{O}_3)_2$ | (ii) BeO_2 |
| (iii) FeH_3 | (iv) Si_6 |
| (v) CuI_2 | (vi) HNO_4 |
| (vii) ClBr_3 | (viii) AsH_5 |
| (ix) HgH | |

Ans. (8)

59. Count the total number of X–O bonds are having equal length in HSO_4^- and $\text{S}_3\text{O}_6^{2-}$ respectively & number of X – X in $\text{S}_3\text{O}_6^{2-}$.

Ans. (362)

60. (A) In $\text{P}(\text{CH}_3)_3(\text{CF}_3)_2$ molecule, the maximum number of atoms may lie in the same plane is and the number of such plane/s may have maximum number of atoms

(B) Number of species in following showing same hybridisation in central atom as found in XeF_4 are :

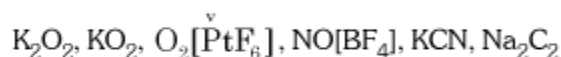


Ans. (732)

61. (a) Out of given ten molecules total molecules which have dipole moment zero is, $\text{IOF}_5, \text{H}_2\text{O}_2, \text{ClF}_3, \text{CO}_2, \text{SO}_2, \text{P}_4, \text{S}_8, \text{PF}_2\text{Cl}_3, \text{N}_2\text{O}_4, \text{SCl}_2$

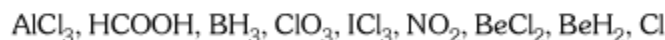
Answer is x.

(b) Find the total number of compound(s) which is/are repelled by magnetic field.



Answer is y.

(c) All the given species undergoes dimer formation, dimerization of which molecule does not require vacant orbital.



Answer is z.

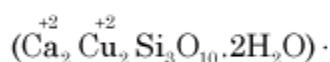
[Give your answer as xyz]

Ans. (544)

62. If x is the number of C–C sigma bonds in C_{60} (fullerene) and y is number of pentagonal rings in C_{60} then find out the value of $(x + 5y)$.

Ans. (150)

63. How many monovalent oxygen atoms are present in the mineral kunoite



Ans. (8)

64. How many maximum number of equivalent P – O bonds are present in hydrogen hypo-phosphate ion.

Ans. (3)

65. Consider $\text{Al}_2(\text{OH})_6$ compound and calculate the value of $(X + Y) \div Z$

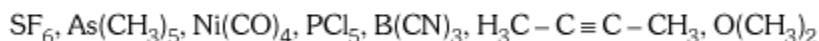
Where X = Total number of $(2c - 2e^-)$ bond.

Y = Total number of $(3c - 2e^-)$ bond

Z = Total number of $(3c - 4e^-)$ bond

Ans. (5)

66. In how many of the given compounds 5 or more atoms (per molecule) lie in single plane?



Ans. (6)

67. Find correct relations.

(i) Acidic oxides $\Rightarrow \text{CO}_2, \text{SO}_2, \text{SO}_3, \text{N}_2\text{O}_5, \text{V}_2\text{O}_5, \text{P}_4\text{O}_{10}$

(ii) Basic oxides $\Rightarrow \text{Na}_2\text{O}, \text{CaO}, \text{BaO}$

(iii) $\text{CO}, \text{NO}, \text{N}_2\text{O} \Rightarrow$ Neutral oxide.

(iv) Amphoteric $\Rightarrow \text{ZnO}, \text{Al}_2\text{O}_3, \text{BeO}, \text{PbO}, \text{Cr}_2\text{O}_3$

(v) $\text{Pb}_3\text{O}_4, \text{Fe}_3\text{O}_4, \text{Mn}_3\text{O}_4 \Rightarrow$ Mixed oxide

(vi) $\text{Na}_2\text{O}_2, \text{K}_2\text{O}_2, \text{CrO}_5, \text{BaO}_2 \Rightarrow$ Peroxide

(vii) $\text{KO}_2, \text{RbO}_2 \Rightarrow$ Super oxide

(viii) $\text{C}_3\text{O}_2, \text{PbO}_2 \Rightarrow$ Suboxide

Ans. (7)

Gaseous State

1. A vessel of volume V is evacuated by means of a piston air pump. One piston stroke captures the volume ΔV . How many strokes are needed to reduce the pressure in the vessel n times? The process is assumed to be isothermal and the gas ideal.

(A) $\frac{\ell n n}{\ell n \left(1 + \frac{\Delta V}{V}\right)}$

(B) $\frac{n \ell n n}{\ell n \left(1 + \frac{\Delta V}{V}\right)}$

(C) $\frac{\ell n n}{\ell n (V + \Delta V)}$

(D) $\frac{n \ell n n}{\ell n (V + \Delta V)}$

Ans. (A)

2. Find the pressure of air in a vessel being evacuated as a function of evacuation time t . The vessel volume is V , the initial pressure is P^0 . The process is assumed to be isothermal and the evacuation rate equal to C and independent of pressure.

[**Note:** The evacuation rate is the gas volume being evacuated per unit time, with that volume being measured under the gas pressure attained by that moment.]

(A) $P = P^0 e^{-\frac{Ct}{V}}$

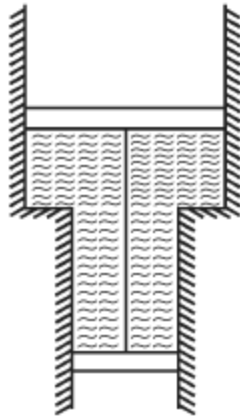
(B) $P = CP^0 e^{-\frac{t}{V}}$

(C) $P = P^0 e^{-\frac{t}{CV}}$

(D) $P = P^0 e^{-\frac{CV}{t}}$

Ans. (A)

3. A smooth vertical tube having two different sections is open from both ends and equipped with two pistons of different area. Each piston slides within a respective tube section. One mole of ideal gas is enclosed between the pistons tied with a non-stretchable thread. The cross-sectional area of the upper piston is 10 cm^2 greater than that of the lower one. The combined mass of the two pistons is equal to 5 kg . The outside air pressure is 1 atm . By how many kelvins must the gas between the pistons be heated to shift the piston through $\ell = 5.0 \text{ cm}$?



- (A) 0.9 K
- (B) 1 K
- (C) 0.7 K
- (D) 0.45 K

Ans. (A)

4. For a real gas that follows the equation of state:

$$\left(P + \frac{a}{TV_m^2}\right)(V_m - b) = RT,$$

where, P = Pressure of the gas,

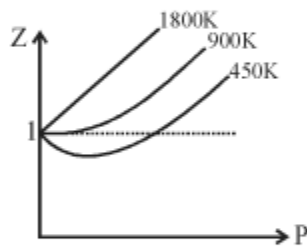
V_m = Molar volume of the gas,

R = Universal gas constant,

T = Absolute temperature,

a & b = Temperature independent constants.

Three isotherms at different temperatures depicting the variation of the compressibility factor Z with the pressure P of the gas are as shown :



For such a gas the ratio b/a is :

- (A) $\frac{1}{900R}$
 (B) $\frac{1}{(900R)^2}$
 (C) $900R$
 (D) $\frac{1}{R(900)^2}$

Ans. (D)

5. At low pressure, if $RT = 2\sqrt{Pa}$, then the volume occupied by a real gas is,

- (A) $\frac{2RT}{P}$
 (B) $\frac{2P}{RT}$
 (C) $\frac{RT}{2P}$
 (D) $\frac{3RT}{P}$

Ans. (C)

6. The correct statement in the following is-
- (A) At Boyle's temperature a gas is ideal at all pressure
 - (B) At Boyle temperature a gas can not liquified
 - (C) Boyle temperature is smaller than critical temp for a given real gas
 - (D) Boyle's temperature of O_2 gas is more then NH_3 gas.

Ans. (B)

7. For a real gas obeying Vanderwaal's equation which of the following is/are correct -

- (A) Internal energy of gas is dependent on temperature only
- (B) If $z > 1$ then forces of repulsion are dominant
- (C) At high pressure $Z = 1 + \frac{Pb}{RT}$
- (D) Work done during expansion of a real gas is more in magnitude than for an ideal gas under isothermal condition in low pressure range under similar condition.

Ans. (BC)

8. A gas obeys the equation : $P = \frac{RT}{V_m - B} - \frac{A}{V_m \cdot T}$,

where A and B are constants ($A > 0$, $B > 0$) and V_m is molar volume of the gas at pressure P and temperature TK. Which of the following is/are correct statement(s) about the gas ?

- (A) The critical temperature of gas is $\frac{8A}{27RB}$
- (B) There is no critical condition in the gas
- (C) The gas is a permanent gas (It can never be liquified)
- (D) The gas may be liquified at any temperature, with proper adjustment of pressure

Ans. (BC)

9. Which of the following is/are **CORRECT**.

- (A) For a gas obeying following the equation : $P[V - nb] = nRT$, internal energy of gas does not change in isothermal expansion.
- (B) For ideal gas under going any process, $C_{p,m} - C_{v,m} = R$
- (C) When different gases He, O_2 , CO_2 , CH_4 are subjected to adiabatic compression from same initial state to same final volume then P-V work is maximum for CH_4 .
- (D) In compression of ideal gas, entropy of system may increase.

Ans. (A, B, D)

10. Vander waal's gas equation may be expressed as $Z=1+\frac{B}{V_m}+\frac{C}{V_m^2}+\dots$ where

V_m = molar volume of gas. If $B = -0.105 \text{ L mol}^{-1}$ and $C = 400 \text{ ml}^2 \text{ mol}^{-2}$ at 127°C the value of vander waal constant 'a' (in $\text{atm L}^2 \text{ mol}^{-2}$) is ($R = 0.08 \text{ L-atm/K-mol}$)

Ans. (4)

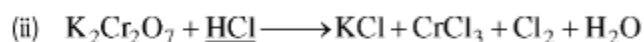
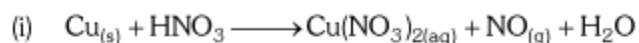
11. Air contains oxygen and nitrogen only and oxygen present 20% by volume in air. If density of air is $d \text{ gm/litre}$ at 0.821 atm pressure and 400 K temperature then calculate

the value of $\frac{100d}{18}$.

Ans. (4)

Redox and Equivalent

1. Equivalent wt. of the underlined substances in the given reaction is respectively (approx).



(A) 63, 36.5

(B) 63, 73

(C) 84, 85

(D) 84, 42.5

Ans. (C)

2. 100 ml of 1 M HCN is titrated with 5 M KMnO_4 in acidic medium, a gas (B) is liberated which turns lime water milky along with an anion which gives brown ring test (central atom in the highest oxidation state) when this gas is passed through lime water 10 gms of a white ppt. is obtained. Find the normality of HCN and volume of KMnO_4 .

(A) 10 N, vol = 40 ml

(B) 5 N, vol = 20 ml

(C) 10 N, vol = 20 ml

(D) 10 N, vol = 80 ml

Ans. (A)

3. 40 ml, 0.05 M solution of sodium sesquicarbonate dihydrate $[\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 \cdot 2\text{H}_2\text{O}]$ is titrated against 0.05 M HCl solution, x ml of acid is required to reach the phenolphthalein end point, while y ml. of same acid was required when methyl orange indicator was used in a separate titration. Which of the following statements are correct ?
- (A) $y - x = 80$ ml
 (B) $y + x = 160$ ml
 (C) If the titration is started with phenolphthalein indicator and methyl orange is added at the end point, 2x ml. HCl would be required further to reach end point.
 (D) $y + x = 120$ ml.

Ans. (ABC)

4. 250 ml of 1M I_2 is separately reacted with 0.5M Cu_2S solution, 0.5M CuS solution and 0.5 M $\text{S}_2\text{O}_3^{2-}$ solution causing production of Cu^{+2} and SO_4^{2-} in the first two and $\text{S}_4\text{O}_6^{2-}$ in the last case along with iodide ions. Which of the following option is/are correct assuming 100% completion of the reaction.
- (A) 100 ml of Cu_2S solution will be consumed.
 (B) 125 ml of CuS solution will be consumed.
 (C) 500 ml of $\text{S}_2\text{O}_3^{2-}$ solution will be consumed.
 (D) Equivalent weight of I_2 in each of the reaction will be 127.

Ans. (A, B, D)

5. Which of the following reaction(s) is/are redox ? Assume aqueous conditions (Ignore balancing)
- (A) $\text{K}_2\text{Cr}_2\text{O}_7 + \text{KOH}_{(\text{aq.})} \longrightarrow$
 (B) $\text{K}_2\text{MnO}_4_{(\text{aq.})} + \text{CO}_2 \longrightarrow$
 (C) $\text{ZnS} + \text{NaCN}_{(\text{aq.})} \longrightarrow$
 (D) $\text{FeCl}_3_{(\text{aq.})} + \text{H}_2\text{S} \longrightarrow$

Ans. (BD)

Paragraph-1

A 3.00 g sample containing Fe_3O_4 , Fe_2O_3 and an inert impure substance, is treated with excess of KI solution in presence of dilute H_2SO_4 . The entire iron is converted into Fe^{2+} along with the liberation of iodine. The resulting solution is diluted to 100 mL. A 20 mL of the diluted solution require 11 mL of 0.5 M $\text{Na}_2\text{S}_2\text{O}_3$ solution to reduce the iodine present. A 50 mL of diluted solution after complete extraction of the iodine requires 12.80 mL of 0.25 M KMnO_4 solution in dilute H_2SO_4 medium for the oxidation of Fe^{2+} .

6. Calculate the percentage (approx) of Fe_2O_3 .

- (A) 48
- (B) 30
- (C) 35
- (D) 65

Ans. (A)

7. Calculate the percentage (approx) of Fe_3O_4 .

- (A) 34
- (B) 68
- (C) 17
- (D) 51

Ans. (A)

8. If 1 mole of Fe_2O_3 and 1 mole of Fe_3O_4 completely reduced by hydrazine and get converted into Fe^{2+} and N_2 . If all Fe^{2+} treated with V ml of 0.1 M KMnO_4 in acidic medium, then find out the V.

- (A) 10 litre
- (B) 100 litre
- (C) 50 litre
- (D) None of these

Ans. (A)

9. A sample of oleum is labelled as 113.5%.

a = mass% of SO_3 in the sample of oleum.

b = mass of H_2SO_4 obtained when 400 gm of oleum is mixed with 60 gm of H_2O .

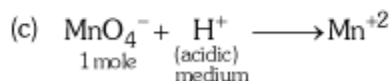
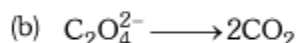
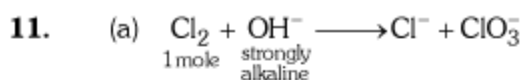
c = Minimum mass of H_2O required to completely react all the SO_3 present in 200 gm of sample.

Calculate sum of the digits in $\left(\frac{a+b+c}{2}\right)$ and fill OMR.

Ans. (5)

10. Na_2O_2 is a strong oxidising agent. It oxidise chromic compound. How many moles of Na_2O_2 are required to convert two moles of $\text{Cr}(\text{OH})_3$.

Ans. (3)



(1) For step 'a' number of equivalent for 1.5 mole Cl_2 is 'x'

(2) For step 'b' number of equivalent for 1 mole CO_2 is 'y'

(3) For step 'c' number of equivalent for 1 mole MnO_4^- is 'z'

Then find $x + \frac{y}{2} + z$

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

Ans. (8)

12. 5.0 g of an alloy of copper was dissolved in one litre of dilute sulphuric acid. In 20 ml of this solution, excess of potassium iodide was added and final solution required 25 ml hypo solution for titration. In 20 ml of potassium dichromate solution (containing 2.4 g of potassium dichromate per litre of solution), sulphuric acid and excess KI were added. This solution required 30 ml of same hypo solution. Calculate the percentage by weight of copper present in the alloy.

[Give your answer as sum of digits before decimal and fill OMR sheet]

Ans. (6)

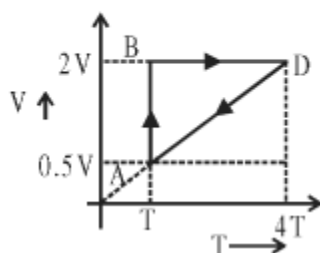
Thermodynamics

1. In a certain polytropic process the volume of argon was increased 4 times. Simultaneously, the pressure decreased 8 times. Find the molar heat capacity of argon in this process, assuming the gas to be ideal.

- (A) 4.2 J/(K.mol)
 (B) -4.2 J/(K.mol)
 (C) -2.1 J/(K.mol)
 (D) 2.1 J/(K.mol)

Ans. (B)

2. Calculate the work done when 1 mole of ideal gas is used for the cyclic process ABD.
(Use : $\ln 2 = 0.7$, $\ln 3 = 1.1$)



- (A) $\frac{8RT}{5}$
(B) $10 RT$
(C) 0
(D) $\frac{7}{10} RT$

Ans. (A)

3. The solubility of $H_2(g)$ and $CO(g)$ in molten $KNO_3/NaNO_3$ is given by

$$\log_{10}(S_{H_2}) = -54 - 78 (T/K)^{-1}$$

$$\log_{10}(S_{CO}) = -6 - 980 (T/K)^{-1}$$

where S is the solubility in $\text{mole cm}^{-3} \text{ bar}^{-1}$. Based on above write up identify the correct statement(s) among the following. (Assume ΔS° and ΔH° do not vary with T)

- (A) $\Delta H_{\text{Solution}}^\circ[CO(g)] > \Delta H_{\text{Solution}}^\circ[H_2(g)]$ (at 500 K)
(B) $\Delta S_{\text{Solution}}^\circ[CO(g)] > \Delta S_{\text{Solution}}^\circ[H_2(g)]$ (at 500 K)
(C) At very high temperature the process $CO(g) + \text{solvent} \rightarrow CO(\text{solution})$ becomes spontaneous process in standard conditions.
(D) At 500 K the molar enthalpy of solution of $H_2(g)$ is 14.7 kJ/mole.

Ans. (AD)

4. Select incorrect statement(s).

- (A) S° values for all elements at 298 K & 1 bar are positive.
(B) For the reaction $2H(g) \longrightarrow H_2(g)$, ΔH and ΔS are positive.
(C) ΔS_{System} is negative for reversible adiabatic compression of an ideal gas.
(D) $|w|_{\text{adiabatic}} < |w|_{\text{isothermal}}$ in compression process for an ideal gas.

Ans. (BCD)

5. Pick out the incorrect statement among the following :
- (A) Heat absorb during a change of state (P, V_1, T_1) to (P, V_2, T_2) in a closed system involving ideal gas must be equal to ΔH .
 - (B) Heat exchanged during a reversible isothermal process between two fixed states is a definite quantity.
 - (C) Complete conversion of heat into work is not possible in a reversible isothermal expansion of ideal gas.
 - (D) Complete consersion of heat into work is not possible in an irreversible isothermal expansion of ideal gas.

Ans. (ACD)

6. Select the incorrect statement(s) :
- (A) ΔH_f° of $\text{CO}_2(\text{g})$ is same as the $\Delta H_{\text{Combustion}}^\circ$ of carbon graphite.
 - (B) The larger and more complex a molecule, the greater is it's absolute entropy.
 - (C) $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g}) + 185\text{kJ}$. On the basis of above information ΔH_f° (HCl, g) is 92.5 kJ/mole.
 - (D) More volatile substance has high boiling point.

Ans. (CD)

7. Assume ideal gas behaviour for all the gases considered & vibrational degree of freedom to be active separate equimolar samples of He , H_2 , SO_2 & CH_4 were subjected to a two step process as mentioned. Initially all are at same state of temperature & pressure.

Step I $\rightarrow$ All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decreases in their temperature.

Step II $\rightarrow$ After step I all are given appropriate amount of heat isochorically to restore the original temperature.

Mark the correct option(s).

- (A) Due to step I only, the decreases in temperature will be maximum for CH_4 .
- (B) During step II, heat given will be maximum for CH_4 .
- (C) There will be no change in internal energy for any of the gas after both the steps of process are completed.
- (D) The P-V graph of CH_4 & SO_2 will be same.

Ans. (BC)

Paragraph-1

From the following information for a A Vander waal's gas answer the following questions.

$$P = \frac{nRT}{V - nb} - \frac{an^2}{V^2}$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P$$

9. $\left(\frac{\partial U}{\partial V}\right)_T$ for the Vander Waal's gas is

(A) Zero

(B) $\frac{an^2}{V^2}$

(C) $-\frac{an^2}{V^2}$

(D) None of these

Ans. (B)

10. Calculate ΔU for the isothermal expansion from $(P_1 V_1 T)$ to $(P_2 V_2 T)$.

(A) Zero

(B) $an^2\left(\frac{1}{V_2} - \frac{1}{V_1}\right)$

(C) $-an^2\left(\frac{1}{V_2} - \frac{1}{V_1}\right)$

(D) None of these

Ans. (C)

11. Calculate ΔH for the above change. (**More than one may correct**)

(A) $\Delta H = -an^2 \left(\frac{1}{V_2} - \frac{1}{V_1} \right) + (P_2V_2 - P_1V_1)$

(B) $\Delta H = (P_2V_2 - P_1V_1)$

(C) $\Delta H = -2an^2 \left(\frac{1}{V_2} - \frac{1}{V_1} \right) + nRT \left[\frac{V_2}{V_2 - nb} - \frac{V_1}{V_1 - nb} \right]$

(D) $\Delta H = nRT \left[\frac{V_2}{V_2 - nb} - \frac{V_1}{V_1 - nb} \right]$

Ans. (A, C)

Paragraph-2

An ideal gas ($C_p/C_v = \gamma$) is expanded so that the amount of heat transferred to the gas is equal to the decrease in its internal energy.

12. What is the molar heat capacity of gas in this process ?

- (A) C_v
(B) $-C_v$
(C) C_p
(D) $2C_v$

Ans. (B)

13. What is the relationship between temperature and volume of gas in this process ?

- (A) $T.V^2(\gamma - 1) = \text{constant}$
(B) $T.V^{\gamma - 1} = \text{constant}$
(C) $T.V^{(\gamma - 1)/2} = \text{constant}$
(D) $T.V^{\gamma} = \text{constant}$

Ans. (C)

14. What is the magnitude of work performed by one mole of the gas when its volume increases eight times if the initial temperature of the gas is 300 K ? C_v for the gas is $1.5 R$. ($R = 2 \text{ Cal/mol/K}$)

- (A) 900 Cal
(B) 450 Cal
(C) 1247.7 Cal
(D) 623.8 Cal

Ans. (A)

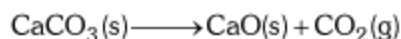
15. Match the following.

Column-I

- (A) Reversible adiabatic compression
 (B) Reversible vaporisation
 (C) Free expansion of ideal gas in vacuum
 (D) Dissociation of

Column-II

- (P) $\Delta S_{\text{System}} > 0$
 (Q) $\Delta S_{\text{System}} < 0$
 (R) $\Delta S_{\text{Surrounding}} < 0$
 (S) $\Delta S_{\text{Surrounding}} = 0$

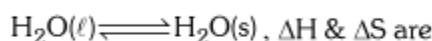


Ans. [(A) → (S); (B) → (P, R); (C) → (P, S); (D) → (P, R)]

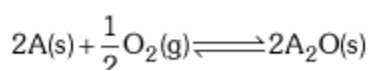
16. Match the following.

Column-I

- (A) For the process



- (B) For the endothermic reaction



ΔS & ΔG are

- (C) $\text{C}(\text{diamond}) \rightleftharpoons \text{C}(\text{graphite})$, favourable conditions for formation of diamond are high pressure and high temperature then ΔH for formation of diamond and ΔS for formation of graphite from diamond are

- (D) For the given reaction



$E_{\text{a (forward)}} = 57.2 \text{ kJ}$ and

$E_{\text{a (backward)}} = 3.2 \text{ kJ}$, ΔH & ΔS for the given reaction

Column-II

- (P) -ve, +ve

- (Q) +ve, +ve

- (R) +ve, +ve

- (S) -ve, -ve

Ans. [(A) → (S); (B) → (P); (C) → (R); (D) → (R)]

17.

Column-I

- (A) Adiabatic expansion of an ideal gas against 1 atm
 (B) Condensation of vapour at normal boiling point
 (C) Adiabatic reversible compression involving an ideal gas
 (D) Isothermal reversible expansion of an ideal gas

Column-II

- (P) $\Delta S_{\text{sys}} > 0$

- (Q) $\Delta U = 0$

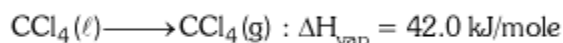
- (R) $\Delta S_{\text{uni}} = 0$

- (S) $w = 0$

- (T) $q = 0$

Ans. [(A) → (P, T); (B) → (R); (C) → (R, T); (D) → (P, Q, R)]

18. The heat of vaporisation : ΔH_{vap} , of CCl_4 at 27°C is 42 kJ/mole



If 1 mole of liquid CCl_4 at 27°C has entropy of 214 J/K mole , what is the entropy (in J/K-mole) of 1 mole of vapour in equilibrium with liquid at this temperature

Ans. (354)

19. One mole of an ideal gas at 300 K is heated at constant volume (V_1) until its temperature is doubled, then it is expanded isothermally till it reaches the original pressure. Finally the gas is cooled at constant pressure till system reached to the half of original volume ($V_1/2$).

Determine total work done ($|w|$) in cal.

[Use : $\ln 2 = 0.70$; $R = 2 \text{ Cal/K.mol}$]

Ans. (060)

20. A system of 100 kg mass undergoes a process in which its specific entropy increases from $0.3 \text{ kJ/kg}^{-1}\text{K}^{-1}$ to $0.4 \text{ kJ/kg}^{-1}\text{K}^{-1}$. At the same time, the entropy of the surrounding decreases from 80 kJ/K^{-1} to 75 kJ/K^{-1} . Find the $(\Delta S)_{\text{universe}}$ in kJ/K^{-1} .

Ans. (5)

Thermochemistry

1. Silicon hydrides $\text{Si}_n\text{H}_{2n+2}$ are called silanes. Most of them contain Si – Si bonds, but they become increasingly unstable as the number of silicon atoms increases. Based on following data calculate bond enthalpy of Si – Si bond of $\text{Si}_n\text{H}_{2n+2}$ ($n = 2$).

Bond enthalpy of H – H bond = 430 kJ/mole

Bond enthalpy of Si – H bond = 300 kJ/mole

$\Delta_f H$ of $\text{Si}(\text{g}) = 450 \text{ kJ/mole}$

$\Delta_f H$ of $\text{Si}_2\text{H}_6(\text{g}) = 80 \text{ kJ/mole}$

(A) 2110 kJ/mole

(B) 310 kJ/mole

(C) 1210 kJ/mole

(D) 1810 kJ/mole

Ans. (B)

Chemical Equilibrium

1. For the reaction takes place at certain temperature $\text{NH}_4\text{HS(s)} \rightleftharpoons \text{NH}_3\text{(g)} + \text{H}_2\text{S(g)}$, if equilibrium pressure is X bar, then $\Delta_r G^\circ$ would be
- (A) $-2 RT \ln X$
 (B) $-RT \ln (X - \ln 2)$
 (C) $-2RT (\ln X - \ln 2)$
 (D) None of these

Ans. (C)

2. **Statement – 1** : Conversion of liquid water to vapour at pressure below it's boiling point pressure is a spontaneous process.

Statement – 2 : All spontaneous process proceed in a direction of increase in entropy of the system.

- (A) Statement–1 is true, statement–2 is true and statement–2 is correct explanation for statement–1.
 (B) Statement–1 is true, statement–2 is true and statement–2 is NOT the correct explanation for statement–1.
 (C) Statement–1 is true, statement–2 is false.
 (D) Statement–1 is false, statement–2 is true.

Ans. (C)

3. If heat of reaction at constant volume exceeds that at constant pressure for an endothermic reaction at 25°C by 1192 Cal, what is the value of ratio K_p / K_c ($R = 2 \text{ cal} = 0.0821 \text{ L atm}$).
- (A) $(0.0821 \times 298)^{-2}$
 (B) (0.0821×298)
 (C) $(0.0821 \times 298)^2$
 (D) Can't be calculated

Ans. (A)

4. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O(s)} \rightleftharpoons \text{CuSO}_4\text{(s)} + 5\text{H}_2\text{O(g)}$ $K_p = 10^{-1} \text{ (atm)}$. 10^{-2} moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O(s)}$ is taken in a 2.5L container at 27°C then at equilibrium

[Take : $R = \frac{1}{12} \text{ litre atm mol}^{-1}\text{K}^{-1}$]

- (A) Moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ left in the container is 9×10^{-3}
 (B) Moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ left in the container is 9.8×10^{-3}
 (C) Moles of CuSO_4 left in the container is 10^{-3}
 (D) Moles of CuSO_4 left in the container is 2×10^{-4}

Ans. (B, D)

5. $2\text{CaSO}_4(\text{s}) \rightleftharpoons 2\text{CaO}(\text{s}) + 2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}), \Delta H > 0$

Above equilibrium is established by taking sufficient amount of $\text{CaSO}_4(\text{s})$ in a closed container at 1600 K. Then which of the following may be correct option (Assume that solid CaSO_4 is present in the container in each case)

- (A) Moles of $\text{CaO}(\text{s})$ will increase with the increase in temperature
- (B) If the volume of the container is doubled at equilibrium then partial pressure of $\text{SO}_2(\text{g})$ will change at new equilibrium
- (C) If the volume of the container is halved particle pressure of $\text{O}_2(\text{g})$ at new equilibrium will remain same
- (D) If two moles of the He gas is added at constant pressure then the mole of $\text{CaO}(\text{s})$ will increase

Ans. (A, C, D)

6. Identify the **correct** statements ?

- (A) Low pressure is favourable for evaporation of $\text{H}_2\text{O}(\text{l})$
- (B) On adding graphite to an equilibria involving $\text{C}_{(\text{s})}$, $\text{O}_{2(\text{g})}$ and $\text{CO}_{2(\text{g})}$ more $\text{CO}_{2(\text{g})}$ will be formed.
- (C) For the reaction : $3\text{Fe}_{(\text{s})} + 4\text{H}_2\text{O}_{(\text{g})} \rightleftharpoons \text{Fe}_3\text{O}_{4(\text{g})} + 4\text{H}_{2(\text{g})}$ K_p and K_C will be same.
- (D) Normal boiling point of water will be more than standard boiling point of water.

Ans. (AD)

7. For the exothermic reaction $\text{A}_{(\text{g})} + \text{B}_{2(\text{g})} \rightleftharpoons 3\text{C}_{2(\text{g})}$ which of the following statements regarding change and effect is /are true

- (A) If inert gas is added at constant pressure condition without affecting temperature then concentration of C_2 at new equilibrium will be more as compared to that at old equilibrium.
- (B) On increase of temperature extent of dissociation of A will decrease.
- (C) On increase in pressure at constant temperature, the reaction will shift in backward direction.
- (D) If $\Delta G_{300}^0 = 0$ for the reaction then $\Delta G_{200\text{K}}^{0.5\text{bar}}$ will be a negative value.

Ans. (B, C, D)

8. In the reaction $\text{C}(\text{s}) + \text{CO}_2(\text{g}) \rightleftharpoons 2\text{CO}(\text{g})$ the equilibrium pressure is 12 atm. If 50% of the CO_2 reacts, then

- (A) K_p will be equal to 4
- (B) K_p will be equal to 16
- (C) The initial pressure = 8
- (D) The partial pressure of CO is 8 atm at equilibrium

Ans. (B, C, D)

9. A vessel at 1000 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on adding graphite. The total pressure was found to be 0.8 atm.
- (A) The equilibrium is achieved as soon as $Q = K$, K value = 0.1 atm^2
- (B) The equilibrium is achieved as soon as $Q = K$, K value = 1.8 atm
- (C) The $p_{\text{CO}_2} = 0.2 \text{ atm}$ at equilibrium
- (D) The $p_{\text{CO}} = 0.6 \text{ atm}$ at equilibrium

Ans. (B, C, D)

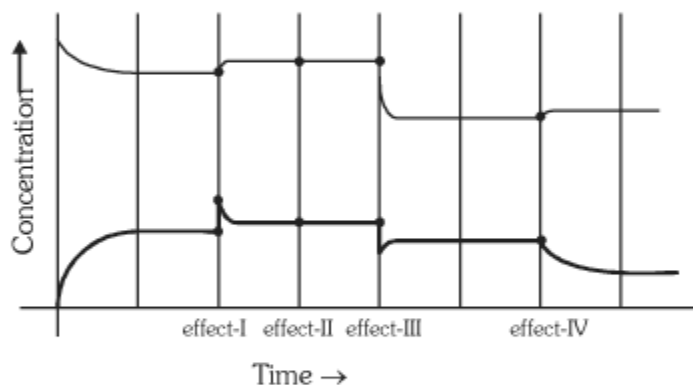
10. The degree of dissociation of PCl_5 at 523 K under atmospheric pressure is 0.798. Then
- (A) $K_p = 1.76 \text{ atm}$
- (B) $K_c = 1.714 \text{ mol/L}$
- (C) $K_c = 3.99 \times 10^{-2} \text{ mol/L}$
- (D) $K_c = 4.1 \times 10^{-2} \text{ mol/L}$

Ans. (A, D)

11. For the given endothermic reaction



The variation in concentration due to different changes is plotted. In column II various changes are given and you are supposed to match the entries in column-I with possible changes the system have undergone. Neglect the slope of change in concentration when system approaches equilibrium. Assume that changes are carried out very fast.



Column-I

- (A) Effect-I
- (B) Effect-II
- (C) Effect-III
- (D) Effect-IV

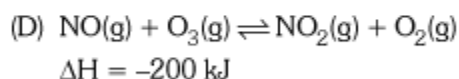
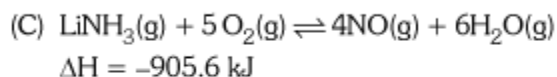
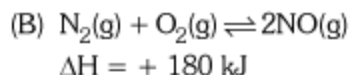
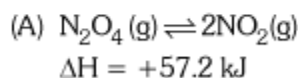
Column-II

- (P) Addition of inert at constant pressure
- (Q) Addition of inert gas at constant volume
- (R) Decreases in temperature at constant volume
- (S) Addition of B only at constant volume

Ans. [(A) → (S); (B) → (Q); (C) → (P); (D) → (R)]

12. Match the following

Column-I



Column-II

(Favourable condition for forward reaction)

(P) Rise of temperature

(Q) Low of temperature

(R) Rise of pressure

(S) Addition of inert gas of constant pressure.

Ans. [(A) → (P, S); (B) → (P); (C) → (Q, S); (D) → (Q)]

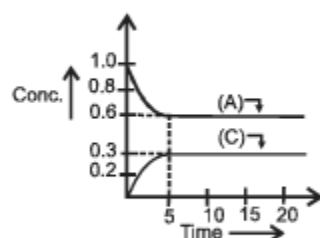
13. A saturated solution of iodine in water contains 0.254 g I_2 / L. More than this can dissolve in a KI solution because of the following equilibrium :

$\text{I}_2(\text{aq}) + \text{I}^-(\text{aq}) \rightleftharpoons \text{I}_3^-(\text{aq})$. A 0.10 M KI solution (0.10 M I^-) actually dissolves 12.954 g of iodine/L, most of which converted to I_3^- . Assuming that the concentration of I_2 in all saturated solutions is the same, calculate the equilibrium constant for the above reaction. What is the effect of adding water to a clear saturated of I_2 in the KI solution ?

[Give your answer sum of digits and Fill OMR sheet]

Ans. (1)

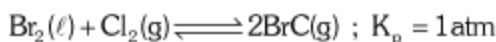
14. For the reaction $\text{A} \xrightleftharpoons[K_b]{K_f} \text{C}$, the concentration and time is shown as



Calculate the value of K_C for the reaction $2\text{C} \rightleftharpoons 2\text{A}$

Ans. (4)

15. For the reaction



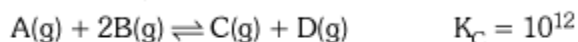
In a closed container of volume 164 L initially 10 moles of Cl_2 are present at 27°C . What minimum mass (in gram) of $\text{Br}_2(\ell)$ must be introduced into this container so that above equilibrium is maintained at total pressure of 2.25 atm. Vapour pressure of $\text{Br}_2(\ell)$ at 27°C is 0.25 atm. Assume that volume occupied by liquid is negligible.

($R = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1}$, Atomic mass of Br = 80)

[Give your answer as sum of digits and fill OMR sheet]

Ans. (8)

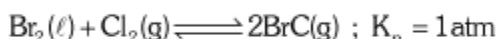
16. For the reaction,



If initial moles of A, B, C and D are 0.5, 1, 0.5 and 3.5 moles respectively in one litre vessel. If equilibrium concentration of B is ($y \times 10^{-4} \text{ M}$) then find y.

Ans. (2)

17. For the reaction



In a closed container of volume 164 L initially 10 moles of Cl_2 are present at 27°C . What minimum mass (in gram) of $\text{Br}_2(\ell)$ must be introduced into this container so that above equilibrium is maintained at total pressure of 2.25 atm. Vapour pressure of $\text{Br}_2(\ell)$ at 27°C is 0.25 atm. Assume that volume occupied by liquid is negligible.

($R = 0.082 \text{ L atm mol}^{-1} \text{ K}^{-1}$, Atomic mass of Br = 80)

[Give your answer as sum of digits and fill OMR sheet]

Ans. (8)

18. For the reaction : $\text{A}(\text{g}) \rightleftharpoons 2\text{B}(\text{g})$, the equilibrium partial pressures of gases A and B is 4 bar and 2 bar, respectively. If the equilibrium mixture is slowly & isothermally compressed to 12 bar, maintaining equilibrium, then the new equilibrium pressure (in bar) of B(g) becomes (Take : $\sqrt{184} = 13.5$)

Ans. (3)

19. One mole of N_2O_4 is present in a vessel at 25°C . At a pressure of 1 atm equilibrium $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$ is attained and the equilibrium constant $K_p = 0.12$. If x moles of helium gas is introduced at constant pressure, then degree of dissociation become 0.3. Calculate the value of x (approximate to nearest integer)

Ans. (3)

1. Equal volume of 0.06 M $\text{Na}_2\text{SO}_4(\text{aq})$ and 0.06 M $\text{KHSO}_4(\text{aq})$ are mixed together. Calculate pH of the solution. Given : $K_{\text{aHSO}_4^-} = 2 \times 10^{-2}$
- (A) 1.7
(B) 2
(C) 2.4
(D) 1.8

Ans. (B)

2. The solubility of $\text{Co}(\text{OH})_3$ in water will be given by which of the following options if its solubility product is $2.7 \times 10^{-43} \text{ M}^4$
- (A) $3 \times 10^{-11} \text{ M}$
(B) 10^{-11} M
(C) $\frac{10^{-7}}{3} \text{ M}$
(D) $2.7 \times 10^{-22} \text{ M}$

Ans. (D)

3. Vinegar is solution of acetic acid. A sample of vinegar contains 3% V/V acetic acid. 100 ml of this vinegar solution is diluted to 250 ml and 25 ml of this diluted solution is titrated against 0.1 M NaOH solution. pH of the resultant solution when 20 ml of titrant was added will be

[Given : Density of pure liquid acetic acid is 1.2 g/ml, K_{a} of acetic = 2×10^{-5} and $\log 2 = 0.3$]

- (A) 4.7
(B) 3.2
(C) 4.4
(D) 5

Ans. (C)

4. What will be the effect of adding 100 ml of 10^{-3} M HCl solution to a solution of HA having concentration 0.1M? (K_{a} of HA = 10^{-5})
- (A) degree of dissociation of HA will decrease pH will remain constant
(B) degree of dissociation of HA will increase and pH decreases.
(C) neither pH nor degree of dissociation of HA will change.
(D) both degree of dissociation and pH will decrease

Ans. (C)

5. 20 ml of 0.8 M-HA solution is neutralised by 0.2 M-NaOH solution, at 25°C. pH of resulting solution at equivalent point is (pK_a of $A^- = 8$, $\log 2 = 0.3$)
- (A) 2.6
(B) 9.6
(C) 4.4
(D) 10.4

Ans. (B)

6. A 2.5 gm impure sample containing weak monoacidic base (Mol. wt. = 45) is dissolved in 100 ml water and titrated with 0.5 M HCl when $\left(\frac{1}{5}\right)^{\text{th}}$ of the base was neutralised the pH was found to be 9 and at equivalent point pH of solution is 4.5. Given : All data at 25°C & $\log 2 = 0.3$.
Select correct statement(s).
- (A) K_b of base is less than 10^{-6}
(B) Concentration of salt (C) at equivalent point is 0.25 M.
(C) Volume of HCl is used at equivalent point is 100 ml.
(D) Weight percentage of base in given sample is 80%.

Ans. (B, C)

7. When 100 ml $Na_3A(0.1\text{ M})$ is titrated against $(0.1\text{ M})\text{ HCl}$
 $K_{a_1} = 10^{-5}$, $K_{a_2} = 10^{-8}$, $K_{a_3} = 10^{-12}$

Tick the **correct** optionS:

- (A) After addition of 150 ml HCl pH = 8
(B) After addition of 200 ml HCl pH = 6.5
(C) After addition of 250 ml HCl pH = 5
(D) After addition of 300 ml HCl pH = 3

Ans. (A, B, C)

8. Equal volumes (10 cm^3) of 0.01 molar solutions of CH_3COOH & $HClO$ were mixed & then diluted to a total volume of 100 cm^3 . Ionisation constant of CH_3COOH is equal to 1.8×10^{-5} & that for $HClO$ is 3.7×10^{-8} . Find correct statements.
- (A) Degree of ionisation for CH_3COOH acid is 0.325.
(B) Degree of ionisation if $HClO$ of the diluted solution would not contain CH_3COOH is 6.08×10^{-3} .
(C) pH value for the solution containing at the same time CH_3COOH & $HClO$ is 3.9.
(D) Degree of ionisation of CH_3COOH is 0.225.

Ans. (B, C)

9. At 25°C, the dissociation constant (K_a) of a weak monoprotic acid, HA is **numerically equal** to the hydrolysis constant (K_h) of its conjugate base, A^- . Which of the following statement(s) is/are correct ?

(A) The dissociation constant of the acid (HA) is 10^{-7} .
 (B) The pH of a 0.1 M aqueous solution of the acid (HA) is 6.0.
 (C) The pH of a 0.1 M aqueous solution of the conjugate base (A^-) is 8.0.
 (D) The pH of an aqueous solution containing 0.1 M – HA and 0.01 M – HCl is 2.0.

Ans. (A, D)

10. K_{sp} of $PbCl_2 = 4 \times 10^{-15}$ at 25°C. Which of the following statement(s) is/are correct ?

(A) Solubility of $PbCl_2$ at 25°C is 10^{-5} M.
 (B) The relation, $[Pb^{2+}] \times 2 = [Cl^-]$, holds good for a pure aqueous solution of $PbCl_2$ at 25°C.
 (C) Solubility of $PbCl_2$ in a 1 M $Pb(NO_3)_2$ solution at 25°C is $10^{-7.5}$ M
 (D) 10^{-6} M solution of pure $PbCl_2$ is unsaturated.

Ans. (A, C, D)

11. How many of the following mixtures can act as buffers when their components are taken in equal concentrations ?

(A) $NaH_2PO_4 + Na_3PO_4$
 (B) $CH_3COONa + HNO_3$
 (C) $H_2CO_3 + Na_2CO_3$
 (D) $CH_3COOH + CH_3COONa$

Ans. (A, B, C, D)

12. For which of the following reactions is the equilibrium constant called an acidity constant ?

(A) $2H_2O \rightleftharpoons H_3O^+ + OH^-$
 (B) $H_2S + H_2O \rightleftharpoons H_3O^+ + HS^-$
 (C) $HONO + H_2O \rightleftharpoons H_3O^+ + NO_2^-$
 (D) $H_3O^+ + CH_3NH_2 \rightleftharpoons CH_3 - \overset{+}{N}H_3 + H_2O$

Ans. (B, C)

13. Choose the correct statement.

(A) pH of acidic buffer solution decrease if more salt is added.
 (B) pH of acidic buffer solution increases if more salt is added.
 (C) pH of basic buffer decreases if more salt is added.
 (D) pH of basic buffer increases if more salt is added.

Ans. (B, C)

14. Which of the following is/are **CORRECT**.

- (A) When 5 mole of H_3PO_4 and 1 mole Na_3PO_4 are taken in 300 ml solution, pH of solution is equal to p^{Ka_1}
- (B) For AS_2S_3 sol, the coagulating power of MgCl_2 is more than that of KCl .
- (C) When radius ratio of Rock salt type structure is equal to its limiting value, then one of the anion is in direct contact with 18 ions.
- (D) Solubility of gas in liquid increases with increase in temperatures at constant pressure

Ans. (A, B, C)

15. Given $K_{\text{sp}} \text{Fe}(\text{OH})_3 = 4 \times 10^{-38}$, $K_{\text{a}}(\text{NH}_4^+) = 10^{-9}$

Which is / are correct statements ?

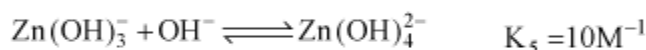
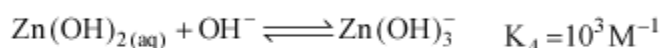
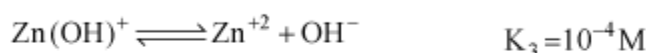
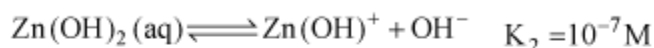
- (A) Solubility of $\text{Fe}(\text{OH})_3$ in water is $4 \times 10^{-17} \text{ M}$
- (B) Solubility of $\text{Fe}(\text{OH})_3$ in 0.01 M NaOH solution is $4 \times 10^{-32} \text{ M}$
- (C) Solubility of $\text{Fe}(\text{OH})_3$ in a buffer solution of 0.05 M NH_4OH and $0.025 \text{ M}(\text{NH}_4)_2\text{SO}_4$ is $4 \times 10^{-28} \text{ M}$
- (D) Solubility of $\text{Fe}(\text{OH})_3$ in 0.01M $\text{Ca}(\text{OH})_2$ solution is $4 \times 10^{-32} \text{ M}$

Ans. (A, B)

Paragraph-1

$\text{Zn}(\text{OH})_2$ is amphoteric in nature. Its solubility at different pH are different :

Given



The total concentration of zinc hydroxide that dissolve in water is given by Dissolved

$$\text{Zn}(\text{OH})_2 = [\text{Zn}^{+2}] + [\text{Zn}(\text{OH})^+] + [\text{Zn}(\text{OH})_2(\text{aq})] + [\text{Zn}(\text{OH})_3^-] + [\text{Zn}(\text{OH})_4^{2-}]$$

16. Which is correct ?

(A) $[\text{Zn}(\text{OH})_2(\text{aq})] = 10^{-6} \text{ M}$

(B) $[\text{Zn}(\text{OH})^+] = 10^{-10} \text{ M}$ at pH = 11

(C) $[\text{Zn}^{+2}] = 10^{-9} \text{ M}$ at pH = 10

(D) All of these

Ans. (D)

17. Solubility of solid $\text{Zn}(\text{OH})_2$ at pH = 5 is :

(A) $2 \times 10^{-4} \text{ M}$

(B) $1.12 \times 10^{-6} \text{ M}$

(C) 10 M

(D) 10^{-4} M

Ans. (C)

Paragraph-2

Consider three metal ion M_a^{+2} , M_b^{+2} , M_c^{+2} and K_{sp} of their sulphides and chlorides

$$K_{sp} \text{ of } M_aS = 5 \times 10^{-30}$$

$$K_{sp} \text{ of } M_aCl_2 = 1.5 \times 10^{-5}$$

$$K_{sp} \text{ of } M_bS = 1.4 \times 10^{-15}$$

$$\left. \begin{array}{l} M_bCl_2 \\ M_cCl_2 \end{array} \right\} \text{ both are freely soluble in water}$$

$$K_{sp} \text{ of } M_cS = 2 \times 10^{-47}$$

18. During qualitative analysis of cations if original solution contain M_a^{+2} , M_b^{+2} and M_c^{+2} ion. Select **CORRECT** options.

- (I) Only M_a^{+2} is precipitated by Ist group reagent
- (II) If only one cation is not precipitated when original solution directly reacts with 2nd group reagent that must be of M_b^{+2}
- (III) After the removal of Ist group cation, M_c^{+2} is selectively precipitated in the form of sulphide from the remaining ions.

- (A) I, II, III
- (B) I, II
- (C) II, III
- (D) I, III

Ans. (A)

19. A solution contain $[M_b(CN)_4]^{2-}$ and $[M_c(CN)_4]^{2-}$ ions, both in 0.5 M concentration and 0.1 mole/litre free cyanide ions. Which metal ion is precipitated in the form of sulphide by introducing $H_2S_{(g)}$?

$$\text{Given : } K_f \text{ of } [M_b(CN)_4]^{2-} = 1.5 \times 10^{17}$$

$$K_f \text{ of } [M_c(CN)_4]^{2-} = 5 \times 10^{28}$$

- (A) Both M_bS and M_cS , simultaneously
- (B) M_bS first and then M_cS
- (C) M_cS first and then M_bS
- (D) Neither M_bS nor M_cS

Ans. (C)

20. Match the column.

Column-I

Column-II

- (A) $\left(\begin{array}{l} 10 \text{ litre of } 0.03 \text{ N } \text{X}(\text{OH})_2 \text{ (strong diacidic base)} \\ + \\ 5 \text{ litre of } 0.08 \text{ M } \text{HNO}_3 \\ + \\ 485 \text{ litre of } 0.01 \text{ M } \text{NaNO}_3 \end{array} \right)$ (P) $\text{pH} \approx 3.7$
- (B) $\left(\begin{array}{l} 10 \text{ ml of } 0.5 \text{ M } \text{RNH}_3\text{Cl} (\text{K}_h = 10^{-9}) \\ + \\ 40 \text{ ml of } 0.125 \text{ M } \text{KOH} \end{array} \right)$ (Q) $\text{pH} \approx 11$
- (C) $\left(\begin{array}{l} 100 \text{ ml of } 0.8 \text{ M } \text{HCO}_3^- \\ + \\ 100 \text{ ml of } 0.4 \text{ M } \text{CO}_3^{2-} \\ \text{(for } \text{H}_2\text{CO}_3, \text{ use } \text{K}_{a1} = 4 \times 10^{-7} \text{ \& } \text{K}_{a2} = 4 \times 10^{-11}) \end{array} \right)$ (R) $\text{pH} \approx 7$
- (D) Saturated aqueous of $\text{Co}(\text{OH})_3$ ($\text{K}_{\text{sp}} = 2.7 \times 10^{-43}$) (S) $\text{pH} \approx 10$

Ans. [(A) $\rightarrow$ (P); (B) $\rightarrow$ (Q); (C) $\rightarrow$ (S); (D) $\rightarrow$ (R)]

21. Match the Column.

Column-I

Column-II

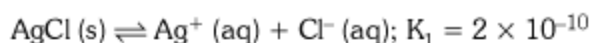
(Solution)

(pH value)

- (A) pH at which $\text{Mg}(\text{OH})_2$ just start precipitating from $2.0 \times 10^{-3} \text{ M}$ Mg^{2+} solution. (P) 5.0
(K_{sp} of $\text{Mg}(\text{OH})_2 = 2 \times 10^{-6} \text{ M}^3$)
- (B) Maximum pH at which 0.1 mole of $\text{Al}(\text{OH})_3$ dissolves completely as Al^{3+} in 1L solution. (Q) 9.0
(K_{sp} of $\text{Al}(\text{OH})_3 = 1 \times 10^{-28} \text{ M}^4$)
- (C) pH at which 0.1 M CH_3COOH dissociates $\frac{1}{10}\%$ (R) 12.5
(K_a of $\text{CH}_3\text{COOH} = 1.0 \times 10^{-5} \text{ M}$)
- (D) pH of $3 \times 10^{-5} \text{ M}$ CaA_2 solution (S) 4
(CaA_2 dissociates completely in water)
(K_a of $\text{HA} = 5 \times 10^{-9}$)

Ans. [(A) $\rightarrow$ (R); (B) $\rightarrow$ (P); (C) $\rightarrow$ (S); (D) $\rightarrow$ (Q)]

22. AgCl is dissolved in some NaCl solution and the following equilibria exist.



Neglect hydrolysis of Ag^+ .

Calculate sum of solution codes for those statements, which are correct.

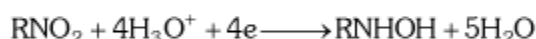
Solution codes Statements

- | | |
|----|--|
| 08 | The solubility of AgCl will be always less in NaCl solution compared to its solubility in water. |
| 13 | If solubility of AgCl is 'S' then $S = \frac{K_1}{[\text{Cl}^-]} + K_2 [\text{Cl}^-]$ |
| 27 | For solubility of AgCl to be minimum, $[\text{Cl}^-] = 5 \times 10^{-5}$ |
| 35 | If complex formation is not occurring, then solubility of AgCl will increase as $[\text{Cl}^-]$ increases. |

[Give your answer as sum of digits and Fill OMR sheet]

Ans. (4)

23. The electrolytic reduction of an organic nitro compound was carried out in a solution buffered by acetic acid and sodium acetate. The reaction was



300 ml of a 0.0100 M solution of RNO_2 buffered initially at pH 5.00 was reduced, with the reaction above going to completion. The total acetate concentration, $[\text{HOAc}] + [\text{OAc}^-]$ was 0.50 M. Calculate $[\text{H}^+]$ of the solution after the reduction is

complete. Express of answer in terms of $[\text{H}^+] = \frac{19 \times 10^{-x}}{28}$. Calculate the value of x.

[Given $K_{\text{aHOAc}} = 2 \times 10^{-5}$]

Ans. (5)

24. $^+\text{NH}_3 - \text{CH(R)} - \text{CO}_2\text{H}$ has its $K_{\text{a1}} = 10^{-7}$. The isoelectric point of the α -amino acid occur at pH = 8.2. If $\text{p}K_{\text{a2}}$ of the conjugate acid of α -amino acid is equal to X, what will be 10 X.

[Hint : If $\text{p}K_{\text{a2}}$ is 2.4 write 0024 in OMR sheet]

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

Ans. (4)

25. How many millimoles of NH_3 must be added to a one litre of 0.004 M- Ag^+ solution to just prevent the precipitation of AgCl, when $[\text{Cl}^-]$ reaches 0.001 M?

[Give : K_{sp} of AgCl = 1.0×10^{-10} and $K_{\text{instability}}$ of $\text{Ag}(\text{NH}_3)_2^+ = 1.0 \times 10^{-8}$]

[Divide your answer by 7 and Fill OMR sheet]

Ans. (4)

- 26.** In the titration of a solution of a weak base BOH with HCl the pH is 8.2 after 10 ml of HCl solution has been added and 7.6 after 20 ml of HCl solution has been added. Calculate the value of pK_b of the base and express your answer as ab.cd.

[For example if your answer is 3.8 the fill OMR as 4]

Ans. (6)

- 27.** 500 ml 0.02M $AgNO_3(aq)$ solution and 500 ml 0.02 M $HCN(aq)$ solution are mixed together. If concentration of Ag^+ at equilibrium is $y \times 10^{-z}$ M then calculate the value of $y + z$.

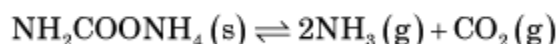
Auto protolysis constant of water = 10^{-14}

Given : $K_{sp}(AgCN) = 4 \times 10^{-16}$

Hydrolysis constant of NaCN = 10^{-4}

Ans. (6)

- 28.** Solid NH_2COONH_4 is heated in a closed rigid container of volume 8.21 Litre at 500 K temperature as following



If equilibrium pressure is 1.5 atm and all moles of NH_3 present at equilibrium dissolved in one litre water and this solution is titrated with 2000ml 0.1 M HCl solution. Now 1000 ml 0.1M NaOH added. Calculate pH of final solution

(Given : $pK_a(NH_4^+) = 9$)

Ans. (9)

- 29.** The solubility of CH_3COOAg ($K_{sp} = 10^{-10}$) in a buffer solution of pH equal to 5 is

$(\sqrt{x}) \times 10^{-y}$. Calculate the value of $(x + y)$ Given : $K_b(CH_3COO^-) = 10^{-9}$. x and

y are integer where $0 < x < 10$.

Ans. (7)

1. **S₁** : Concentrated solution of an alkali metal in liquid ammonia is bronze coloured and diamagnetic.

S₂ : Increasing stability of the peroxide or super oxide of alkaline earth metal down the group, is due to the stabilisation of large anions by large cations through lattice energy effects.

S₃ : In the vapour phase BeCl_2 tends to form a chloro - bridged dimer.

(A) T T T

(B) FFF

(C) TFT

(D) FTF

Ans. (A)

2. **Consider the following statements,**

S₁ : Ozone reacts with dry KOH forming a deep red coloured compound and liberating oxygen gas at low temperature.

S₂ : KO_2 is better oxidant than K_2O_2 both in aqueous and acidic medium.

S₃ : KOH is better absorber of CO_2 than NaOH.

S₄ : Quick lime on heating with carbon in an electric furnace forms CaCO_3 .
and arrange in the order of true/false.

(A) T T T F

(B) T T F T

(C) F F T F

(D) F T T F

Ans. (A)

3. Which of the following statement is incorrect :

(A) Temporary hardness is due to presence of bicarbonate of Ca and Mg.

(B) Temporary hardness can be removed by adding CaO

(C) Permanent hardness is due to presence of sulphates and chlorides of both Ca and Mg.

(D) Permanent hardness can be removed by adding CaO

Ans. (D)

4. The solubility of I_2 is more in KI than water because

(A) It is due to formation of KI_3

(B) KI_3 is soluble in water

(C) It is due to formation of K_2I_4

(D) None of these

Ans. (A, B)

5. A compound 'X' on heating gives a colourless gas which is not supporter of combustion. The residue is dissolved in water and excess of SO_2 is passed through it. First we get white ppt which dissolves in excess of SO_2 . The compound 'X' cannot be
- (A) CaCO_3
 - (B) NaNO_3
 - (C) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
 - (D) K_2CO_3

Ans. (B, C, D)

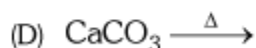
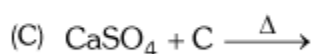
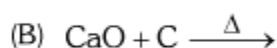
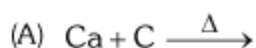
6. Which of the following elements form various types of oxides only when reacted with air ?
- (A) Li, Na
 - (B) K, Na
 - (C) Rb, K
 - (D) Li, K

Ans. (B, C)

7. Which of the following is known as fusion mixture ?
- (A) $\text{Na}_2\text{CO}_3 + \text{K}_2\text{CO}_3$
 - (B) $\text{Na}_2\text{CO}_3 + \text{KOH}$
 - (C) $\text{K}_2\text{CO}_3 + \text{KHCO}_3$
 - (D) $\text{NaOH} + \text{KOH}$

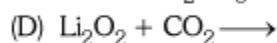
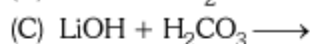
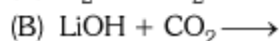
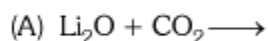
Ans. (A)

8. In which of the following methods, CaC_2 is obtained as product or one of the products ?



Ans. (A, B)

9. Which of the following are suitable methods for the preparation of Li_2CO_3 ?



Ans. (AD)

10. When KO_2 reacts with water, the products are :

- (A) KOH
- (B) H_2O_2
- (C) K_2O_2
- (D) O_2

Ans. (ABD)

11. Which of the following give NO_2 on heating :

- (A) LiNO_3
- (B) NH_4NO_3
- (C) KNO_3
- (D) $\text{Ca}(\text{NO}_3)_2$

Ans. (AD)

Paragraph-1

Alkali metals oxide are obtained by combustion of the metals. Although Na normally gives Na_2O , it will take up further oxygen at elevated pressure and temperatures to form NaO_2 . The per and superoxides of the heavier alkalis can also be prepared by passing stoichiometric amounts of oxygen into their solution in liquid ammonia.

The different alkali metal oxides can be distinguished by reaction with water. The superoxides reacts with CO_2 and give oxygen gas. The stability of per and superoxides is based upon that larger cation can stabilise larger anion, due to larger lattice energy.

Alkali metals dissolve in liquid ammonia. Dilute solutions are dark blue in colour but as the concentration increases above 3M, the colour changes to copper bronze and the solution acquires the metallic lustre due to the formation of metal ions clusters. The solution of alkali metals in liquid ammonia are good conductors of electricity due to the presence of ammoniated cations and ammoniated electrons. However, the conductivity decreases as the concentrations increases, since ammoniated electrons and ammoniated cation associate.

12. Solution of sodium metals in liquid ammonia is strongly reducing due to the presence of :

- (A) Sodium hydride
- (B) Sodium atoms
- (C) Sodium amide
- (D) Solvated electrons.

Ans. (D)

13. KO_2 is used in oxygen cylinders in space and submarines because it.

- (A) Eliminates moisture
- (B) Absorbs CO_2 only
- (C) Absorbs CO_2 and increases O_2 contents
- (D) Produces ozone.

Ans. (C)

- 14.** Select the correct choice for alkali metal oxides.
- (A) Metal oxides reacts with water forming only metal hydroxides.
 (B) Metal peroxides reacts with water forming metal hydroxides and oxygen gas.
 (C) Metal superoxides reacts with water forming metal hydroxide, Hydrogen peroxide and O_2 gas.
 (D) All of these

Ans. (D)

- 15. Match the following :**

Column (A) Reaction.	Column (B) Product's character.
(A) $2 Na_2O \xrightarrow{400^\circ C}$	(P) Diamagnetic.
(B) $KOH + O_3 \longrightarrow$	(Q) Paramagnetic.
(C) $Na + O_2 (excess) \xrightarrow{350^\circ C}$	(R) Bond order 1
(D) $K (dissolved\ in\ liquid\ NH_3) \xrightarrow{3O_2}$	(S) Bond order 1.5

Ans. [(A) $\rightarrow$ (P, Q, R); (B) $\rightarrow$ (Q or Q, S); (C) $\rightarrow$ (P, R); (D) $\rightarrow$ (P, Q, R, S)]

- 16.** Total number of moles of water in 1 mole of Epsom salt is :

Ans. (7)

- 17.** Out of Be, Mg, Ca, Sr and Ba, number of elements which do not impart any colour to the flame, is :

Ans. (2)

- 18.** How many total no. of correct statements :

- (a) BeO is almost insoluble but $BeSO_4$ is soluble in water
 (b) KI is more soluble than LiI in ethanol
 (c) $BaSO_4$ insoluble but BaO is soluble in water
 (d) Li can form hydrated salt
 (e) Cs has least melting point among alkali metals
 (f) LiF insoluble in water but LiCl soluble in water and acetone
 (g) K_2CO_3 can be prepared by solvay process

Ans. (5)

19. How many statements **CORRECT** ?

- (i) In the pure state H_2O_2 is almost colorless (very pale blue)
- (ii) Magnesium is an essential constituent of chlorophyll, the green colouring matter of plants.
- (iii) Calcium bicarbonate is known in solid state.
- (iv) BeH_2 is an ionic hydride.
- (v) In the electrolysis of fused calcium hydride, hydrogen is liberated at cathode.
- (vi) $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ on heating forms MgCl_2 .

Ans. (2)

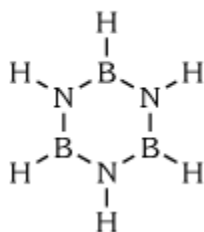
20. How many of following are correctly matched (increasing order of indicated property)

- | | |
|--|--|
| (i) Thermal stability | $\text{BeSO}_4 < \text{MgSO}_4 < \text{CaSO}_4$ |
| (ii) Polarising power | $\text{Ca}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$ |
| (iii) Solubility in H_2O | $\text{Be}(\text{OH})_2 < \text{Mg}(\text{OH})_2 < \text{Ca}(\text{OH})_2$ |
| (iv) Covalent nature | $\text{CaCl}_2 < \text{MgCl}_2 < \text{BeCl}_2$ |
| (v) Hydrolysis nature | $\text{CaCl}_2 < \text{MgCl}_2 < \text{BeCl}_2$ |
| (vi) Lattice energy | $\text{BaF}_2 < \text{CaF}_2 < \text{MgF}_2$ |
| (vii) Hydration energy | $\text{Ba}^{2+} < \text{Mg}^{2+} < \text{Be}^{2+}$ |
| (viii) Solubility in water | $\text{BaF}_2 < \text{MgF}_2 < \text{BeF}_2$ |
| (ix) Basic nature | $\text{Be} < \text{Mg} < \text{Ca} < \text{Sr}$ |

Ans. (9)

P Block

1. The structure of $\text{B}_3\text{N}_3\text{H}_6$ is as follows



How many derivative structure of $\text{B}_3\text{N}_3\text{H}_4\text{X}_2$ can be derived from the basic structure, by the replacement of two hydrogen atoms ?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

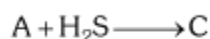
Ans. (C)

2. Which of the following option is correctly matched ?

- (A) S_3O_9 ; contains no S-S linkage
- (B) $S_2O_6^{2-}$; contains -O-O- linkage
- (C) $(HPO_3)_3$; contains P-P linkage
- (D) $S_2O_8^{2-}$; contains S-S linkage

Ans. (A)

3. $SF_4 \xrightarrow{\text{Hydrolysis}} A + B$



The 'C' is

- (A) Solid sulphur
- (B) Colloidal sulphur
- (C) SO_3
- (D) Gaseous sulphur

Ans. (B)

4. Which one of the following is the correct statement?

- (A) Boric acid is a protonic acid
- (B) Both Tl^{3+} and Al^{3+} ion act as oxidising agent in aqueous solution
- (C) Hydrogen bonding in H_3BO_3 gives it a 2D sheet like structure.
- (D) $B(OEt)_3$ imparts blue colour to the burner flame.

Ans. (C)

5. Boric acids is a weak acid in aqueous solution. But its acidity increases significantly in the presence of ethylene glycol, because

- (A) ethylene glycol releases additional H^+ .
- (B) $B(OH)_4^-$ is consumed in forming a compound with ethylene glycol.
- (C) ethylene glycol neutralizes H^+ released by boric acid.
- (D) Boric acid dissociates better in the mixed solvent.

Ans. (B)

6. Which of the following compounds are the common product/s obtained in the hydrolysis of XeF_6 and XeF_4 .

- (A) XeO_2F_2
- (B) HF
- (C) XeO_3
- (D) Both (B) & (C)

Ans. (D)



Y is

- (A) C_3O_2
- (B) CO
- (C) CO_2
- (D) C_{12}O_9

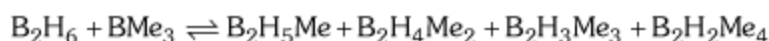
Ans. (D)

8. White phosphorus can be separated from red phosphorus by :

- (A) sublimation
- (B) dissolving in CS_2
- (C) distillation
- (D) heating with alkali solution.

Ans. (B)

9. Consider the methyl substituted diboranes in the given reaction :

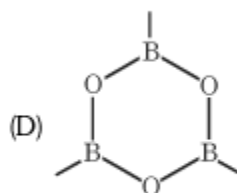
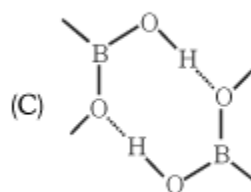
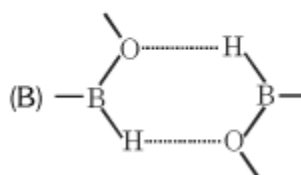
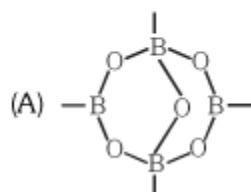


Which of them will give more than one type of boron containing Hydrolysis products ?

- (A) $\text{B}_2\text{H}_5\text{Me}$, $\text{B}_2\text{H}_4\text{Me}_2$, $\text{B}_2\text{H}_3\text{Me}_3$ and $\text{B}_2\text{H}_2\text{Me}_4$
- (B) $\text{B}_2\text{H}_5\text{Me}$ and $\text{B}_2\text{H}_4\text{Me}_2$
- (C) $\text{B}_2\text{H}_5\text{Me}$ and $\text{B}_2\text{H}_3\text{Me}_3$
- (D) $\text{B}_2\text{H}_4\text{Me}_2$ and $\text{B}_2\text{H}_2\text{Me}_4$

Ans. (C)

10. In solid state structure of boric acid which of the following bonding arrangement is observed between closest boron atoms :



Ans. (C)

11. In the structure of borax :
- (A) 5 B–O–B bonds are present
 - (B) The number of σ bonds are 34
 - (C) 19 atoms are sp^3 and another 2 atoms are sp^2 hybridized
 - (D) The number of lone pair's are 34

Ans. (A, B, C, D)

12. $Ca_2B_6O_{11} + Na_2CO_3 \xrightarrow{\Delta} [X] + CaCO_3 + NaBO_2$ (unbalanced equation)
- (A) structure of anion of crystalline [X] has one boron atom sp^3 hybridised and other three boron atoms sp^2 hybridised.
 - (B) X with NaOH (aq.) gives a compound which on reaction with hydrogen peroxide in alkaline medium yields a compound used as brightner in soaps.
 - (C) hydrolysis of [X] with HCl or H_2SO_4 yields a compound which on reaction with HF gives fluoroboric acid
 - (D) [X] on heating with chromium salts in oxidising flame gives green coloured bead in cold.

Ans. (B, C, D)

13. Nitrogen (I) oxide is produced by :
- (A) thermal decomposition of ammonium nitrate.
 - (B) disproportionation of N_2O_4 .
 - (C) thermal decomposition of ammonium nitrite.
 - (D) interaction of hydroxyl ammine and nitrous acid

Ans. (A, D)

14. The compound(s) which disproportionate(s) in alkaline medium is/are :
- (A) $(CN)_2$
 - (B) S_8
 - (C) NO_2
 - (D) P_4

Ans. (A, B, C, D)

15. Which of the following statements is/are incorrect.
- (A) Monoclinic sulphur is soluble in CS_2 .
 - (B) Rhombic sulphur is yellow in color & insoluble in water.
 - (C) S_2 is paramagnetic like O_2 .
 - (D) $4HCl + O_2 \xrightarrow{CuCl_2} 2Cl_2 + 2H_2O$

Ans. (A, B, C, D)

16. Fluorine reacts with water and may give :

- (A) HF
- (B) O₂
- (C) O₃
- (D) HFO₃

Ans. (A, B, C)

17. XeOF₄ is prepared by

- (A) partial hydrolysis of XeF₄
- (B) partial hydrolysis of XeF₆
- (C) reaction of XeO₃ with XeF₆
- (D) reaction of oxygen with XeF₄

Ans. (B, C)

18. The oxide(s) of chlorine is/are colourless oily liquid and acidic

- (A) Cl₂O
- (B) ClO₂
- (C) Cl₂O₅
- (D) Cl₂O₇

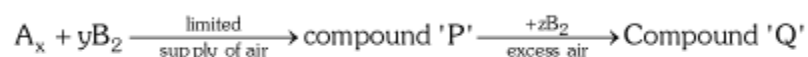
Ans. (D)

19. Which of the following statements is/are incorrect.

- (A) Moist ammonia gas can be dried by using anhydrous CaCl₂.
- (B) When metallic copper react with very dilute nitric acid, ammonium nitrite is one of the products.
- (C) Phosphine is a weaker base than ammonia because phosphorous atom is less electronegative than nitrogen.
- (D) NaH₂PO₂ is an acidic salt and it is named as sodium hypophosphate.

Ans. (A, B, C, D)

20. Consider the following reactions :



If atomic number of elements A and B are 15 and 8 respectively, then according to the given information the correct statement(s) is/are

- (A) (B – A – B) bond angle in compound 'Q' > (B – A – B) bond angle in compound 'P'
- (B) (A – B) bond length in compound 'Q' < (A – B) bond length in compound 'P'
- (C) Value of x + y + z is 9
- (D) Compound 'P' is P₂O₅ and compound 'Q' is P₄O₁₀

Ans. (A, B, C)

21. Which of the following statements is/are **CORRECT** w.r.t Boron for its compounds.
- (A) Boric acid with NaOH does not give a sharp end point but when taken along with sugars it gives a sharp end point with NaOH.
 - (B) Aqueous solution of Borax shows buffer properties.
 - (C) B_2H_6 on reaction with excess NH_3 at low temperatures forms an ionic compound which on heating gives an aromatic compound which is less reactive than benzene.
 - (D) On hydrolysis of BF_3 , boric acid along with fluoborates are obtained.

Ans. (A, B, D)

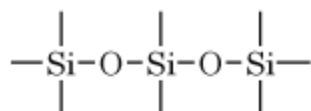
22. Select the correct reaction for H_2O_2 :

- (A) $Ag_2O + H_2O_2 \longrightarrow 2Ag + H_2O + O_2$
- (B) $Cl_2 + H_2O_2 \longrightarrow 2HCl + O_2$
- (C) $O_3 + H_2O_2 \longrightarrow H_2O + 2O_2$
- (D) $H_2O_2 + 2KI \longrightarrow 2KOH + I_2$

Ans. (A, B, C, D)

23. Silicones are a group of organo silicon polymer, select correct statement(s) about them.

- (A) R_3SiCl produce very complex cross linked polymeric silicones on hydrolysis.
- (B) Their water repellency arises because a silicone chain is surrounded by organic side groups.
- (C) Their strength and interness are related to their stable silica like skeleton of



- (D) Hydrolysis product of CH_3SiCl_3 will block the end of the straight chain produce by hydrolysis of $(CH_3)_2SiCl_2$

Ans. (B, C)

24. Which of the following reaction(s) is/are written correctly (w.r.t products) ?

- (A) $2FeSO_4 + H_2SO_4 + Cl_2 \longrightarrow Fe_2(SO_4)_3 + 2HCl$
- (B) $Na_2SO_4 + Cl_2 + H_2O \longrightarrow Na_2SO_4 + 2HCl$
- (C) $SO_2 + 2H_2O + Cl_2 \longrightarrow H_2SO_4 + 2HCl$
- (D) $I_2 + 6H_2O + 5Cl_2 \xrightarrow{\text{(excess)}} 2HIO_3 + 10HCl$

Ans. (A, B, C, D)

25. Consider the following table :

Compound	Enthalpy of formation (kJ mol ⁻¹)	Bond angle (H-C.A.-H)	Boiling point (°C)
H ₂ O	x ₁	y ₁	z ₁
H ₂ S	x ₂	y ₂	z ₂
H ₂ Se	x ₃	y ₃	z ₃
H ₂ Te	x ₄	y ₄	z ₄

According to given information the correct order is/are :

(A) $x_1 > x_2 > x_3 > x_4$

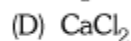
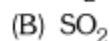
(B) $y_1 > y_2 > y_3 > y_4$

(C) $z_1 > z_2 > z_3 > z_4$

(D) $x_1 > x_4 > x_3 > x_2$

Ans. (A, B, C)

26. Parmanent bleaching is possible by :



Ans. (A, C)

27. When Na₂S₂O₃ is treates with Cl₂ and I₂ separately then identify true statements.

(A) Same compound of 'sulphur is formed in both the reaction as the product.

(B) Reaction with I₂ is disproportionation reaction

(C) In the product formed on reaction with I₂, two S-atoms are in + 5 oxidation state and 2 S-atoms are in zero oxidation state

(D) In the product formed by the reaction with Cl₂, S-atom is in its maximum possible oxidation state.

Ans. (C, D)

28. Which is/ae true for sulphur

(A) $\Delta_f H^\circ$ of monoclinic sulphur > 0

(B) Both α -S and β -S has S₈ molecules

(C) on heating about 1000 k it turns into paramagnetic gas

(D) At 369 K both α and β forms are stable.

Ans. (A, B, C, D)

- 29.** An organic acid (**A**) reacts with concentrated H_2SO_4 to give a neutral oxide (**B**), acidic oxide (**C**) and a diatomic oxide (**D**). When (**D**) reacts with chlorine gas, a poisonous gas (**E**) is evolved. This gas (**E**) with ammonia gives an organic compound (**F**). Then identify the true statements.

- (A) (**A**) = $\text{H}_2\text{C}_2\text{O}_4$ and (**F**) = NH_2CONH_2
 (B) (**A**) = HCOOH and (**F**) = NH_2CONH_2
 (C) In (**E**) central atom is in sp^2 hybridisation
 (D) (**C**) is a linear molecule

Ans. (A, C, D)

- 30.** Which of the following statement(s) is/are correct ?

- (A) Chlorine dioxide (ClO_2) is powerful oxidising agent but bleaching action is lower than Cl_2
 (B) Cl_2 in alkaline solution undergoes disproportionation
 (C) ClO_2 is diamagnetic in nature
 (D) ClO_2 is a yellow gas but deep red liquid

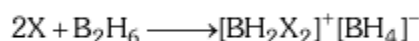
Ans. (B, D)

- 31.** Which of the following order(s) is/are incorrect ?

- (A) $\text{H}_3\text{PO}_4 > \text{H}_3\text{PO}_3 > \text{H}_3\text{PO}_2$ (reducing character)
 (B) $\text{N}_2\text{O} < \text{NO} < \text{N}_2\text{O}_5$ (oxidation state on nitrogen atom)
 (C) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$ (basicity)
 (D) $\text{SbH}_3 > \text{NH}_3 > \text{AsH}_3 > \text{PH}_3$ (reducing character)

Ans. (A, D)

- 32.** In the reaction



the amine(s) X is (are) :

- (A) NH_3
 (B) CH_3NH_2
 (C) $(\text{CH}_3)_2\text{NH}$
 (D) $(\text{CH}_3)_3\text{N}$

Ans. (B, C)

- 33.** Be and Al show diagonal relationship and thus :

- (A) Their oxides are soluble in alkali solution forming $[\text{Be}(\text{OH})_4]^{2-}$
 (B) They form complex anion BeF_4^{2-} and AlF_6^{3-}
 (C) BeCl_2 and AlCl_3 are Lewis acids
 (D) Oxides are only basic

Ans. (A, B, C)

34. The reactions which produce N_2 gas are :

- (A) $H_2N-SO_3H + HNO_2 \rightarrow$
- (B) $CO(NH_2) + HNO_2 \rightarrow$
- (C) $CS(NH_2)_2 + HNO_2 \rightarrow$
- (D) $NO_2^- + N_3^- + 2H^+ \rightarrow$

Ans. (A, B, C, D)

35. The compound(s) which disproportionate(s) in alkaline medium is/are :

- (A) $(CN)_2$
- (B) S_8
- (C) NO_2
- (D) P_4

Ans. (A, B, C, D)

36. Match the product of following reactions.

List-I

- (I) $NH_2CONH_2 + HNO_2 \rightarrow$
- (II) $CaCN_2 + H_2O \rightarrow$
- (III) $CaOCl_2 + NH_3 \rightarrow$
- (IV) $N_2H_4 + HNO_2 \rightarrow$

List-II

- (i) N_2
- (ii) CO_2
- (iii) H_2O
- (iv) NH_3
- (v) CaC_2
- (vi) $CaCO_3$
- (vii) N_3H
- (viii) $Ca(OH)_2$
- (ix) $CaCl_2$

- (A) (I) – (i), (ii), (iii)
- (B) (II) – (iv), (vi)
- (C) (III) – (i), (iii), (ix)
- (D) (IV) – (iii), (vii)

Ans. (A, B, C, D)

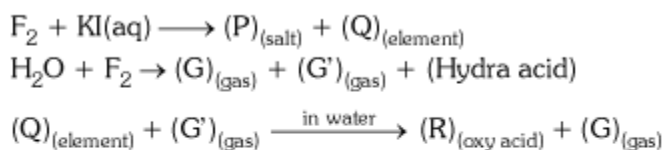
37. $NO_2^- + Zn \xrightarrow{OH^-} X_{(g)}$

Which of the following statement(s) is/are **CORRECT** for $X_{(g)}$:

- (A) It produces white fumes with HCl
- (B) Hybridisation of central atom is sp^3
- (C) It has pyramidal geometry
- (D) It is non planar

Ans. (A, B, C, D)

38. Which of the compound(s) is/are involved in given reaction sequence ?



- (A) KF
(B) I₂
(C) O₂
(D) HIO₃

Ans. (A, B, C, D)

39. Which of the following options is **CORRECT** ?

- (A) Tailing of mercury related to test of Ozone.
(B) Carbon suboxide (C₃O₂) is linear molecule and prepared by dehydration of Malonic acid
(C) K₂O and K₂O₂ are diamagnetic but KO₂ and KO₃ are paramagnetic
(D) Mn₂O₇ is acidic oxide but MnO is amphoteric

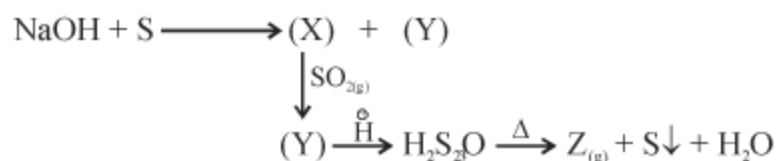
Ans. (A, B, C)

40. One of the important compounds containing P–N bonds is phosphazenes. One such phosphazene has the molecular formula (NPCl₂)_n. All the chlorines are attached to phosphorous in its structure. Select the correct statements about (NPCl₂)₃.

- (A) Both N and P are sp³ hybridized
(B) N is sp² hybridised.
(C) It forms a planar ring
(D) There is a total of 9 lone pairs in its molecule

Ans. (B, C)

Paragraph-1



41. Where (X), (Y) respectively are ?

- (A) Na₂S₂O₃, Na₂SO₃
(B) Na₂S₅, Na₂S₂O₃
(C) Na₂S, Na₂S₂O₃
(D) Na₂SO₃, Na₂S₂O₃

Ans. (C)

42. Which of the following statement is **CORRECT** for $Z_{(g)}$?
- (A) It produce black ppt with $Pb(NO_3)_2$
 - (B) It produce purple colour with sodium nitroprusside, $\{Na_2[Fe(CN)_5NO]\}$.
 - (C) It produce yellow turbidity with SO_2 (aq)
 - (D) It change purple solution of acidic $KMnO_4$ to colourless solution.

Ans. (D)

Paragraph-2

Hydrolysis of a Xenon flouride '**Z**' is not the redox reaction. On partial hydrolysis it forms **X** and **Y** two different compounds of Xe. Complete hydrolysis of **Z** gives **W**. **W** with NaOH forms **V** which further in dil NaOH disproportionates slowly forming compound **U** and two other gases **S** and **T**. Then

43. Which is true about **X** and **Y** ?
- (A) Both have Xe-atom in same oxidation state
 - (B) Both have same hybridisation of central atom
 - (C) Shape of both the molecules are same
 - (D) One is polar and one is non-polar

Ans. (A)

44. Identify the false statement.
- (A) One of S and T is paramagnetic
 - (B) V is octahedral ion
 - (C) In U all bond length are equal
 - (D) In U central atom is in sp^3d^2 hybridisation

Ans. (B)

Paragraph-3

When $Pb(NO_3)_2$ is heated then it forms two gaseous products **A** and **B**. When nitrogen containing gaseous product '**B**' is reacted with another neutral oxide of Nitrogen then it forms a coloured liquid **C** at low temperature. Non nitrogen containg gas '**A**' when heated in silent electric dischange then forms **D** which is strong bleaching agent. Then

45. Which is **not true** about compound '**C**'
- (A) it contains 2 N-atoms
 - (B) Both the N - atoms are in sp^2 hybridisation
 - (C) It has 8 non bonded 'lp' of e^-
 - (D) All 'N - O' bond length are same

Ans. (D)

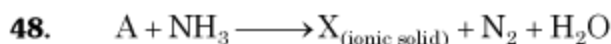
46. Identify true statement
(A) Both A and D are paramagnetic
(B) In D all bond lengths are equal
(C) entropy of system increases in the conversion of A to D
(D) D is colourless in solid and liquid state.

Ans. (B)

47. Which is **NOT CORRECT** about **A** ?
(A) It is highly soluble in water and produced monobasic oxyacid.
(B) Central atom is sp^3 hybridised.
(C) It is paramagnetic.
(D) Bond angle is more than normal tetrahedral bond angle.

Paragraph-4

Freshly precipitated mercuric oxide when heated with Cl_2 gas diluted with dry air gives an oxide of chlorine (**A**), which is anhydride of a oxyacid of chlorine. It dissolves in NaOH to form (B) which is used to bleach wood pulp and fabrics. All bond length parameters are identical in (**A**).



Which is **CORRECT** about X ?

- (A) When heated with alkali produces N_2 gas.
(B) It's aqueous solution is basic.
(C) If it is added to sodium 'aluminate solution and heated white precipitate is observed.
(D) It can act as a base in liq. NH_3 .

Ans. (C)

Paragraph-5

p-Block elements exhibit certain phenomena which is found absent in s-block elements such as inert s-pair effect. The hydrolysis of halides and oxohalides of these elements may proceed through different mechanisms.

49. All of the following give at least one common hydrolysis product except :
(A) $POCl_3$
(B) BCl_3
(C) NCl_3
(D) PCl_3

Ans. (C)

50. Select **CORRECT** statement :

- (A) BCl_3 reacts with gaseous NH_3 to give $\text{B}(\text{NH}_2)_3$
- (B) In TlI_3 , oxidation state of thallium is +3
- (C) Conversion of ozone to oxygen proceeds with decrease in entropy.
- (D) Complete hydrolysis of XeF_6 produces an explosive solid.

Ans. (D)

51. Match the compounds listed in Column-I with characteristic(s) / type of reaction(s) listed in Column-II.

Column-I

- (A) XeF_2
- (B) XeF_4
- (C) XeF_6
- (D) XeO_3

Column-II

- (P) Undergoes hydrolysis with water.
- (Q) Acts as oxidising agent.
- (R) Undergoes addition reaction.
- (S) Has lone pair(s) of electrons.

Ans. [(A) → (P, Q, R, S); (B) → (P, Q, R, S); (C) → (P, Q, R, S); (D) → (Q, S)]

52. Match the following :

List-I

- (A) $\text{O}_3 + \text{H}_2\text{O}_2 \longrightarrow$
- (B) $\text{O}_3 + \text{starch KI} \longrightarrow$
- (C) $\text{O}_3 + \text{Hg} \longrightarrow$
- (D) $\text{O}_3 + \text{I}_2 + \text{H}_2\text{O} \longrightarrow$

List-II

- (P) Blue
- (Q) Tailing
- (R) HIO_3
- (S) $\text{O}_2 + \text{H}_2\text{O}$

Ans. [(A) → (S); (B) → (P); (C) → (Q); (D) → (R)]

53. Match the matrix.

Column-I

- (A) $0.2 \text{ mole } \text{H}_3\text{BO}_3 + 0.1 \text{ mole } \text{NaOH}_{(\text{aq.})} \longrightarrow$
- (B) $\text{Na}_2\text{B}_4\text{O}_7(\text{s}) + \text{H}_2\text{O} \longrightarrow \text{solution}$
- (C) $\text{K}_3\text{BO}_3 \text{ (limiting reagent)} + \text{H}_2\text{SO}_4 \longrightarrow$
- (D) $\text{B}_2\text{H}_6 + \text{H}_2\text{O}_{(\text{excess})} \longrightarrow$

Column-II

- (P) Resulting product contain H_3BO_3
- (Q) Alkaline solution
- (R) $[\text{B}(\text{OH})_4]^-$ in solution
- (S) Volatile product
- (T) Buffer solution

Ans. [(A) → (P, Q, R, T); (B) → (P, Q, R, T); (C) → (P); (D) → (P, R, S)]

54. Column-I

- (A) POCl_3
 (B) SOF_2
 (C) XeOF_4
 (D) $\text{H}_2\text{S}_2\text{O}_8$

Column-II

- (P) Oxyacid formed during hydrolysis undergoes Tautomeric change
 (Q) Oxidation state of central atom does not change during hydrolysis
 (R) Complete as well as partial hydrolysis is possible
 (S) Hydrolysed product reacts with glass
 (T) Hybridization of central atom in the final product remains same as in the substrate on hydrolysis

Ans. [(A) → (Q, T); (B) → (P, Q, S, T); (C) → (Q, R, S); (D) → (Q, R, T)]

55. Match the following :

Column-I

- (A) $\text{ClO}_2 \longrightarrow \text{Cl}_2\text{O}_6$
 (B) $[\text{Al}(\text{OH})_4]^- \longrightarrow \text{Al}(\text{OH})_3 \downarrow$
 (X) $\text{P}_4 \longrightarrow \text{PH}_3 + \text{H}_2\text{PO}_2^-$
 (D) $\text{XeF}_2 \longrightarrow \text{Xe}$

Column-II

- (P) Boiling with NaOH solution.
 (Q) On passing ozone.
 (R) Reaction with hydrogen.
 (S) On passing CO_2 gas.

Ans. [(A) → (Q); (B) → (S); (C) → (P); (D) → (P, R)]

56. Match the following

Column-I

- (A) BCl_3
 (B) H_3BO_3
 (C) BH_4^-
 (D) $\text{AlCl}_3(\text{g})$

Column-II

- (P) Central atom in sp^2 hybridisation
 (Q) Central atom in sp^3 hybridisation
 (R) acts as Lewis acid
 (S) Gives white fume with moisture

Ans. [(A) → (P, R, S); (B) → (P, R); (C) → (Q); (D) → (P, R, S)]

57. Column-I

- (A) Mg + very dil. HNO_3
 (B) Zn + conc. HNO_3
 (C) Fe + dil. HNO_3
 (D) Pb + dil. HNO_3

Column-II

- (P) NO
 (Q) H_2
 (R) NO_2
 (S) NH_4NO_3

Ans. [(A) → (Q); (B) → (R); (C) → (S); (D) → (P)]

58. Match the following

Column-I

Column-II

- (A) $\text{Cl}_2 + \text{cold and dilute NaOH} \xrightarrow{0^\circ\text{C}}$ (P) Disproportionation take place
 (B) $\text{Cl}_2 + \text{hot and conc. NaOH} \xrightarrow[\Delta]{>50^\circ\text{C}}$ (Q) Chloride (Cl^-) is produced
 (C) $\text{KClO}_3 \xrightarrow[\text{at low temp. } < 100^\circ\text{C}]{\Delta}$ (R) Chlorate (ClO_3^-) is produced
 (D) $\text{ClO}_2 + \text{NaOH} \longrightarrow$ (S) Hypochlorite (OCl^-) is produced
 (T) Chlorite (ClO_2^-) is produced

Ans. [(A) → (P, Q, S); (B) → (P, Q, R); (C) → (P, Q); (D) → (P, R, T)]

59. Match the Column

Column-I

Column-II

(Allotropic form)

(Physical appearance)

- (A) Graphite (P) Solid
 (B) α -sulphur (Q) Yellow coloured
 (C) $-(\text{P}_4)_n$ (R) Red coloured
 (D) $(\text{BN})_x = \text{inorganic graphite}$ (S) Slippery solid
 (T) White in colour

Ans. [(A) → (P, S); (B) → (P, Q); (C) → (P, R); (D) → (P, S, T)]

60. Match the following Column

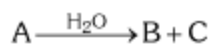
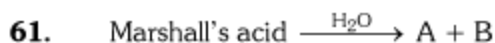
Column-I

Column-II

- (P) B_2H_6 (1) $2p_\pi - 2p_\pi$ (Count back bonding also if present)
 (Q) $\text{B}_3\text{N}_3\text{H}_6$ (2) $2p_\pi - 3d_\pi$ (Count back bonding also if present)
 (R) BF_3 (3) Ring formed in the structure
 (S) PF_3 (4) Non planar
 (5) sp^2 hybridisation of central atom(s)

Codes :	P	Q	R	S
(A)	3,4	1,3,5	1,5	2,4
(B)	1,2,3	3,4,5	1,5	2,3,4
(C)	2,3,4	3,4	1,3	2,3,4
(D)	1,5	2,3	3,4	3,4

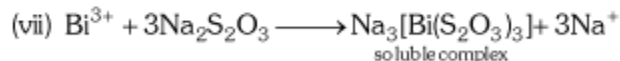
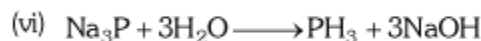
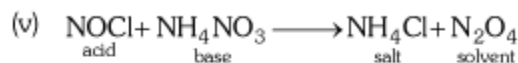
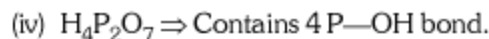
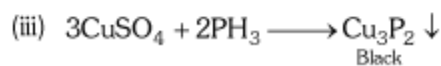
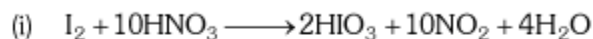
Ans. (A)



If P and Q represent maximum number of atoms that can lie in a plane of compound A and C respectively. Then, find out value of (P – Q)

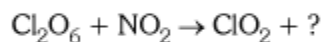
Ans. (2)

62. Find correct number of statements/reactions.



Ans. (7)

63. Let us assume that bond angles in sp , sp^2 , sp^3 hybridisation are respectively 180° , 120° and 109° . If X° is the difference in the cationic and anionic parts in one of the products of the reaction given below.

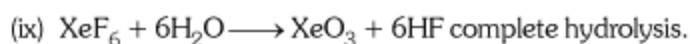
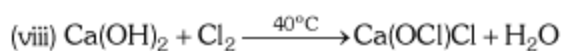
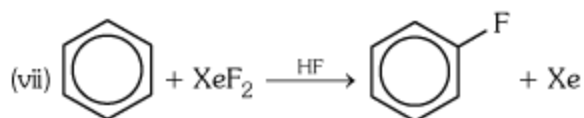
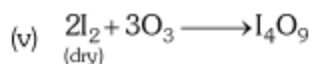
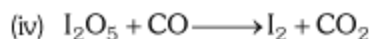
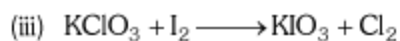
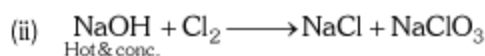
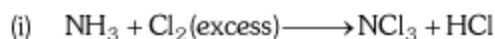


Find 'X'.

[Give your answer as sum of digits and Fill OMR sheet]

Ans. (8)

64. Find correct number of reactions.



Ans. (9)

65. Find out total number of non-metals from following which react with water :

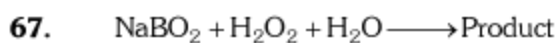
$\text{N}_2, \text{F}_2, \text{I}_2, \text{P}_4, \text{Br}_2, \text{Boron}, \text{Si}, \text{Cl}_2$

Ans. (3)

66. What is the value of X in the given chemical formula of crystalline borax ?



Ans. (8)



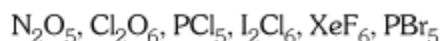
Give number of B—O bond present in the product species.

Ans. (8)

- 68.** How many statements are correct.
- (i) Ethyl borate, $B(OC_2H_5)_3$ burns with green edged flame.
 - (ii) Boron is obtained by the pyrolysis of BI_3 .
 - (iii) One mole of borax reacts with one mole of acid.
 - (iv) In sodium peroxoborate, each boron is sp^3 hybridised.
 - (v) H_3BO_3 does not dissolve in aqueous HF.
 - (vi) $B_2H_6 + NH_3 \xrightarrow[\text{higher temp. (>200^\circ C)}]{\text{Excess } NH_3} B_3N_3H_6$ (borazine).
 - (vii) The graphite is paramagnetic and diamond is diamagnetic in nature.
 - (viii) Graphite acts as a metallic conductor along the layers of carbon atoms and as semi-conductor perpendicular to the layers of the carbon atoms.
 - (ix) Graphite is less denser than diamond.
 - (x) Silicones are strongly water repellent.

Ans. (6)

- 69.** Consider the following covalent compounds in their solid state and find the value of expression $(X + Y + Z)$.



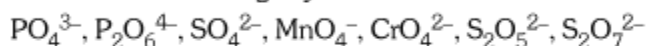
Where X = Total number of compounds in which central atom of cationic or anionic part is sp^3 hybridized.

Y = Total number of compounds having 90° bond angle either in cationic or anionic part.

Z = Total number of compounds having $109^\circ 28'$ bond angle either in cationic or anionic part.

Ans. (9)

- 70.** Consider the following oxyanions :



and find the value of $R + Q - P$

Where P = Number of oxy anions having three equivalent X-O bonds per central atom

Q = Number of oxy anions having two equivalent X-O bonds per central atom

R = Number of oxy anions having four equivalent X-O bond per central atom

Ans. (3)

- 71.** When B_2H_6 is allowed to react with following lewis bases, then how many given lewis bases form adduct through symmetrical cleavage of B_2H_6 ?



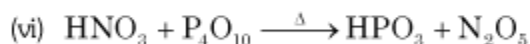
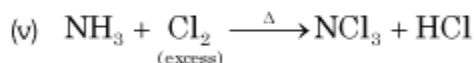
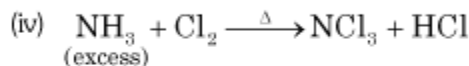
Ans. (6)

72. Choose total number of correct reactions.

(i) When CuSO_4 solution reacts with NH_3 , complex is formed.

(ii) When CuSO_4 solution react with PH_3 , Cu_3P_2 is formed.

(iii) bond energy of Cl_2 is highest among halogens



Ans. (6)

73. Select the total number of substance which is/are given as their most stable allotrope of corresponding element as well as their formation enthalpy is zero :

(i) Graphite

(ii) Ozone

(iii) Black phosphorus

(iv) Rhombic sulphur

(v) $\text{Cl}_2(\text{g})$

(vi) $\text{Br}_2(\text{g})$

Ans. (3)

74. In how many of the following compounds d-orbital is not involved in the hybridisation of the central atom as well as compound is polar.

PH_3 , NH_3 , XeO_3 , XeF_4 , BCl_3 , $\text{ICl}_3(\text{s})$, NO_2 , HNO_3 , SO_3

Ans. (5)

75. How many statements are correct.

(i) B_2H_6 is prepared by reacting NaBH_4 with I_2 .

(ii) +4 oxidation state of Pb is less stable than +2.

(iii) Borax on strong heating forms sodium metaborate as one the product.

(iv) $\text{H}_3\text{BO}_3(\text{s})$ has hydrogen bond.

(v) In cyclic silicate two oxygen atom are shared per tetrahedral unit.

(vi) Oxides of aluminium (Al_2O_3) and gallium (Ga_2O_3) are amphoteric in nature.

(vii) Graphite acts as a metallic conductor along the layers of carbon atoms and as semi-conductor perpendicular to the layers of the carbon atoms.

(viii) Graphite is less denser than diamond.

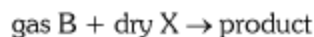
(ix) Silicones are strongly water repellent.

Ans. (9)

76. Consider following statements/reactions :

- (i) $F_2 + H_2O \rightarrow$ fluorine containing compound + gas A + gas B, only gas A is paramagnetic
- (ii) $NaI + \text{conc. } H_2SO_4 \rightarrow X$
- (iii) KI enhances solubility of X in water.

Now consider the following reaction :



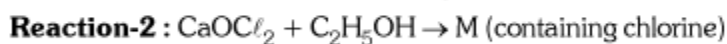
How many moles of oxygen atoms are present in one mole of product ?

Ans. (9)

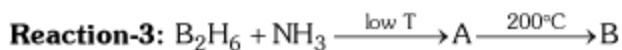
77. Consider following statements/reactions :



Let the oxidation state of bromine in product = x



Let the number of chlorine atom in a molecule of M = y



Let the number of sp^2 hybridised orbitals in a molecule of B = z

Statement-1: In mineral Beryl, $Be_3Al_2Si_6O_{18}$, let number of oxygen atoms shared per tetrahedra = m

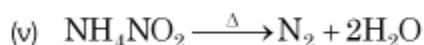
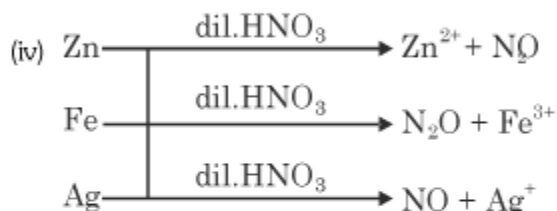
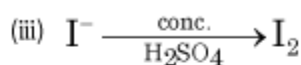
Find value of $\frac{(x + y + z + m + 1)}{8}$ to nearest integer :

Ans. (3)

78. Observe the following reactions/statements & find number of **CORRECT** statements.

(i) SCN^- , CN^- & OCN^- are pseudohalide.

(ii) On oxidation graphite produces $\text{C}_6(\text{COOH})_6$



(vi) $\text{BF}_3 < \text{BCl}_3 < \text{BBr}_3$ (Lewis acidic strength)

(vii) Ionisation energy $\text{O}_2 > \text{O}$.

(viii) $\text{Xe}[\text{PtF}_6]$ is the first compound of Xe

(ix) $\text{Si}_4\text{O}_{11}^{6-}$ is double chain silicate.



Ans. (9)

79. Consider thermal decomposition of following ammonium salts and calculate value of

expression $\frac{|P - Q|}{2}$

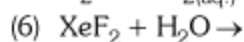
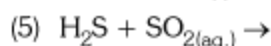
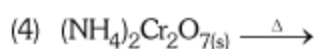
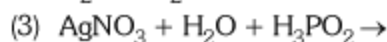
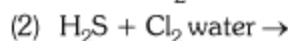
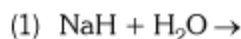
$(\text{NH}_4)_2\text{Cr}_2\text{O}_7$, NH_4NO_2 , NH_4NO_3 , NH_4Cl , $\text{CH}_3\text{COONH}_4$, NH_4ClO_4 , $(\text{NH}_4)_2\text{S}$, $(\text{NH}_4)_2\text{CO}_3$, $(\text{NH}_4)_2\text{C}_2\text{O}_4$.

Where : P : Total number of ammonium salts which liberates basic NH_3 gas on thermal decomposition.

Q : Total number of ammonium salts which liberates neutral N_2 gas on thermal decomposition.

Ans. (1)

80. Consider following reactions :



In how many of the reaction(s) at least one of the product is obtained in elemental state ?

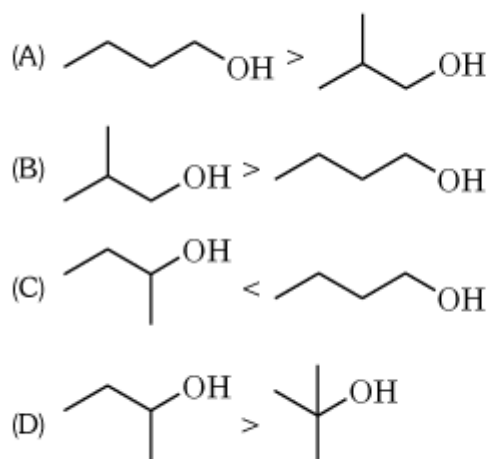
Ans. (6)

General Organic Chemistry

1. The correct order of pKa values for the acids malonic acid (pK_{a1} & pK_{a2}) acetic acid (pK_{a3}) is :
- (A) $pK_{a1} < pK_{a2} < pK_{a3}$
(B) $pK_{a3} < pK_{a1} < pK_{a2}$
(C) $pK_{a1} < pK_{a3} < pK_{a2}$
(D) $pK_{a1} < pK_{a2} = pK_{a3}$

Ans. (C)

2. The correct order of water solubility of alcohol is :

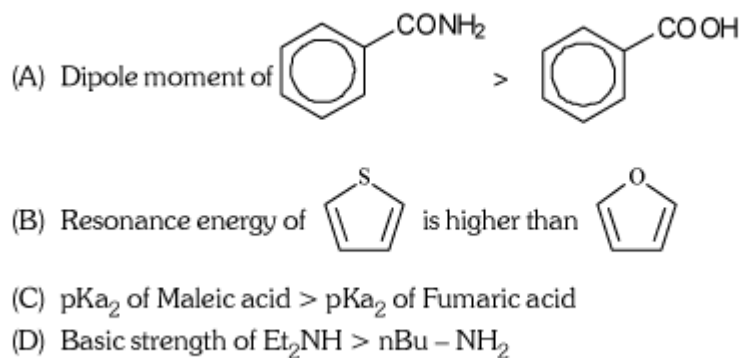


Ans. (B)

3. The formula $C_4H_4O_4$ can represent
- (A) a cyclic ester of dibasic acid
(B) a trans-dibasic acid
(C) a cis-dibasic acid
(D) an α, β - unsaturated dibasic acid which on heating forms a monobasic acid

Ans. (A, B, C, D)

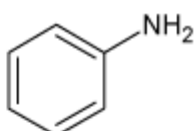
4. Choose the correct statement

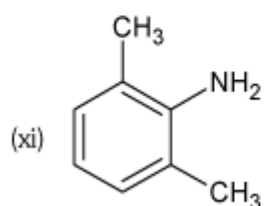
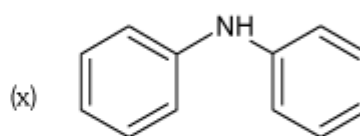
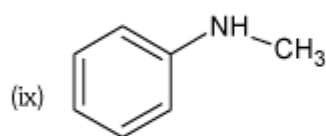
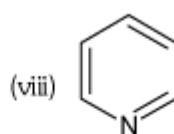
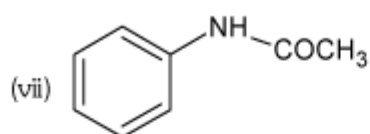
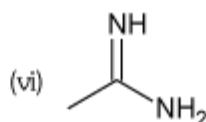
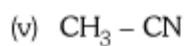
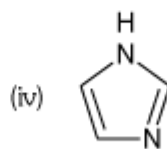
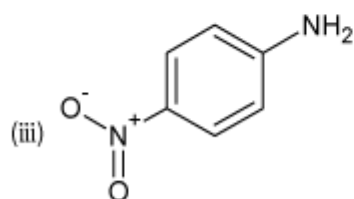
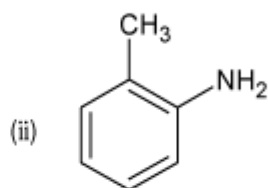
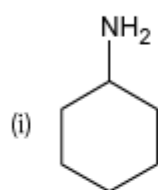


Ans. (A, B, C, D)

5. A chloro compound (A) showed the following properties. It decolourised bromine water. It reacts with H_2 in presence of catalyst. It gave a brick red precipitate with an ammonical Cu_2Cl_2 . When vapourised with 1.49 g of (A) gave 448 ml of vapours at STP. On reaction with excess of hydrogen and nickel gave 'B'. Calculate the molecular weight of 'B' in $g\ mol^{-1}$.

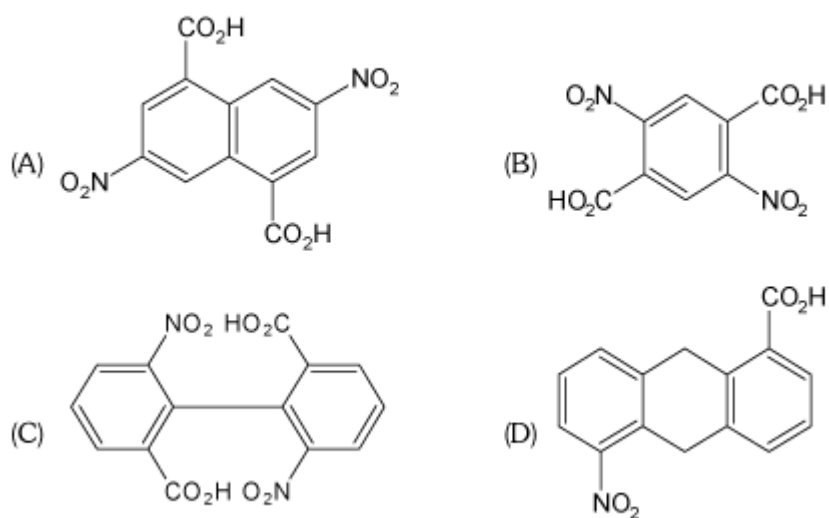
Ans. (44)

6. Find the total number of compounds which are less basic than 



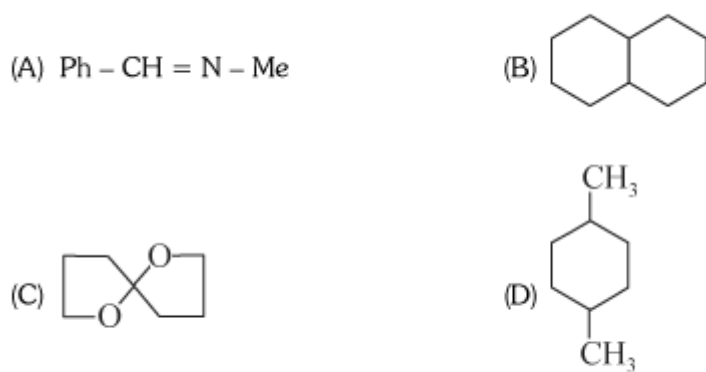
Ans. (6)

1. Which of the following carboxylic acids can be resolved by reaction with an enantiomerically pure chiral amine ?

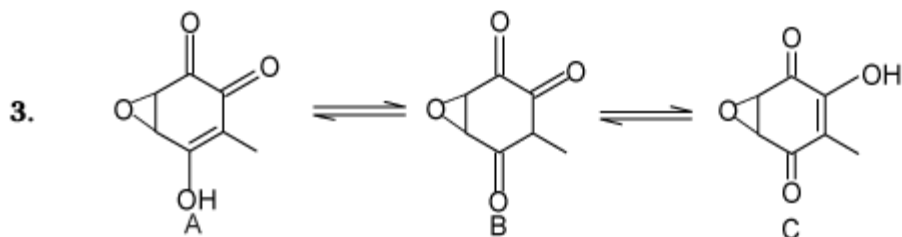


Ans. (C)

2. Which of the following compound will show geometrical isomerism ?



Ans. (A, B, D)



Select the **correct** statement(s)

- (A) A and C are the enol forms of B
- (B) A is more stable than C
- (C) C is more stable than A
- (D) Only A is the enol form of B

Ans. (A, C)



The products formed in the above reaction is/are

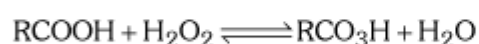
- (A) (2S,3S)–2,3–dichlorobutane
- (B) (2R,3R)–2,3–dichlorobutane
- (C) (S)–1,2–dichlorobutane
- (D) (R)–1,2–dichlorobutane

Ans. (A, D)

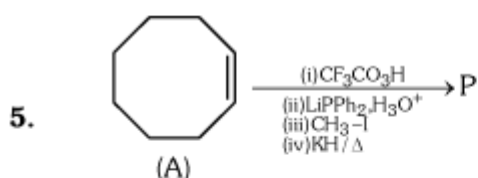
Paragraph-1

Optical rotation or optical activity is the rotation of the plane of polarization of linearly polarized light as it travels through certain material. Optical activity occurs only in chiral materials, those lacking microscopic mirror symmetry.

Several organic peroxyacids are commercially useful. They can be prepared as :



and these peroxyacids are used for epoxidation.



'P' is

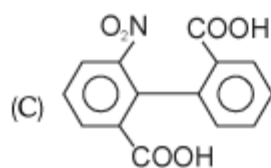
- (A) Non-resolvable
- (B) tert-alcohol
- (C) Non-resolvable
- (D) Diastereomer of A

Ans. (D)

6. Which of them has a plane of symmetry or center of symmetry.

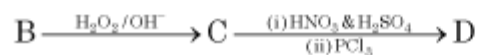
(A) Cis decalin

(B) trans decalin



Ans. (B)

7. In the reaction sequence :

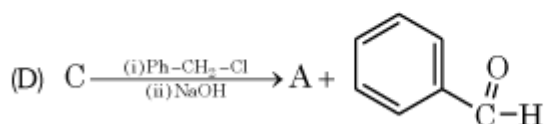


Find the correct option.

(A) A is more basic than B

(B) C is having less dipole moment than B

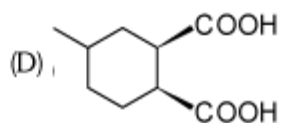
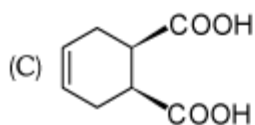
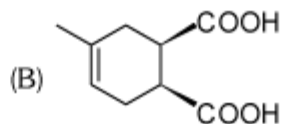
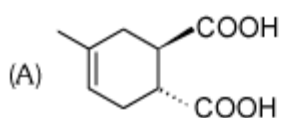
(C) 'D' is more basic than B



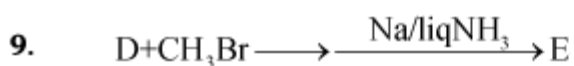
Ans. (D)

Paragraph-2

There are 3 isomers, A, B, & C, having m.f. C_5H_8 . On catalytic hydrogenation all the three give n-pentane. A does not give white precipitate with ammonical $AgNO_3$. But when heated with $NaNH_2$ it gives a salt, D. B & C do not contain any triple bond. B is optically active. C is a conjugated diene and undergoes Diel's-Alder reaction.



Ans. (B)



E is

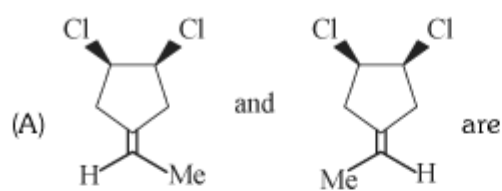
- (A) cis-2-hexene
- (B) trans-2-hexene
- (C) cis-3-hexene
- (D) trans-3-hexene

Ans. (B)

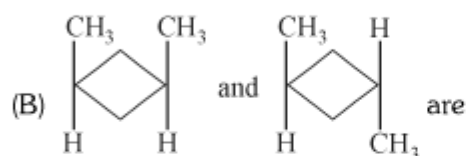
10.

Column-I

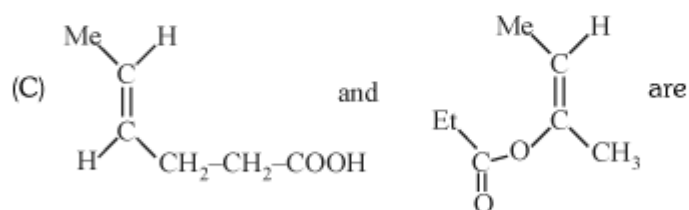
Column-II



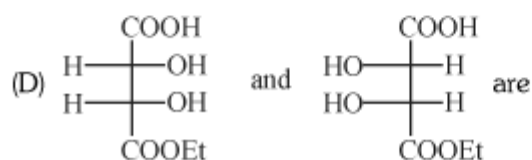
(P) Structural isomers



(Q) Enantiomers



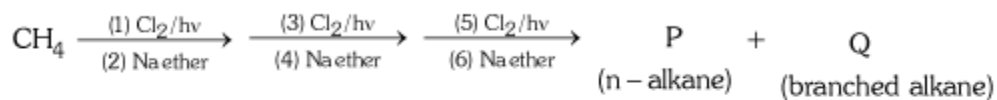
(R) Compounds which are geometrical isomer and diastereoisomers



(S) Compounds are geometrical isomers and enantiomers

Ans. [(A) → (S); (B) → (R); (C) → (P); (D) → (Q)]

11. Observe the following reaction



(y) Total number of fraction obtained $\xleftarrow[\text{distillation}]{\text{fractional}}$

(x) total number of monochloroproducts

If total number of optically active isomers = z in the end product y.

and total number of diastereomeric pairs = w in the end product y.

Column-I

(A) X

(B) Y

(C) Z

(D) W

Column-II

(P) 4

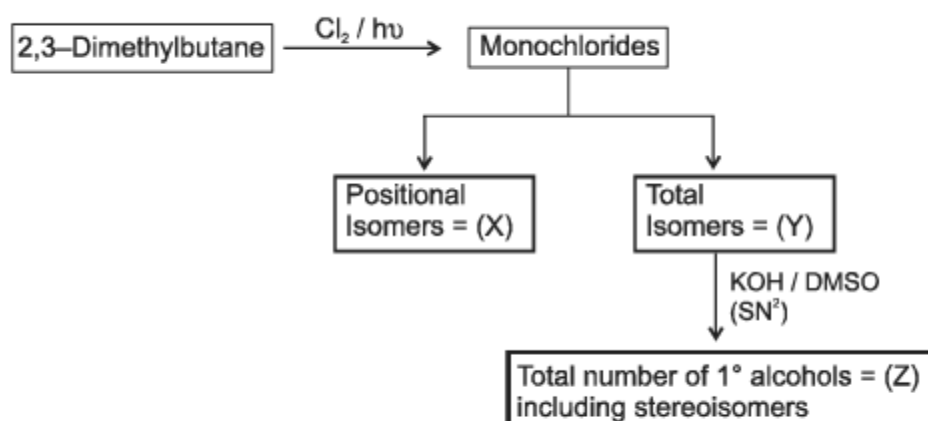
(Q) 6

(R) 7

(S) 0

Ans. [(A) → (R); (B) → (P); (C) → (Q); (D) → (S)]

12. Observe the following flow chart and answer the number of X, Y, Z.



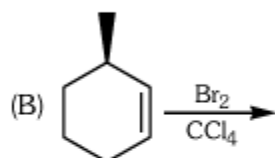
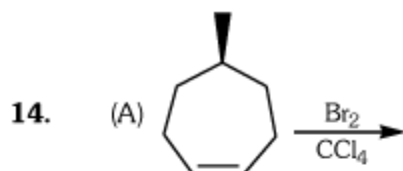
X	Y	Z
---	---	---

Ans. (232)

13. Give number of cyclic isomers (excluding stereoisomers) of formula C_6H_{12} .

[Divide your answer by 2 and fill OMR sheet]

Ans. (6)



(a) number of organic products in reaction A.

(b) number of stereoisomer possible for organic product in reaction A.

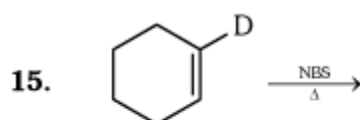
(c) number of organic products in reaction B.

(d) number of stereoisomer possible for organic product in reaction B.

If the answers of Part (a), (b), (c) and (d) are x, y, z and w respectively. Then express

$$\frac{y + z + w}{x} = ? \text{ (In OMR)}$$

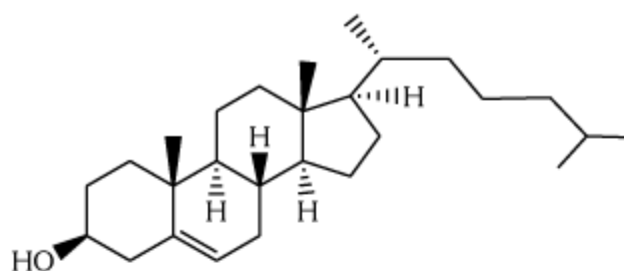
Ans. (7)



How many structural isomers of monobromo cycloalkenes are obtained in this reaction (without counting stereoisomers) ?

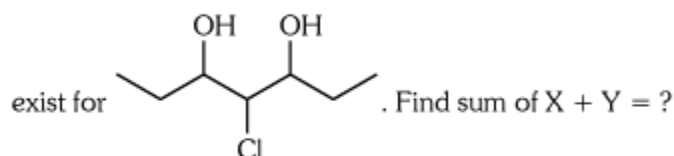
Ans. (3)

16. (a) Cholesterol has eight chiral centres

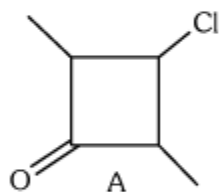


The specific stereoisomer shown above has "X" number of theoretical possible diastereomers and "Y" number of enantiomers. The sum of (X + Y) is :

- (b) There are 'X' number of meso stereoisomers and 'Y' number of enantiomeric pairs

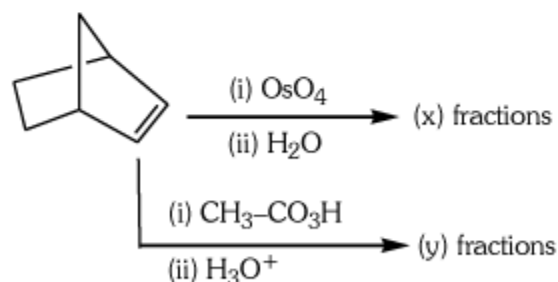


- (c) Compound A below has 'X' chiral carbons, 'Y' meso stereoisomers and 'Z' pair(s) of enantiomers.



Find sum of X + Y + Z ?

- (d) For the following reactions, how many fractions could be collected by fractional distillation or recrystallisation ?



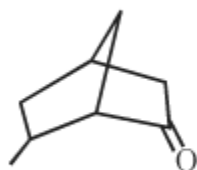
Find answer for (a), (b), (c) and (d) and fill the final answer as sum of digits till you get single digit answer.

For example (a) = 42, (b) = 3, (c) = 11 and (d) = 5, so final answer will be :

42 + 3 + 11 + 5 = 61, so answer is 6 + 1 = 7.

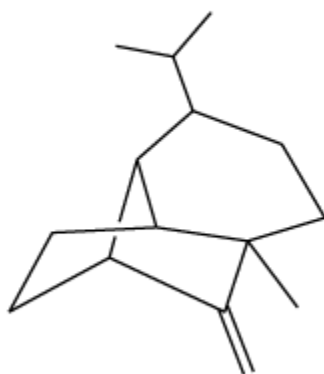
Ans. (6)

17. Total number of stereoisomers for following compound is :

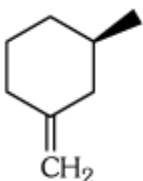


Ans. (4)

18. The structural formula of sativene is shown below. How many stereogenic centers are there in this molecule ?

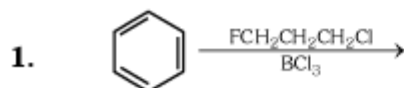


Ans. (5)

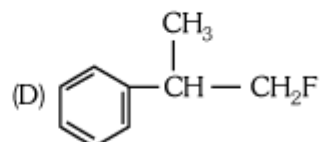
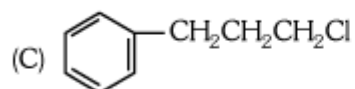
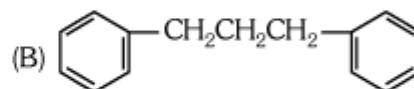
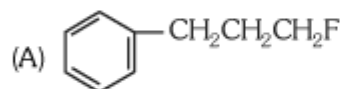
19.  $\xrightarrow{\text{NBS}}$ (X) = Total number of products (including stereoisomers).

Ans. (6)

Aromatic Compound

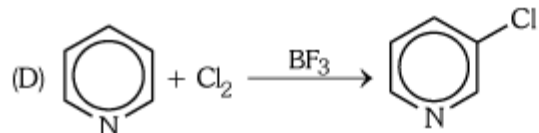
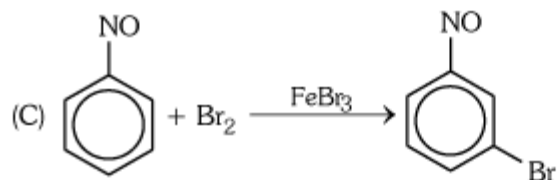
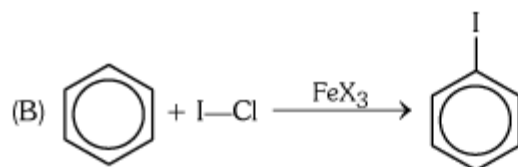
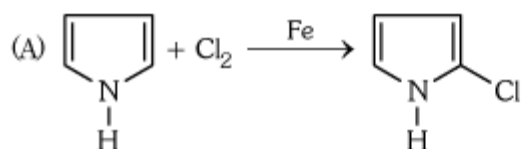


Predominant product is :

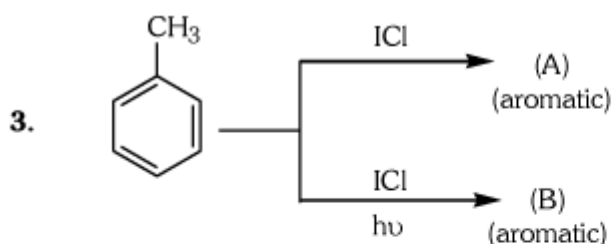


Ans. (C)

2. Which of the following reaction is **incorrect** ?



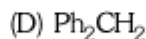
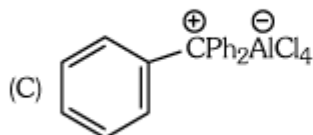
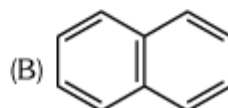
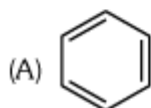
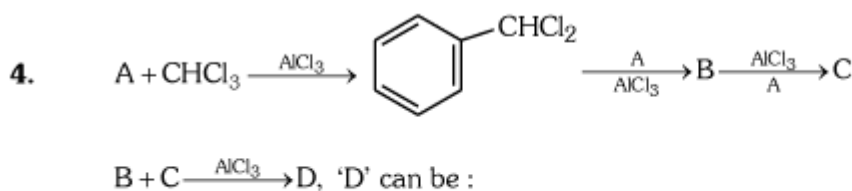
Ans. (C)



Find correct statement.

- (A) (A) is chloroderivative
- (B) (A) is Iododerivative
- (C) (B) is chloroderivative
- (D) (B) is Iododerivative

Ans. (B, C)

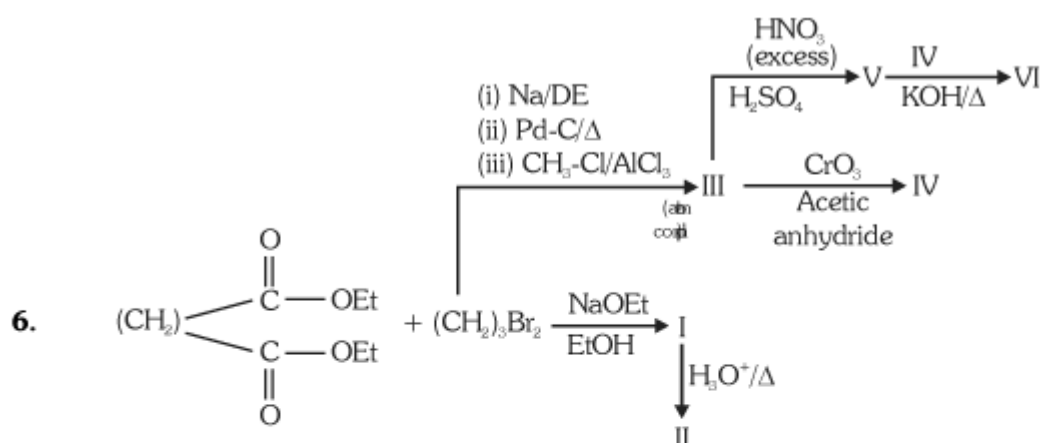


Ans. (C, D)

5. Poisonous gas CH_2N_2 when treated with acetylene gives aromatic compound X, which can undergo electrophilic substitution. Then which of the following are correct about aromatic compound X.

- (A) Nitrogen-1 is more basic than Nitrogen-2 (numbering is IUPAC)
- (B) Electrophilic substitution at 4th position is preferred.
- (C) Position-3 & Position-5 leads to same product by electrophilic substitution.
- (D) Electrophilic substitution does not take place being ring highly deactivated.

Ans. (B, C)



(A) II $\xrightarrow[\text{(iv) H}_3\text{O}^+]{\text{(i) Ag}_2\text{O}}$ product is having b.p. more than cyclopropane.

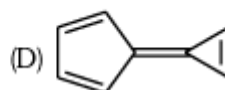
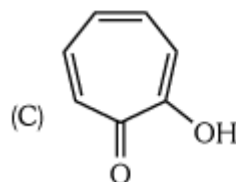
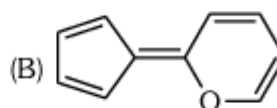
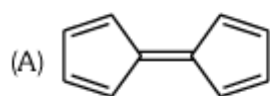
(B) VI gives optically active compound by reaction with D₂(1 eq.)/Pt.

(C) II is resolvable.

(D) IHD of VI is 10.

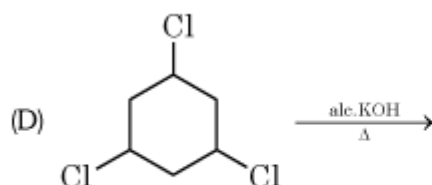
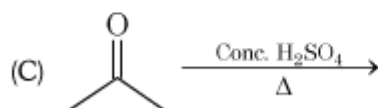
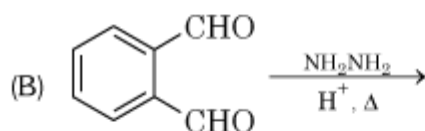
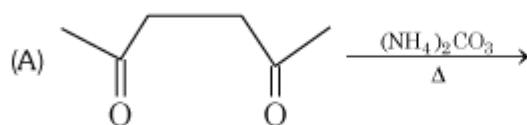
Ans. (A, B)

7. Aromatic compounds among the following is/are



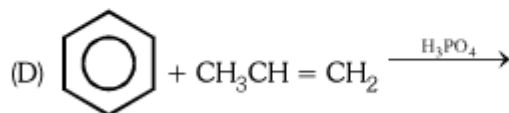
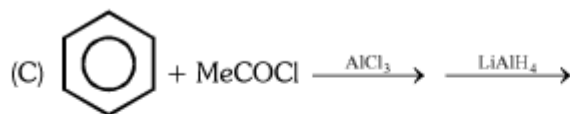
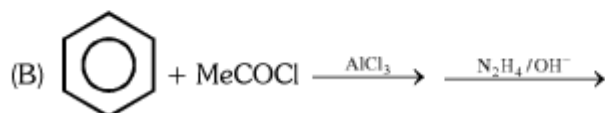
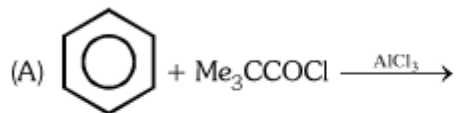
Ans. (B, C, D)

8. Identify the reactions that gives aromatic products.



Ans. (A, B, C, D)

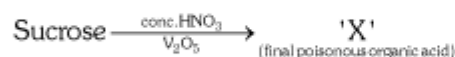
9. Which of the following reaction(s) can make alkyl benzene ?



Ans. (A, B, D)

Paragraph-1

In the reaction

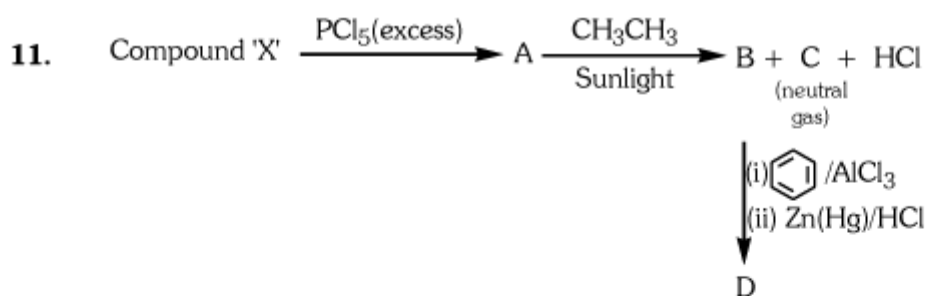


A 1.5 g sample containing acid 'X' and some inert impurity was dissolved in enough water and volume made up to 250 mL. A 20 mL portion of this solution was then mixed with 30 mL of an alkali solution. The resulting solution was then treated with stoichiometric amount of CaCl_2 just needed for precipitation of Calcium salt of acid 'X'. Solution was filtered off and filtrate was finally titrated against 0.1 M HCl solution. 8.0 mL of acid was required to reach the equivalence point. At last, the above neutral solution was treated with excess of AgNO_3 solution and AgCl obtained was washed, dried and weighed to be 0.4305 g.

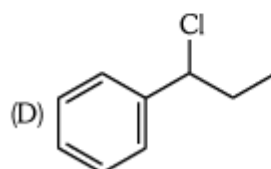
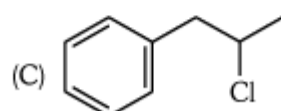
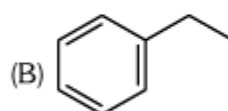
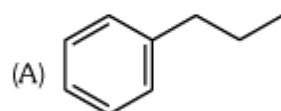
10. The mass percentage of acid 'X' in the initial sample is

(A) 28.5
(B) 71.5
(C) 17.5
(D) 82.5

Ans. (D)

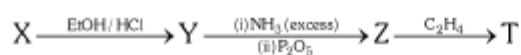


Compound 'D' is



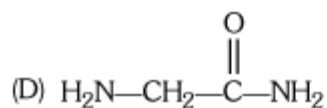
Ans. (A)

12. In the reaction sequence



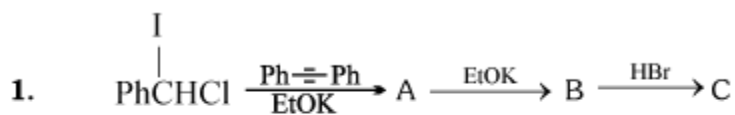
'T' is

- (A) $\text{NC} - \text{CH}_2 - \text{CH}_2 - \text{CN}$
 (B) $\text{H}_2\text{N} - \text{CH}_2 - \text{CH}_2 - \text{NH}_2$
 (C) $\text{NC} - \text{CN}$

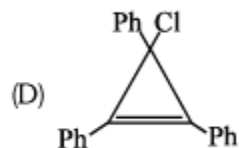
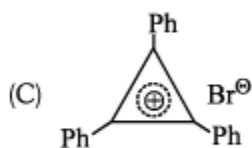
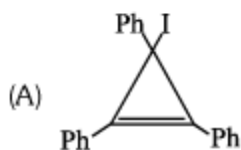


Ans. (A)

Alkyl Halide

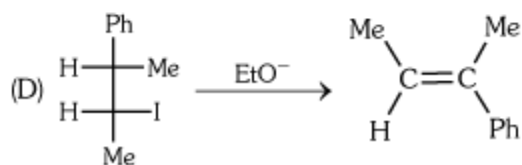
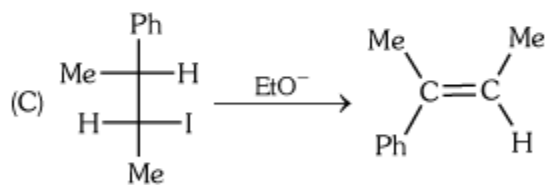
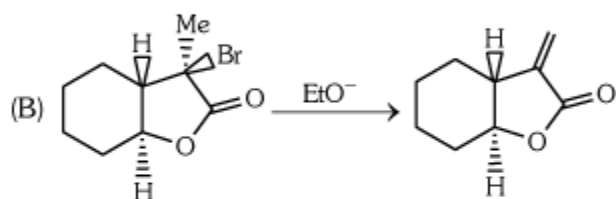
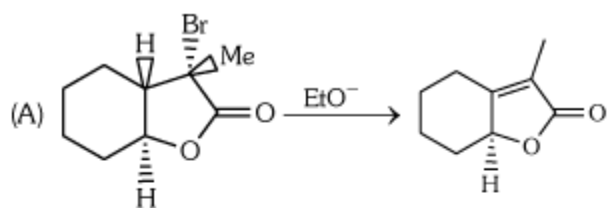


Correct presentation of C is :



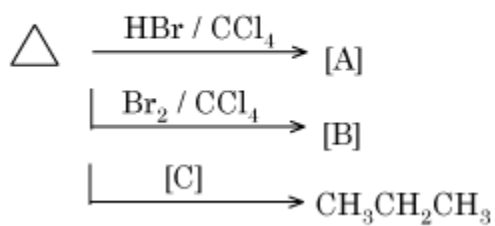
Ans. (C)

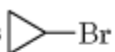
2. Which of the following reaction is/are correct ?

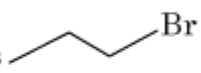


Ans. (A, B, C)

3. In the given reaction identify the unknowns :



(A) Compound [B] is 

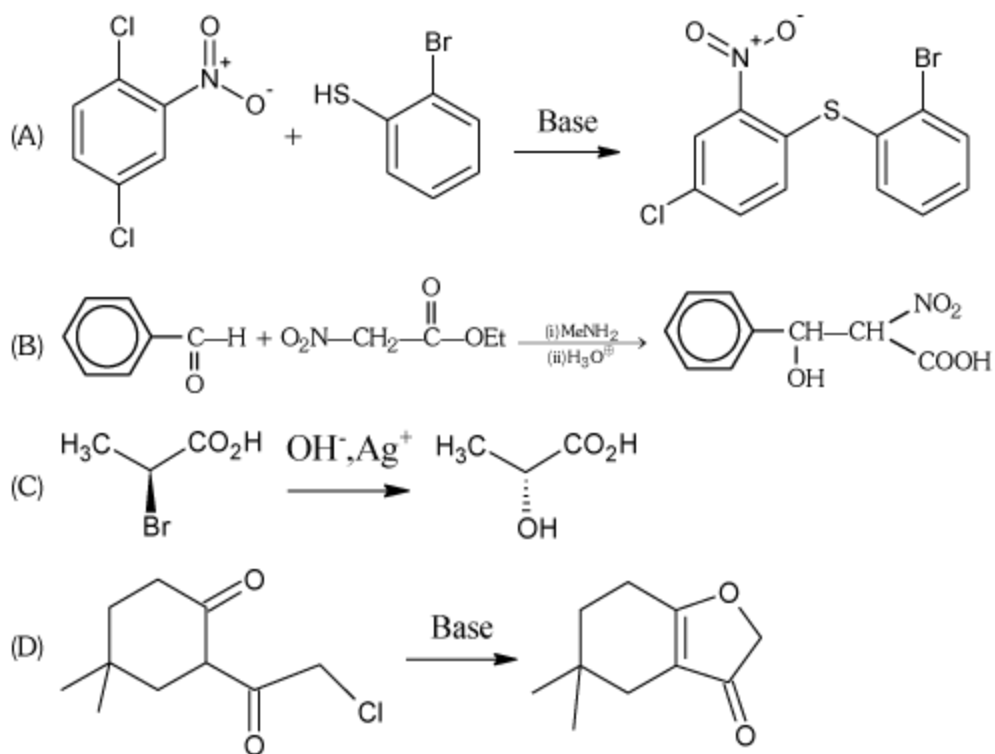
(B) Compound [A] is 

(C) Compound [B] is 

(D) Compound [C] is LAH

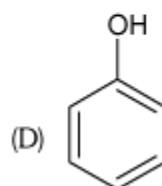
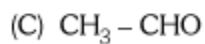
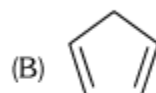
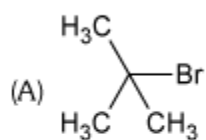
Ans. (B, C)

4. Identify correct reaction set(s) ?



Ans. (A, B, D)

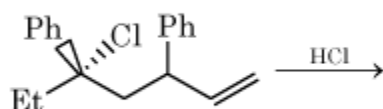
5. (X) $\xrightarrow{\text{CH}_3\text{CH}_2\text{MgBr}}$ $\text{CH}_3 - \text{CH}_3$ + Other products
(X) can be ?



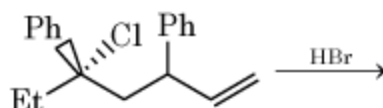
Ans. (A, B, D)

6. Select correct statement for following reactions :

Reaction-I :



Reaction-II :



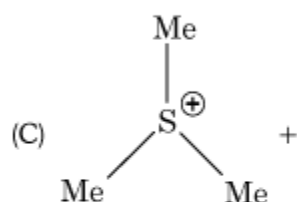
- (A) Reaction I gives mixture of 2 enantiomers & Reaction II gives mixture of two diastereomers.
 (B) Both reactions give mixture of 2 enantiomers.
 (C) Both reactions give mixture of 2 diastereomers.
 (D) Both reactions give 2 fraction of organic compounds.

Ans. (C, D)

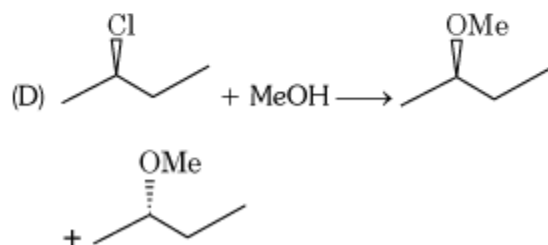
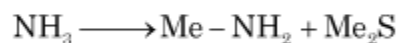
7. **Column-I**

Column-II

- (A) $\text{CH}_3 - \text{Cl} + \text{NaOH} \longrightarrow \text{CH}_3\text{OH} + \text{NaCl}$ (P) Rate of reaction increases in polar protic solvent
 (B) $\text{CH}_3 - \text{Cl} + \text{NH}_3 \longrightarrow \text{CH}_3 - \text{NH}_2 + \text{HCl}$ (Q) Rate of reaction increases in polar aprotic solvent



(R) $\text{S}_{\text{N}}2$ mechanism



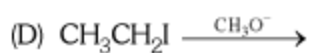
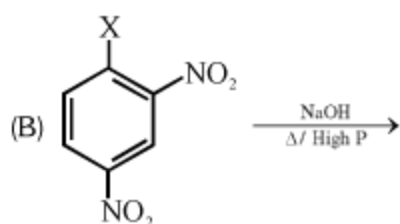
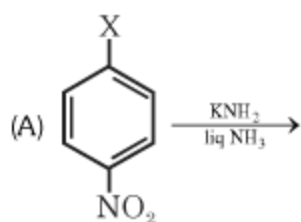
(S) $\text{S}_{\text{N}}1$ mechanism

(T) TS has positive charge

Ans. [(A) $\rightarrow$ (Q, R); (B) $\rightarrow$ (P, R, T); (C) $\rightarrow$ (Q, R, T); (D) $\rightarrow$ (P, S, T)]

8. Match the reactions given in Column-I with their characteristics given in Column-II.

**Column-I
(Reactions)**



**Column-II
(Characteristics)**

(P) Reaction follows reactivity order as



(Q) Reaction mechanism may involve

elimination

(R) Major product formation is done by nucleophilic substitution

(S) Major product formed by 2nd order kinetics.

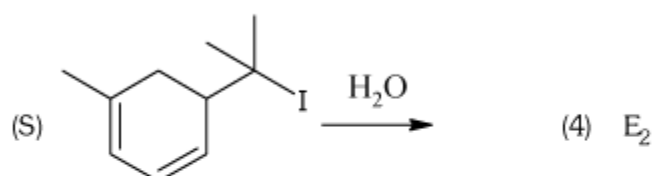
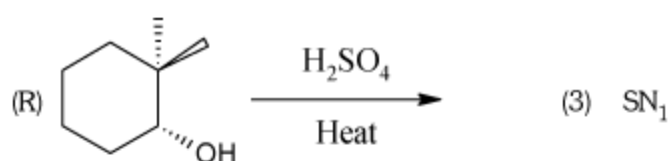
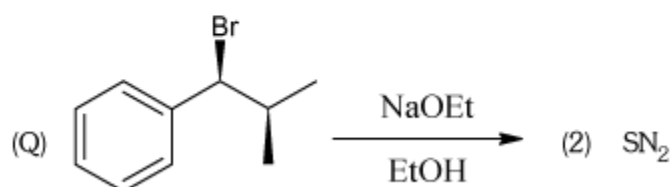
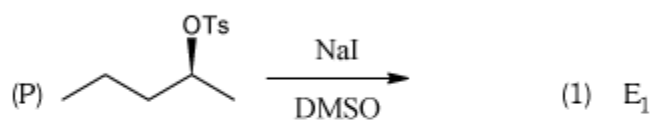
(T) Reaction involves formation of transition state.

Ans. [(A) → (P,Q,R,S,T); (B) → (P,Q,R,S,T); (C) → (Q,S,T); (D) → (Q,R,S,T)]

9. Match the following Column :

List-I
(Reaction)

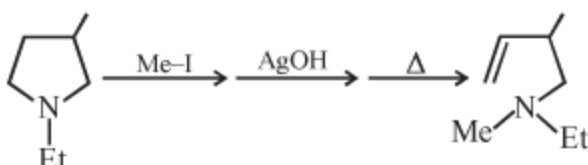
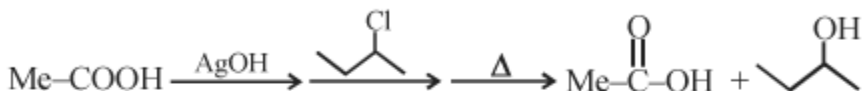
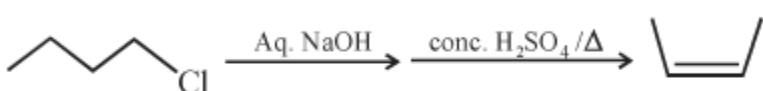
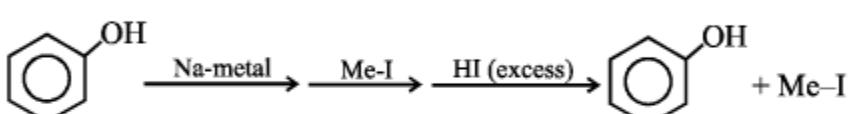
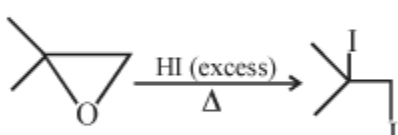
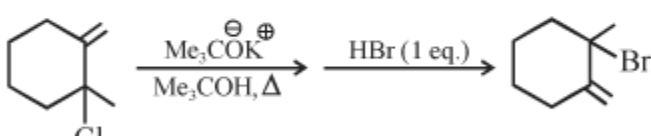
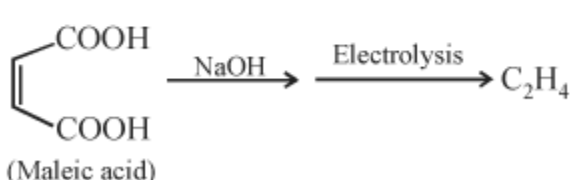
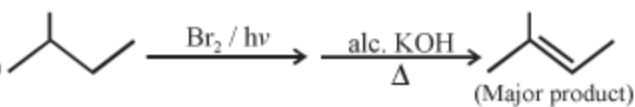
List-II
(mechanism involved in formation of major product)



Codes :	P	Q	R	S
(A)	2	4	3	1
(B)	4	1	3	1
(C)	2	4	1	3
(D)	4	1	2	3

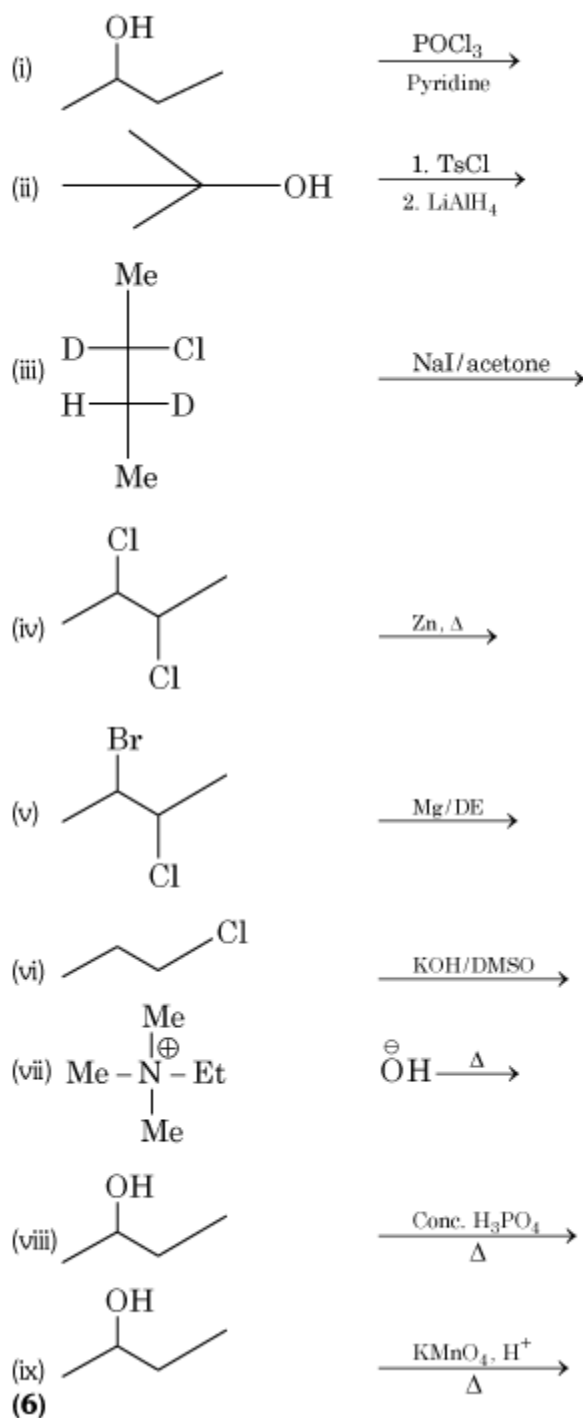
Ans. (C)

10. Decide number of reaction(s) correctly match with their major product :

- (i) 
- (ii) 
- (iii) 
- (iv) 
- (v) 
- (vi) 
- (vii) 
(Maleic acid)
- (viii) 

Ans. (2)

11. Identify total number of β' elimination reactions.



Ans.

(6)

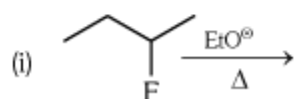
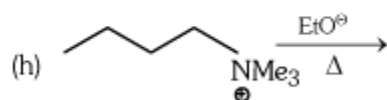
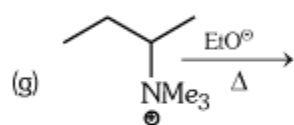
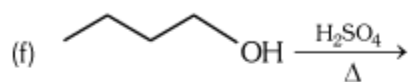
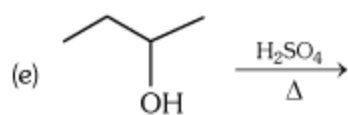
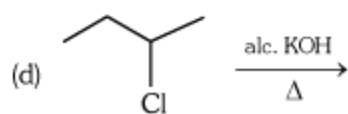
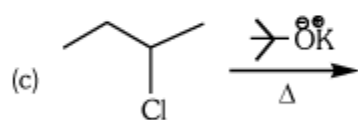
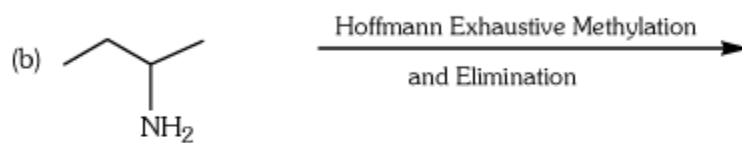
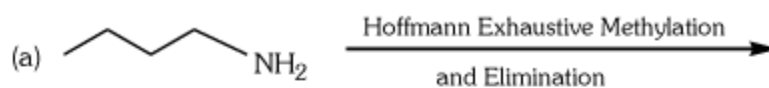
12. Find the total number of substrates more reactive than $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ with KI in acetone.

CH_3Cl , $(\text{CH}_3)_2\text{CHCl}$, $\text{CH}_2 = \text{CH} - \text{CH}_2\text{Cl}$, PhCH_2Cl , $\text{CH}_3\text{OCH}_2\text{Cl}$, PhCOCH_2Cl , $(\text{CH}_3)_3\text{CCl}$, $\text{C}_6\text{H}_5\text{Cl}$

Ans.

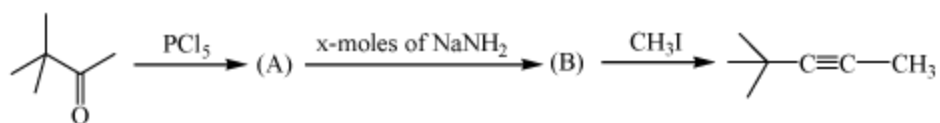
(5)

13. X = number of reaction in which 1-butene is major products

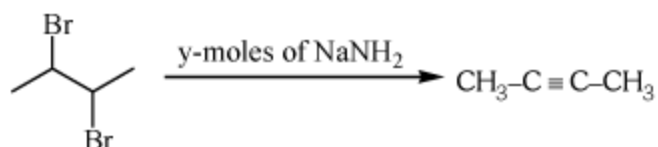


Ans. (6)

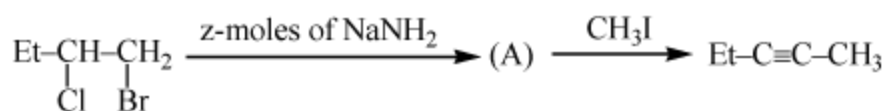
14. Reaction-I :



Reaction-II :



Reaction-III :

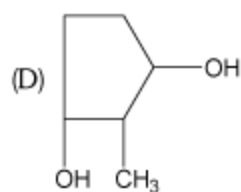
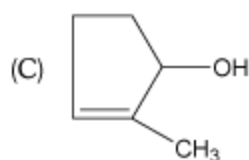
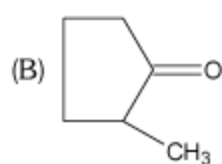
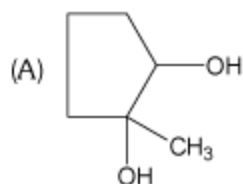
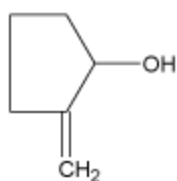


Value of $x + y + z$ is

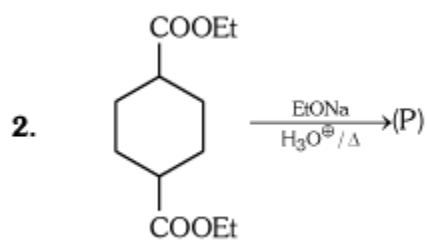
Ans. (8)

Alcohol, Ether and Phenol

1. Which one of the following will be the major product when is treated with H_2SO_4 ?



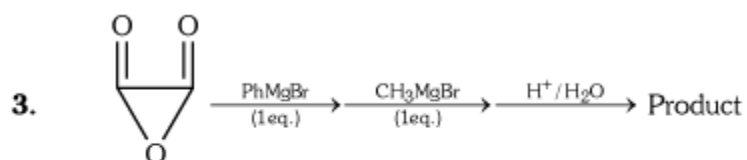
Ans. (B)



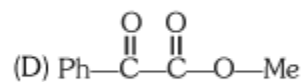
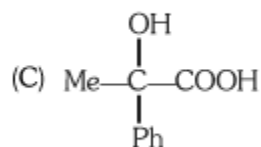
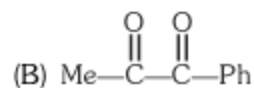
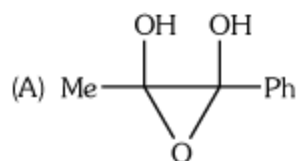
Select incorrect statement.

- (A) P can turn blue litmus red
- (B) P can not give effervescence of CO_2 with NaHCO_3
- (C) It is Dieckmann condensation
- (D) Product is a bicyclo compound

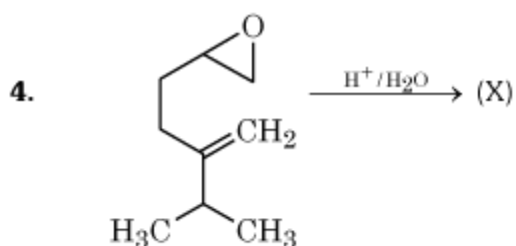
Ans. (B)



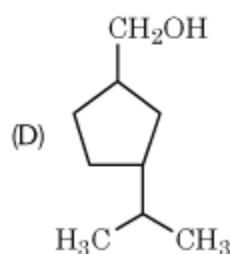
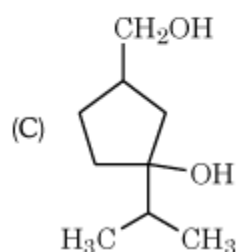
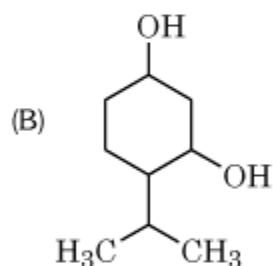
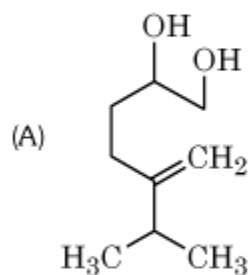
The product formed in the reaction is :



Ans. (C)

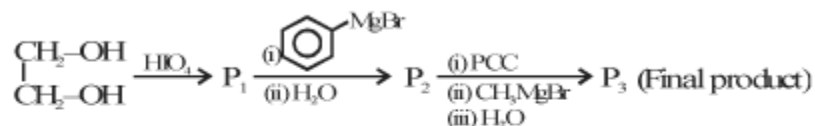


The product (X) is :



Ans. (C)

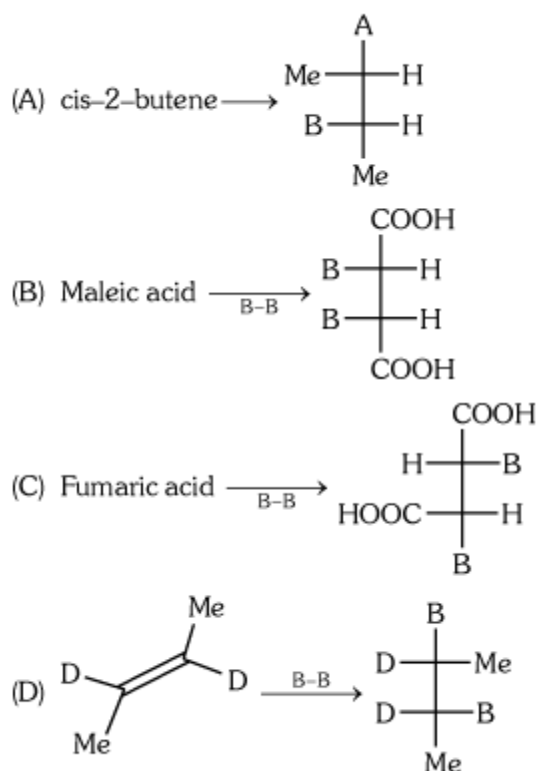
5. Which of the following is correct for the final product of the given sequence of reaction :



- (A) Compound P_3 on oxidation with $(\text{CrO}_3 + \text{H}_2\text{SO}_4)$ gives a compound which gives 2,4 DNP test
- (B) Compound P_3 on reaction with $\text{I}_2 + \text{NaOH}$ gives yellow ppt.
- (C) Compound P_3 on reaction with ceric ammonium nitrate give red colouration
- (D) Compound P_3 on reaction with MnO_2 gives carboxylic acid.

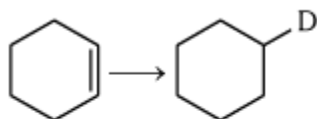
Ans. (A, B, C)

6. Which of the following reaction shows syn addition ?



Ans. (B, D)

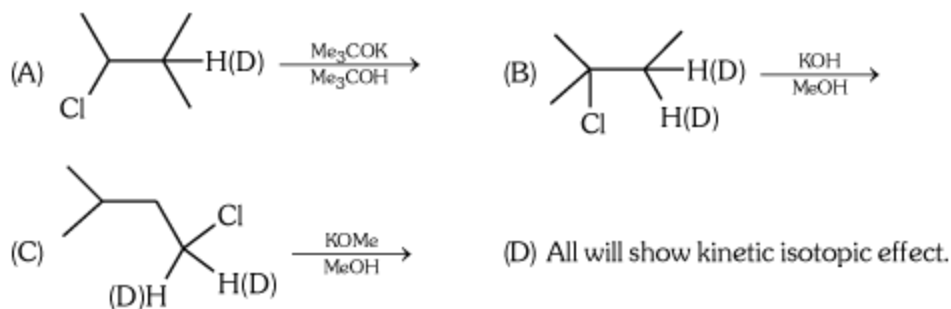
7. Which of the following set of reagents is/are correct for following conversion.



- (A) (i) H_2/Pt , (ii) $\text{Cl}_2, h\nu$, (iii) Li , (iv) D_2O
 (B) (i) DCl , (ii) Mg , (iii) H_2O
 (C) (i) HCl , (ii) Mg , (iii) D_2O
 (D) (i) NBS , (ii) H_2/Pt , (iii) Li , (iv) D_2O

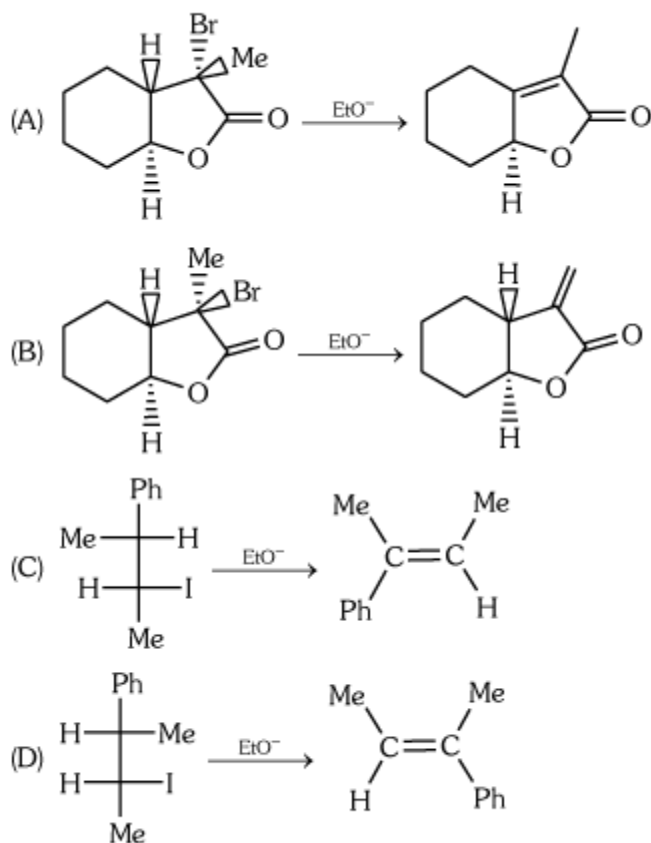
Ans. (A, B, C, D)

8. Which reaction will not show a primary hydrogen kinetic isotopic effect to give major product ?

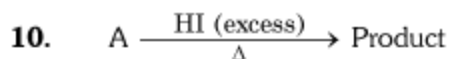


Ans. (A, C)

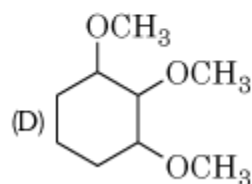
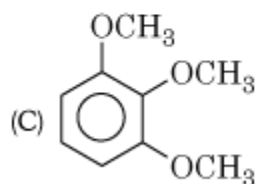
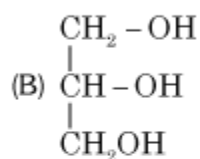
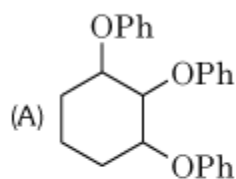
9. Which of the following reaction is/are correct ?



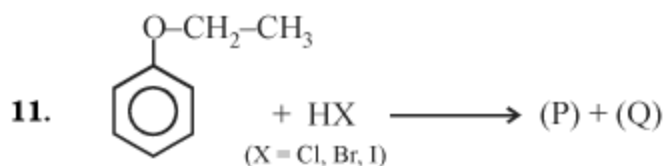
Ans. (A, B, C)



Identify compound A for which more than 4 moles of HI will be consumed



Ans. (A, B, D)

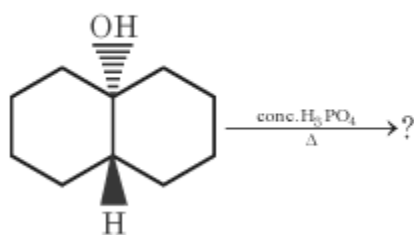


(P) is soluble in polar solvents while (Q) are in organic solvents. Choose the incorrect option(s) ?

- (A) (P) is soluble in aq. NaOH as well as in aq. NaHCO_3
- (B) Order of reactivity is : $\text{HCl} > \text{HBr} > \text{HI}$
- (C) (P) has 5 resonating structures & known as haloarenes
- (D) Both (P) & (Q) can interact with Na metal

Ans. (A, B, C)

12. Consider the reaction given below



Select the correct option(s)

(A) Three products are obtained

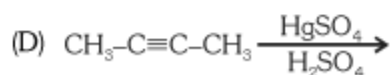
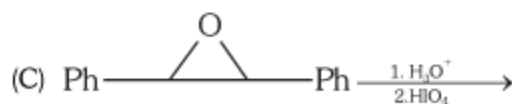
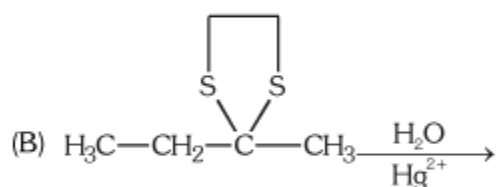
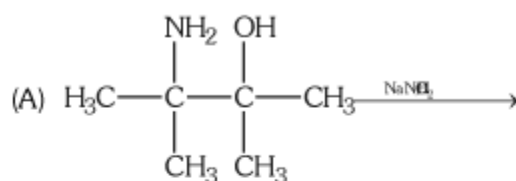
(B) The major product is

(C) Two products are obtained

(D) Two minor products are enantiomers.

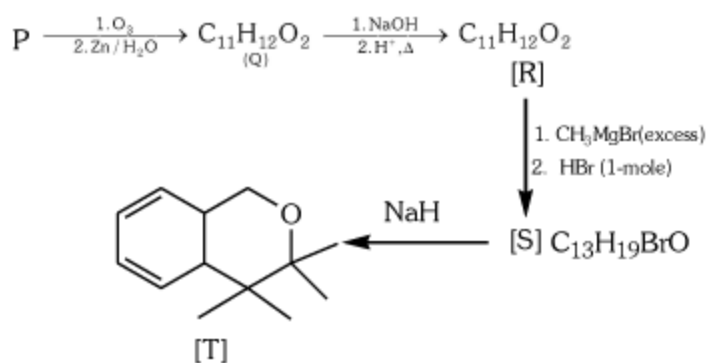
Ans. (A, B, D)

13. In which reaction the product will react with 2, 4 dinitrophenylhydrazine.



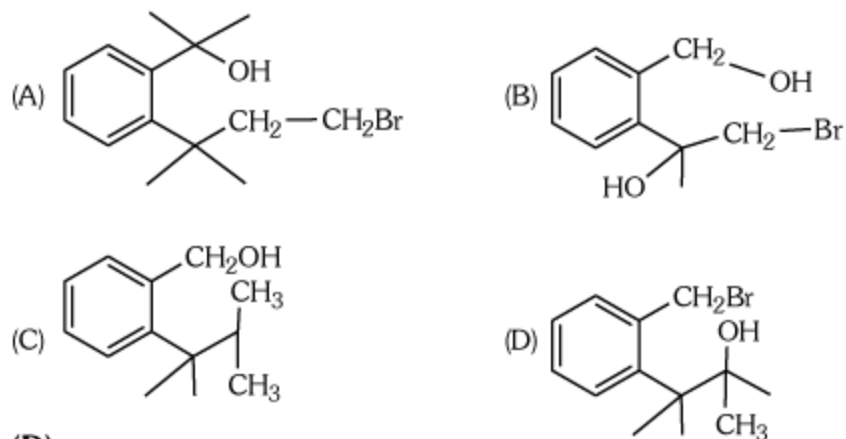
Ans. (A, B, C, D)

Paragraph-1



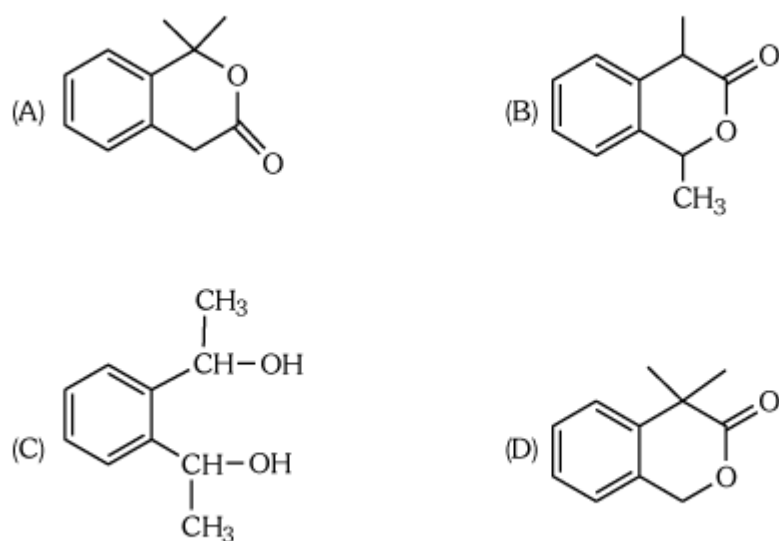
In the following scheme answer the following questions.

14. The compound (S) is



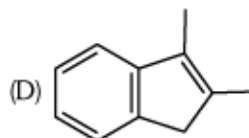
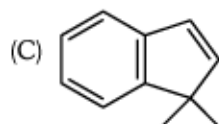
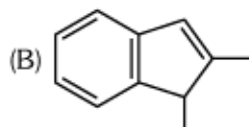
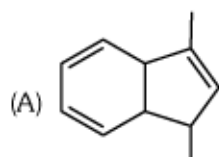
Ans. (D)

15. Compound [R] is



Ans. (D)

16. Compound (P) is

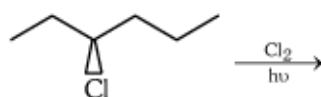


Ans. (C)

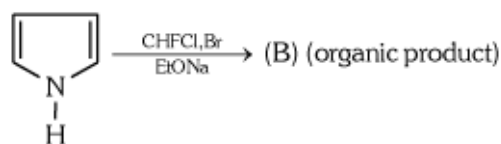
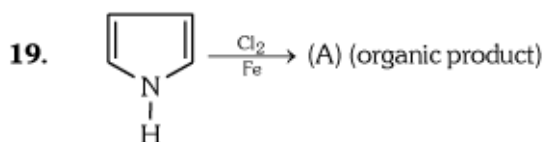
17. 'Diethyl- β , β -dimethyl glutarate' is condensed with diethyl oxalate in the presence of sodium ethoxide and ethanol to form a major product (P), which on acidic hydrolysis followed by heating gives another product (Q). Product (Q) is treated with HIO_4 to give (R). If the degree of unsaturation in (Q) and (R) is m and n respectively then (m + n) is

Ans. (5)

18. How many optically active fractions of dichloro compounds are obtained from the following reaction.



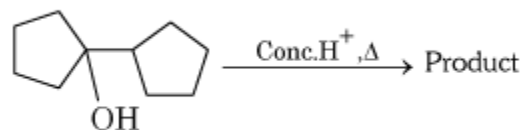
Ans. (7)



Give total number chlorine atom in product A and B.

Ans. (1)

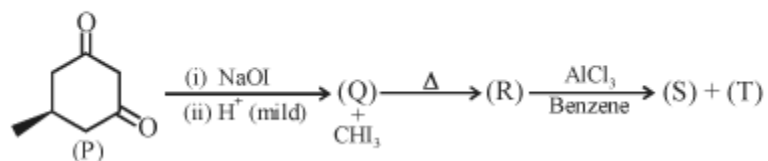
20. How many transition states are formed during the formation of product (major)



Ans. (5)

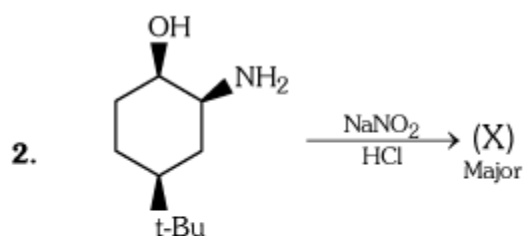
Aldehyde and Ketone

1. If Q, R, S, T are organic product then choose incorrect option ?

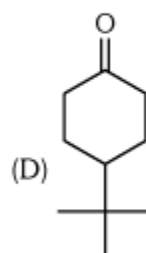
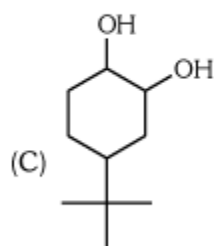
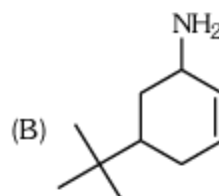
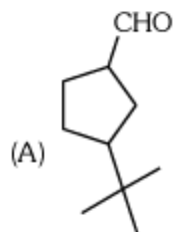


- (A) Q, R, S, T are soluble in hot aq. NaOH
 (B) Formation of (S) & (T) is electrophilic substitution reaction
 (C) Only (S) & (T) can show stereoisomerism
 (D) (R) & (Q) are optically active in nature

Ans. (D)

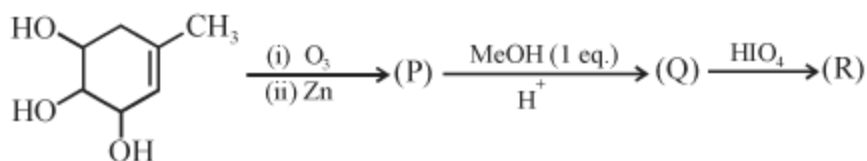


Product 'X' is

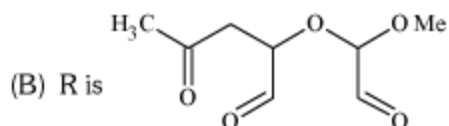


Ans. (A)

3. Correct statement regarding following reaction is/are :



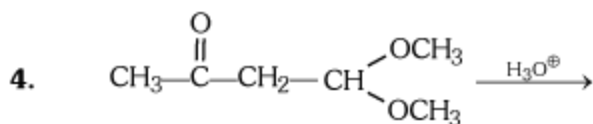
(A) Number of products obtained on hydrolysis of (R) is 4



(C) One of the products obtained on hydrolysis of (R) can give cannizaro reaction

(D) Q contains an acetal linkage

Ans. (A, B, C, D)



Identify correct statement about product.

(A) Product show iodoform test.

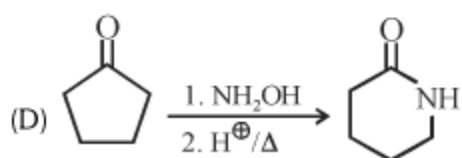
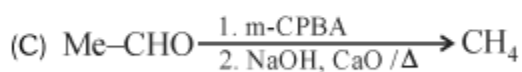
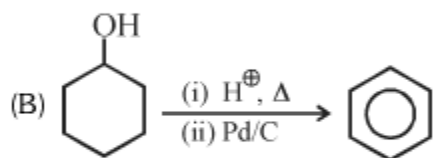
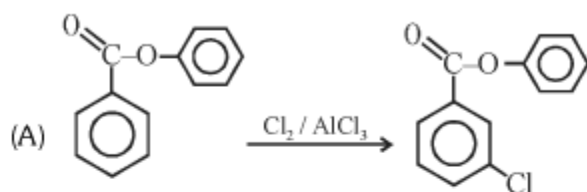
(B) Product show tollen test.

(C) Enol is more stable than keto.

(D) Enol show geometrical isomer.

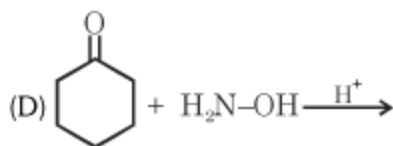
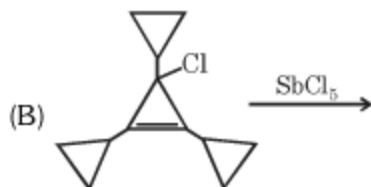
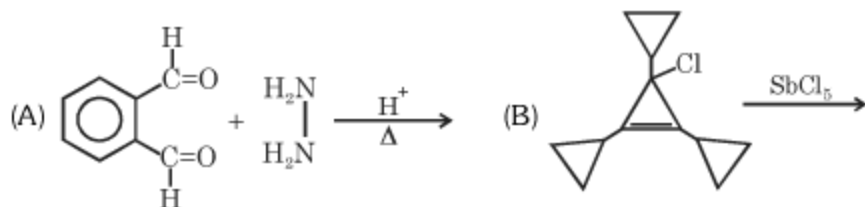
Ans. (A, B, C, D)

5. Select the correctly matched reaction(s) :



Ans. (B, C, D)

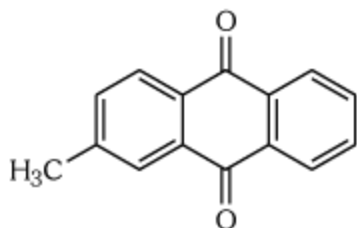
6. Which of the following reaction(s) will give aromatic product ?



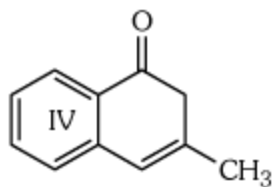
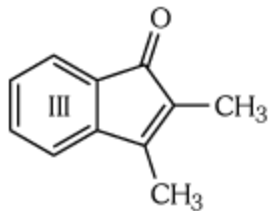
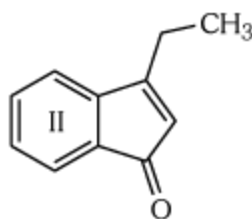
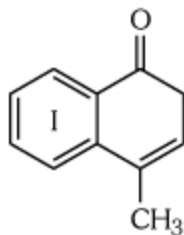
Ans. (A, B, C)

Paragraph-1

A hydrocarbon "**A**" has molecular formula " $C_{11}H_8$ ", and upon treatment with dilute H_2SO_4 in presence of $HgSO_4$ forms **B** ($C_{11}H_{12}O_2$) as the major product. Also, **A** on treatment with B_2H_6/H_2O_2 in alkaline medium gave "**C**", an isomer of **B**. Both **B** and **C** forms yellow precipitate on treatment with alkaline solution of iodine but only **C** forms a light-gray precipitate on treatment with ammonical solution of $AgNO_3$. **A** on boiling with acidic solution of $KMnO_4$ forms $C_8H_6O_4$, which on heating with P_2O_5 yields **D** ($C_8H_4O_3$). **D** on treatment with toluene in presence of $AlCl_3$ catalyst yields **E** ($C_{15}H_{12}O_3$). Major product **E** on refluxing with aqueous HF yielded the following compound as major product.



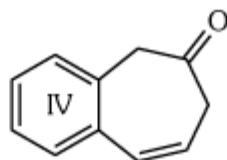
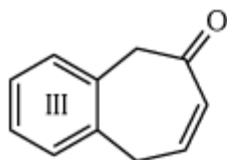
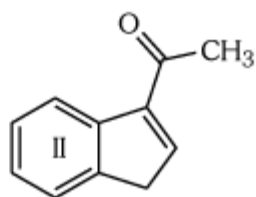
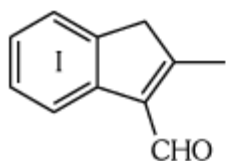
7. If **B** is treated with dil. $NaOH$ solution (aldol condensation), select the possible product(s) from the following list :



- (A) I, II, III & IV
(B) Only I, II & III
(C) Only II & III
(D) Only II, III & IV

Ans. (C)

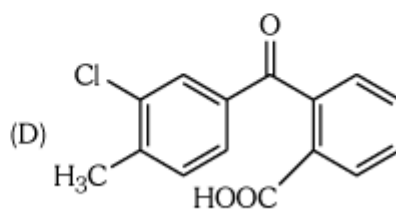
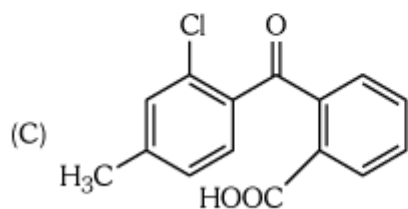
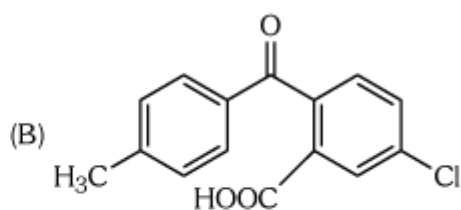
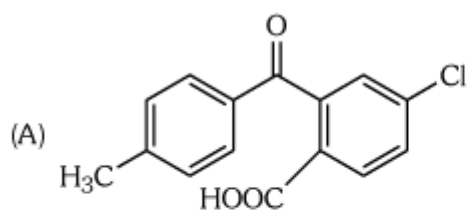
8. If C is treated with dil. NaOH solution, select the possible condensation product(s) from the following list of compound.



- (A) I, II, III & IV
 (B) Only I & III
 (C) Only I & IV
 (D) Only II & IV

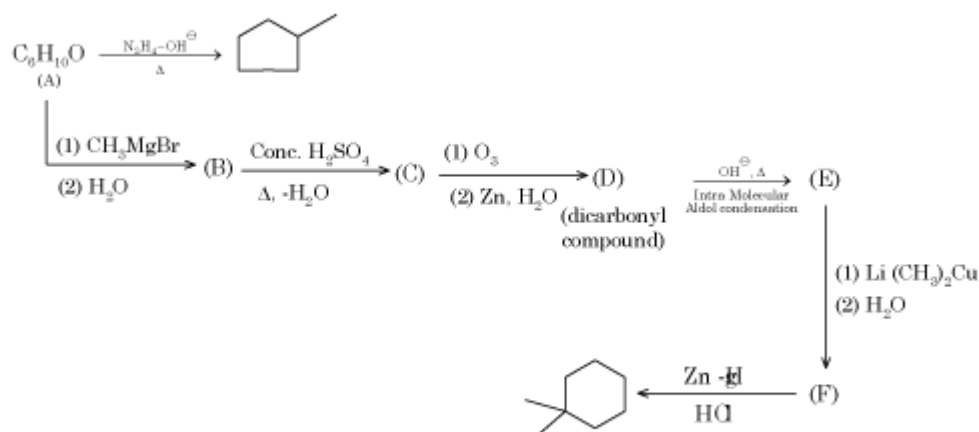
Ans. (A)

9. E on treatment with Cl_2 in presence of AlCl_3 forms predominantly.



Ans. (D)

Paragraph-2

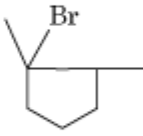
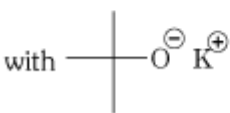


Compound (A) has (S) - configuration at the chiral center.

All organic products are major in above reaction scheme

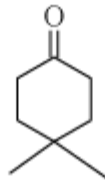
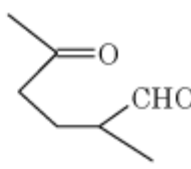
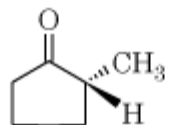
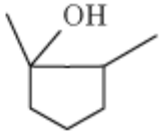
10. Select the correct statement ?

- (A) Formation of Compound (F) involves 1, 2-Addition.
 (B) Compound (C) on reaction with HBr or HBr, R_2O_2 gives same organic product.
 (C) Compound (A) on reaction with NaCN / HCN followed by H_2O gives enantiomers pair as a product.

(D) Compound (C) can also be obtained as major product when  heated with 

Ans. (B)

11. Select the correct one

- (A) Compound (F) =  (B) Compound (C) = 
- (C) Compound (A) =  (D) Compound (B) = 

Ans. (D)

(Various organic reactions)

(1) $2 \text{ H}-\text{C}(=\text{O})-\text{CH}_3 \xrightarrow[\text{NaOH} / \Delta]{\text{Ba(OH)}_2 \text{ or}} \text{CH}_3-\text{C}(=\text{O})-\text{CH}=\text{CH}_2 \rightarrow \text{Aldol condensation}$

(2) $2 \text{ H}-\text{C}(=\text{O})-\text{H} \xrightarrow[\text{NaOH}]{\text{Conc.}} \text{H}-\text{C}(=\text{O})-\text{ONa} + \text{CH}_3\text{OH} \rightarrow \text{Cannizaro's reaction}$

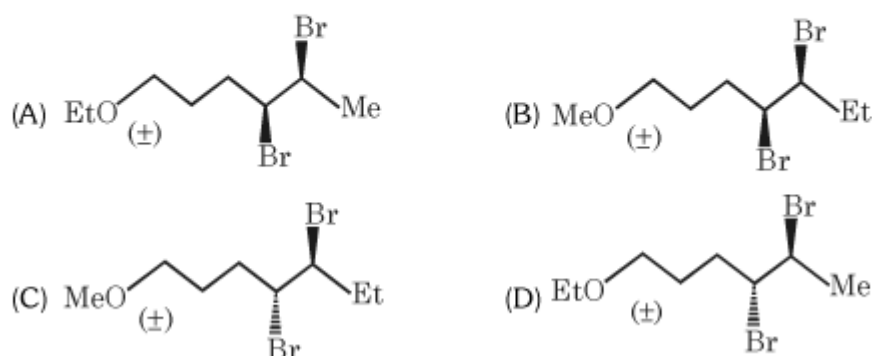
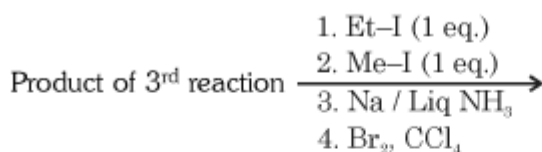
(3) $\text{HO}-\text{CH}_2\text{CH}_2\text{CH}_2-\text{C}\equiv\text{CH} \xrightarrow[\text{NaNH}_2]{\text{excess}} \text{NaO}-\text{CH}_2\text{CH}_2\text{CH}_2-\text{C}\equiv\text{CNa} \rightarrow \text{Acid base reaction}$

(4) $\text{RONa}^\oplus \xrightarrow[\text{R-OTs}]{\text{R-I}} \text{R-O-R} \rightarrow \text{S}_\text{N}^2$
in DMF

(5) $\text{H}_2\text{N}-\text{C}_6\text{H}_3(\text{OH})-\text{NH}-\text{C}(=\text{O})\text{CH}_3 \xrightarrow[\text{Pyridine}]{\text{Ac}_2\text{O, excess}} \text{CH}_3\text{CO}-\text{O}-\text{C}_6\text{H}_3(\text{OH})-\text{NH}-\text{C}(=\text{O})\text{CH}_3 \rightarrow \text{Acylation / Acetylation}$

(A) Reaction (1) involve dehydration & nucleophilic addition
(B) Reaction (2) is redox reaction
(C) During reaction (5), '84 gm' mass increases in product
(D) All of these

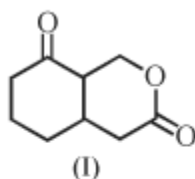
13. Major product of following reaction is :

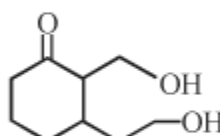


Ans. (C)

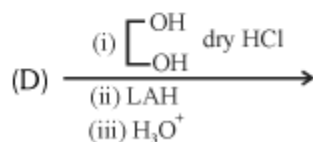
Paragraph-4

A ketonic lactone (I) is used to synthesis varieties of compounds. Observe its structure and answer to following questions given below :

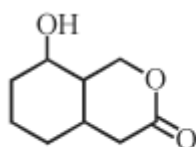


14.  can be prepared from above compound using which of the following reagent

- (A) LiAlH_4
(B) NaBH_4
(C) DIBAL-H



Ans. (D)

15.  can be prepared from above compound using which of the following reagent :

- (A) LiAlH_4
(B) NaBH_4
(C) DIBAL-H

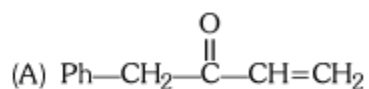


Ans. (B)

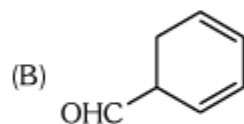
16. Match the Column.

Column-I

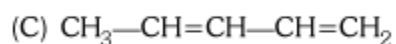
Column-II



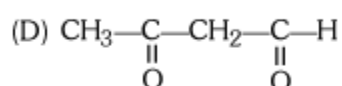
(P) 1,4-addition



(Q) Tautomerism



(R) $\text{AgNO}_3/\text{NH}_4\text{OH}$



(S) 2,4 DNP test

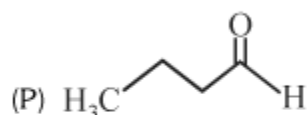
(T) $\text{Br}_2/\text{H}_2\text{O}$ test

Ans. [(A) → (P,Q,S,T); (B) → (P,Q,R,S,T); (C) → (P,T); (D) → (Q,R,S,T)]

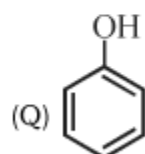
17. Match the following column

List-I
(Compound)

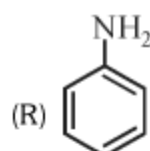
List-II
(Test)



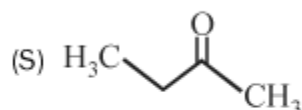
(1) +ve Iodoform test



(2) +ve Isocyanide test



(3) +ve Liebermann nitroso test



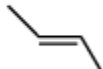
(4) +ve Tollen's test

Codes :	P	Q	R	S
(A)	1	3	2	4
(B)	4	2	3	1
(C)	1	2	3	4
(D)	4	3	2	1

Ans. (D)

18. (a) $\text{Me}-\text{CH}=\text{O} + \text{EtCHO} \xrightarrow[\Delta]{(\text{EtO})_2\text{Al}}$ Total number of cross Tischenko products is **x**.

(b)  + $\text{CHBrI}_2 + \text{Me}_3\text{COK} \longrightarrow$ Total number of cyclic products obtained by major carbene is **y**.

(c)  + $\text{CHF}_2\text{Cl} + \text{EtONa} \longrightarrow$ Total number of fractions obtained after fractional distillation of cyclic products obtained by major carbene is **z**.

Fill your answer as $x + y + z$ in OMR sheet. For example If $x = 2$, $y = 3$ & $z = 4$ then your answer is 9.

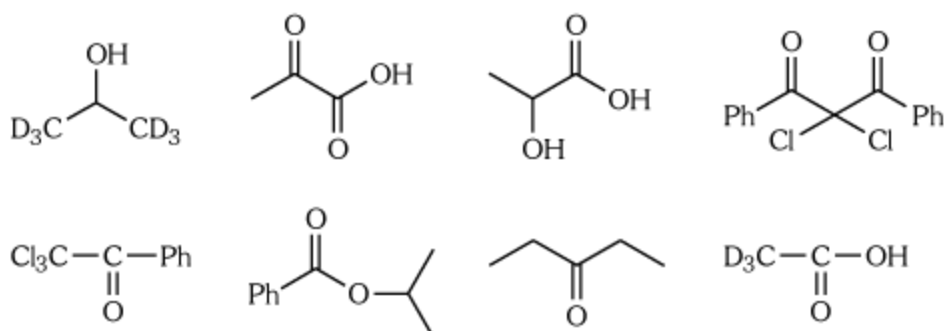
Ans. (7)

19. $\text{CH}_3-\text{CH}=\text{O} + \text{PhCH}_2-\text{CH}=\text{O} \xrightarrow{\text{HO}^\ominus}$ mixture of aldols
(z) total number of aldols including stereoisomers.

[Give your answer as sum of digits and Fill OMR sheet]

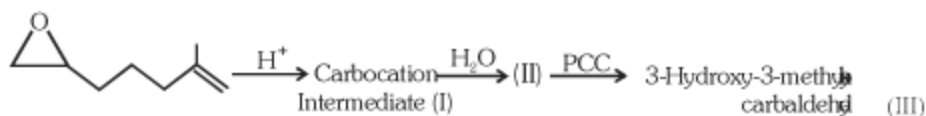
Ans. (3)

20. Number of compounds which can show haloform reaction are :



Ans. (6)

21. Consider the following reaction sequence :



How many of the following statements are correct for the given reaction sequence ?

- (i) Reaction involves cyclization
- (ii) Compound (II) can reduce HIO_4
- (iii) Reaction involves formation of a carbocation with 6 α H
- (iv) Compound (II) can immediately make turbidity with concentrated $HCl + anhydrous\ ZnCl_2$
- (v) Conversion of (II) to (III) is dehydrogenation reaction
- (vi) Compound (III) can react with $Ag(NH_3)_2^+$ solution

Ans. (4)

22. How many of the following reagents can convert $Ph-CHO$ into corresponding $Ph-COO^-$ or $Ph-COOH$

- | | |
|----------------------------|---------------------------------|
| (a) $KMnO_4 / H^+, \Delta$ | (b) PCC |
| (c) MnO_2 | (d) CF_3CO_3H |
| (e) $NaOI$ | (f) Ag_2O |
| (g) Fehling solution | (h) $K_2Cr_2O_7 / OH^-, \Delta$ |
| (i) conc. $NaOH, \Delta$ | (j) CrO_3, H_2SO_4 |

Ans. (6)

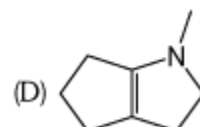
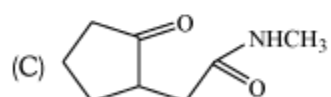
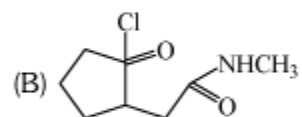
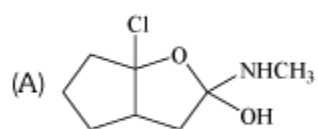
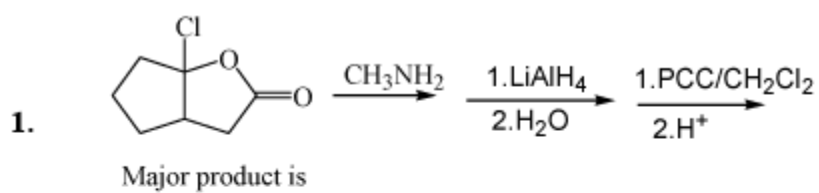
23. If compound which can show positive iodoform test = x and compound which can show positive fehling test = y

Then value of $x + y$ is :

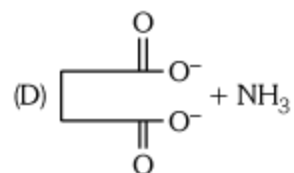
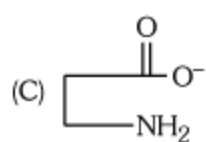
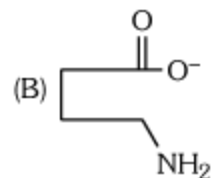
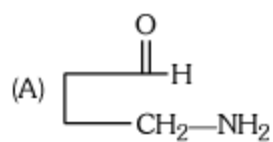
- | | |
|--------------------|-------------------|
| (i) Methanol | (ii) Ethanol |
| (iii) Propan-1-ol | (iv) Propan-2-ol |
| (v) Pentane-3-ol | (vi) Benzaldehyde |
| (vii) Acetaldehyde | (viii) Acetone |
| (ix) Pentan-3-one | (x) Butanone |

Ans. (6)

Amines

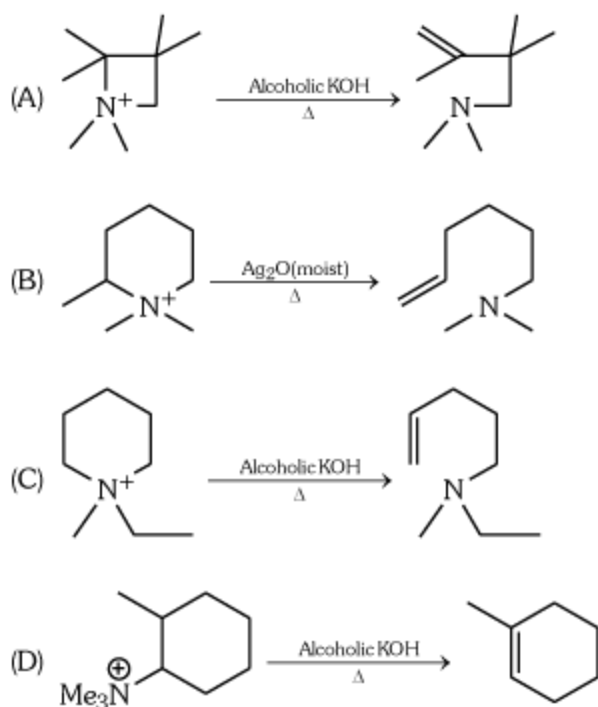


Ans. (D)



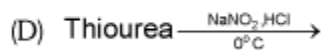
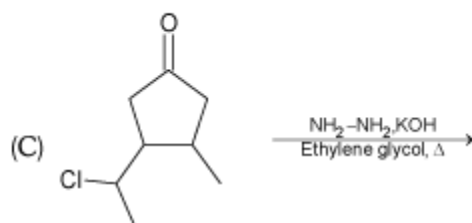
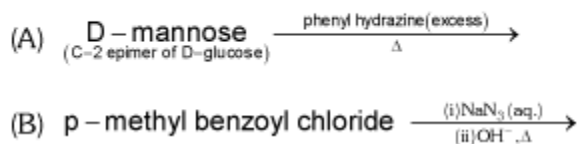
Ans. (C)

3. Identify reactions that correctly match with their major products.



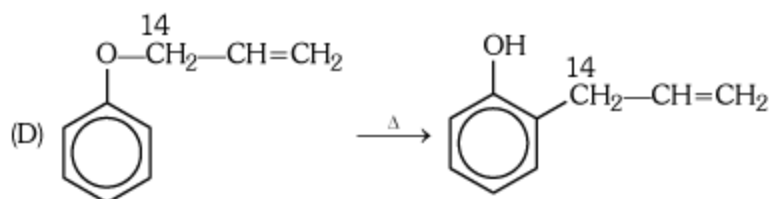
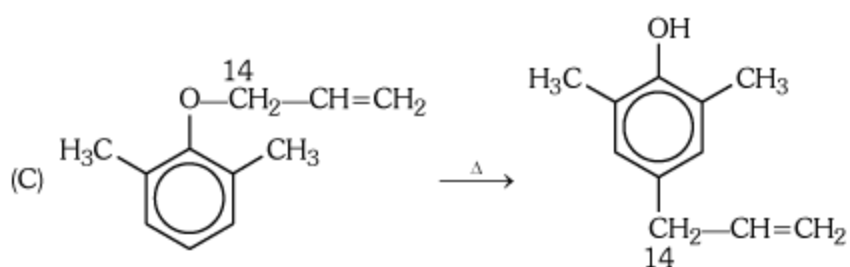
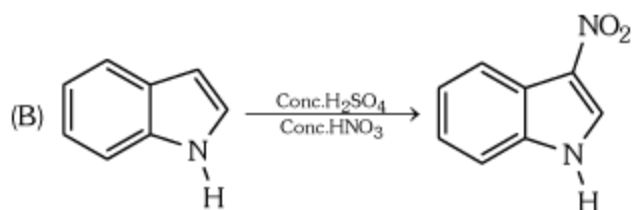
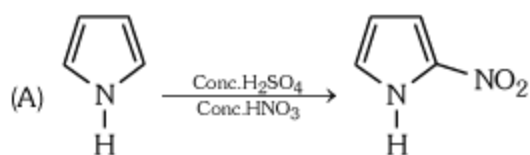
Ans. (A, B)

4. Which of the following reaction(s), takes place with the evolution of nitrogen gas :



Ans. (B, C, D)

5. Which of the following reaction is/are correct ?



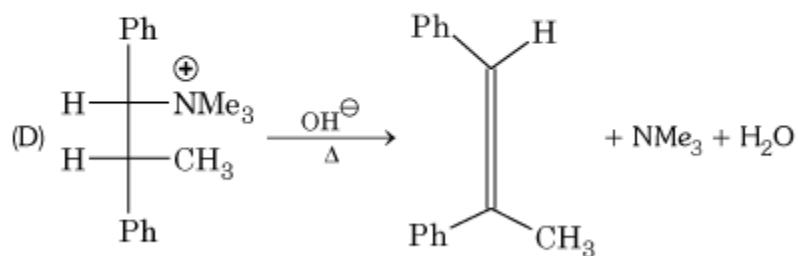
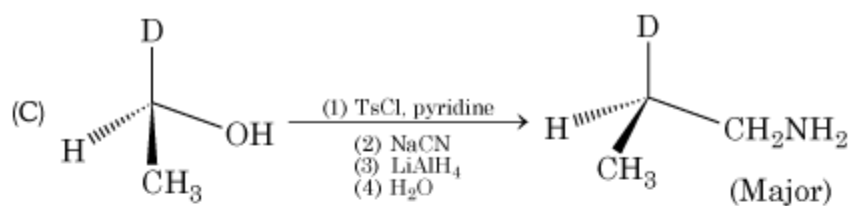
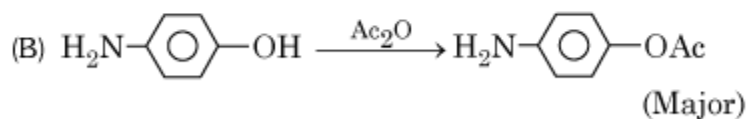
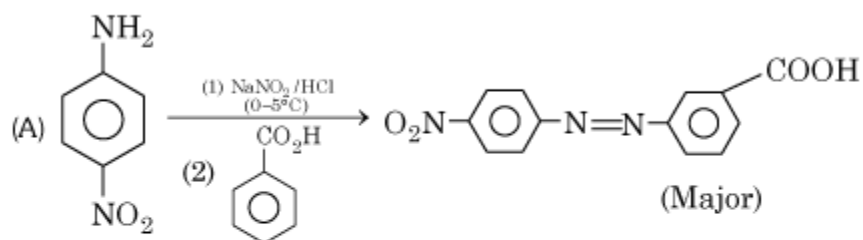
Ans. (A, B, C)

6. Which one of the following is an oxidation product of a primary amines ?

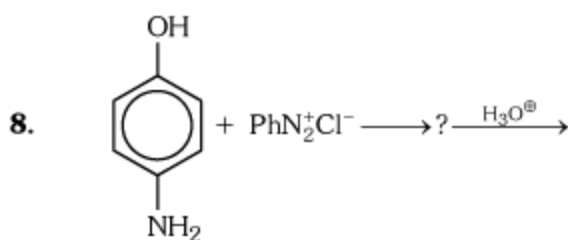
- (A) A hydroxylamine
- (B) A nitroproduct
- (C) A nitroso product
- (D) A alkyl hydrazine

Ans. (A, B, C)

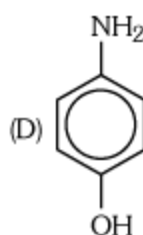
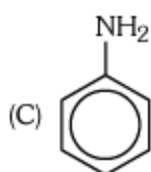
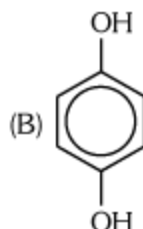
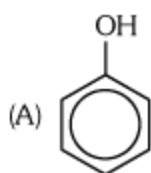
7. Among the following reactions, identify the incorrect ones.



Ans. (A, B, C)



Final product is a mixture containing :



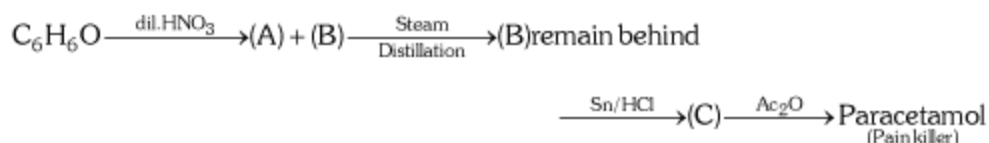
Ans. (A, B, C, D)

9. Find the correct statement(s).

- (A) Methyl amine in water reacts with FeCl_3 to precipitate hydrated ferric oxide.
- (B) Diazonium salt of aromatic amines are more stable than those of aliphatic amines.
- (C) Aniline does not undergo friedel–craft's reaction.
- (D) Gabriel phthalimide synthesis is preferred for 1° amine synthesis.

Ans. (A, B, C, D)

Paragraph-1

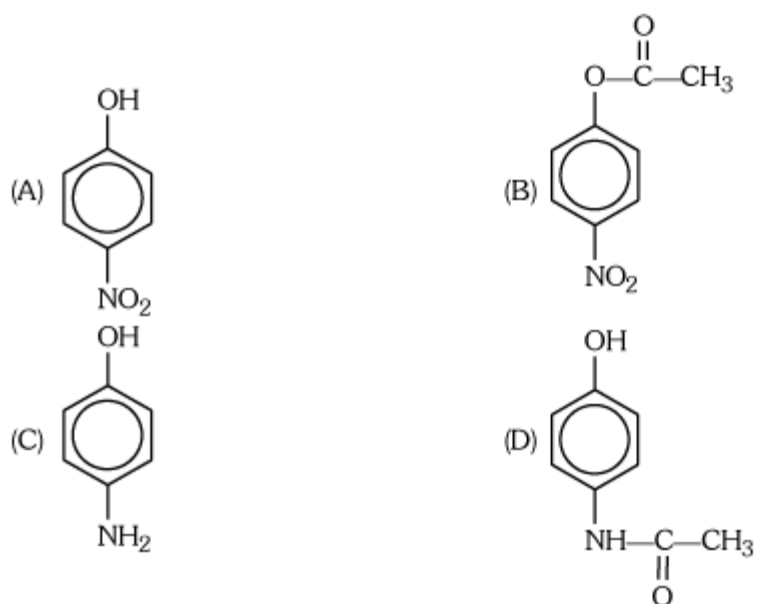


10. Select the correct option.

- (A) Compounds (A) and (B) are functional isomers.
- (B) Compound (A) is steam volatile and has lower boiling point.
- (C) If conc. HNO_3 is used in place of dil. HNO_3 then the yield of (A) and (B) will be high due to high conc. of NO_2^+
- (D) (B) is major product in the first reaction.

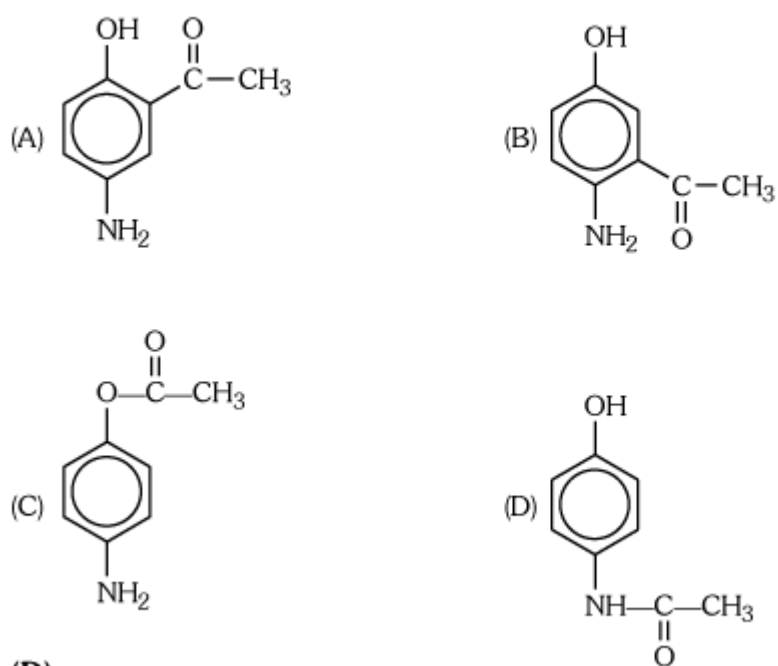
Ans. (B)

11. Structure of compound (C) is :



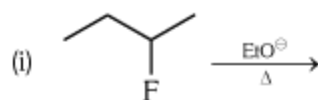
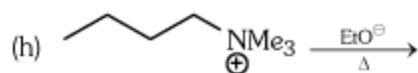
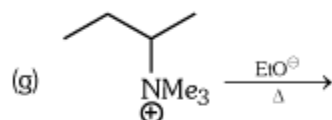
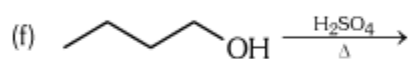
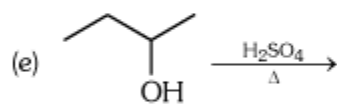
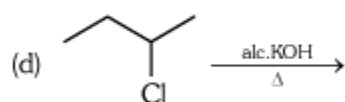
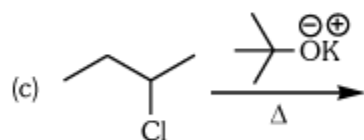
Ans. (C)

12. Paracetamol is :



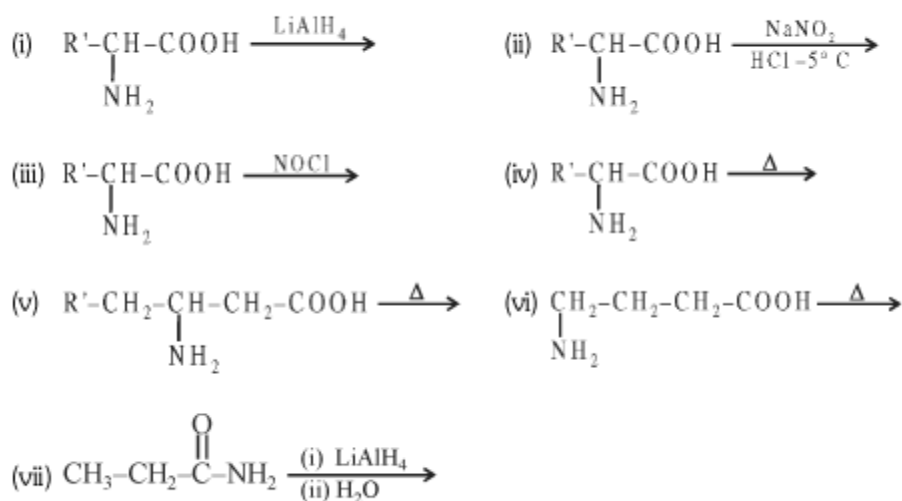
Ans. (D)

13. X = Number of reaction in which 1-butene is major products.



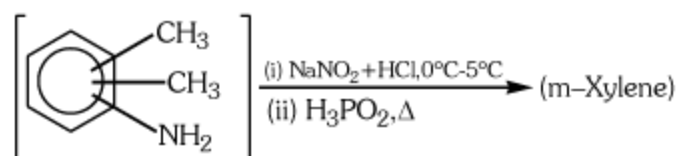
Ans. (6)

- 14.** How many of the following reactions contains nitrogen atom in the major organic product ?



Ans. (4)

- 15.** How many aromatic amino derivative of xylene can be converted to meta-xylene by following sequence of reactions.



Ans. (3)

- 16.** Number of compounds which will give oily layer with $\text{NaNO}_2 + \text{HCl}$:

- | | |
|----------------------------|----------------------------|
| (i) Methyl amine | (ii) Ethyl amine |
| (iii) N-methyl ethyl amine | (iv) N, N-dimethyl aniline |
| (v) N-ethyl aniline | (vi) Triethyl amine |

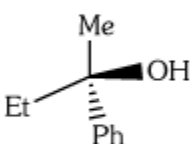
Ans. (2)

- 17.** How many amines are possible with molecular formula $\text{C}_4\text{H}_{11}\text{N}$ which react with Hinsberg reagent but the sulphonamide thus produced is insoluble in KOH .

Ans. (3)

18. (a) $\text{Cl}-\text{CH}_2-\overset{14}{\text{C}}\text{H}_2-\text{S}-\text{CH}_2-\text{CH}_3 \xrightarrow{\text{NH}_3}$ Number of amines produced.

(b)  $\xrightarrow{\text{alc. KOH}}$ Number of alkene produced.

(c)  $\xrightarrow{\text{HI}}$

How many iodides are produced in more than 5% yield.

(d) $\text{PhCOOH} \xrightarrow[\text{Cl}_2]{\text{Red P (limited)}}$

How many Cl atoms are present in the final major product.

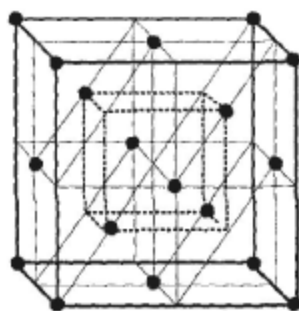
[Give your answer as a + b + c + d and fill OMR sheet]

Ans. (9)

Solid State

1. The structure of diamond and silicon is same and can be described as atoms taking corners and faces of a cube and also taking positions of alternate tetrahedral voids.

What is the value of $\frac{\text{density(diamond)}}{\text{density(silicon)}}$.



At.mass(g / mol)	Covalent Bond Length (Å)
Carbon : 12	C - C = 1.50
Silicon : 28	Si - Si = 2.25

- (A) 1.45
(B) 2.5
(C) 3.5
(D) 4.5

Ans. (A)

2. Copper metal crystallizes in FCC lattice. Edge length of unit cell is 362 pm. The radius of largest atom that can fit into the voids of copper lattice without disturbing it.
- (A) 53 pm
(B) 45 pm
(C) 93 pm
(D) 60 pm

Ans. (A)

3. An ionic solid consists of A^{+2} and B^{-} ion and the radius ratio of $\frac{r_{A^{+2}}}{r_{B^{-}}}$ is 0.8. Identify the **incorrect** statements regarding unit cell and location of ions.
- (A) Simple cubic of B^{-} ions and A^{+2} occupying all the body centered voids or cubic voids.
(B) Face centered cubic of B^{-} ions and A^{+2} occupying all tetrahedral voids.
(C) Face centered cubic of B^{-} ions and A^{+2} occupying 50% tetrahedral voids.
(D) Face centered cubic of A^{+2} ions and B^{-} occupying all tetrahedral voids.

Ans. (A, B, C)

4. Which of the following statements regarding defects in crystalline solids is/are **incorrect** ?
- (A) Schottky defects exist in non-ionic substances and does not affect density of substance
(B) Frenkel defects increases the dielectric constant.
(C) Introduction of impurity defect in Si by insertion of Al will create n-type semiconductors.
(D) The formation of wustite is an example of metal excess defect.

Ans. (A, C, D)

5. Which of the following statements regarding ionic crystals (assuming ideal behaviour) is/are correct.
- (A) Distance between two fluoride ions in CaF_2 = half the edge length of the crystal.
(B) In case of NaCl crystal, edge length = $\frac{4r^{-}}{\sqrt{2}}$ where r^{-} represents radius of Cl^{-} .
(C) In ZnS crystal, Zn^{2+} ion occupy alternate tetrahedral voids.
(D) In Na_2O structure, co-ordination number of O^{2-} is 8.

Ans. (A, C, D)

6. Which of the following statements regarding solids is correct ?
- (A) Electrical resistance of quartz glass will be different in different directions.
(B) Solid rubber melts at a sharp melting point
(C) Ionic solids are good conductors of electricity
(D) Amorphous solids may get converted to crystalline solids at some temperature.

Ans. (D)

7. 80.0 gm salt of weak base & strong acid XY is dissolved in water and formed 2 litre of aqueous solution. The pH of the resultant solution was found to be 5 at 298 K. If XY forms CsCl type crystal having r_{X^+} (radius of X^+) = 1.6 Å & r_{Y^-} (radius of Y^-) = 1.864 Å then select correct statement(s).

(Given : K_b (XOH) = 4×10^{-5} ; $N_A = 6 \times 10^{23}$)

- (A) Molar mass of salt is 100 g/mol
- (B) % Degree of dissociation of salt is 0.25
- (C) Edge length of AB is 4 Å
- (D) Density of solid salt XY is 2 in gm/cc

Ans. (A, C)

8. In a cubic lattice atom A occupy all corners, atoms B occupy alternate face centres and atoms C occupy alternate edge centres. If a defect is created by removing all atoms present on one of the line passing through opposite edge centres then which of the following formula is/are possible for new lattice.

- (A) A_2B_2C
- (B) $A_4B_4C_3$
- (C) ABC
- (D) $A_3B_3C_2$

Ans. (A, B, C)

9. Select the correct statements :

- (A) Distance between two nearest cations will be same as distance between two nearest anions in rock salt structure.
- (B) The packing efficiency of the hexagonal unit cell (two dimensional) is about 90%.
- (C) Solids having F-centres may have metal excess defect due to missing anions.
- (D) Silicon and germanium are intrinsic semiconductors.

Ans. (A, B, C, D)

10. Which of the following statement(s) is/are correct ?

- (A) In body centered cubic unit cell, edge length is equal to $\frac{4R}{\sqrt{3}}$ [R = Radius of atom]
- (B) In BCC unit cell, the edge length is equal to $\frac{2d}{\sqrt{3}}$ where 'd' is the shortest distance between two lattice points.
- (C) Distance between two closest octahedral void in fcc is $2R$ [R = Radius of atom]
- (D) Hexagonal crystal system has only one type of bravais lattice.

Ans. (A, B, C, D)

- 11.** Which of the following is/are true.
- (A) Due to Schottky defect density of crystal remains unaffected.
 - (B) In zinc blend S^{2-} form FCC while alternate tetrahedral voids are occupied by Zn^{2+} .
 - (C) Number of next nearest neighbours of Cs^+ ion in CsCl structure is 6 Cs^+ ions.
 - (D) A crystal has A ions at the cube corners and B ions at the edge centres. The coordination number of A and B are respectively 6 and 2.

Ans. (B, C, D)

- 12.** Identify correct option(s) among following statements :
- (A) In Na_2O crystal oxide ion is present in a cubic void formed by Na^+ ions
 - (B) AgBr shows both schottky and frenkel defects
 - (C) Metals can conduct electricity in molten state
 - (D) $Fe_{0.95}O$ crystal can work as n-type semiconductor.

Ans. (A, B, C)

Paragraph-1

Spinel is a mineral $MgAl_2O_4$ having general formula MM'_2O_4 , where divalent ions are occupied in tetrahedral voids and trivalent ions in octahedral voids and lattice is made up of oxide ion present in CCP arrangement. Only 1/8 of tetrahedral and 1/2 of octahedral void are occupied. (Assume ideal arrangement of particles)

- 13.** The % of Mg in $MgAl_2O_4$ is
- (A) 16.9%
 - (B) 38%
 - (C) 12.5%
 - (D) None of these

Ans. (A)

- 14.** If 50% octahedral voids occupied by Al^{3+} and 12.5% tetrahedral void occupied by Mg^{2+} , then formula of compound is
- (A) $MgAl_2O_4$
 - (B) $Mg_4Al_3O_4$
 - (C) $Mg_2Al_2O_5$
 - (D) None of these

Ans. (A)

15. A given sample of wustite has the formula $\text{Fe}_{0.95}\text{O}$. It is cubic closed packed crystal with respect to oxide ion and Fe^{2+} and Fe^{3+} ions are present in octahedral voids. Match List I with List II and select the correct answer using the codes given below the lists.

Column-I

Column-II

- | | |
|---|-----------|
| (A) The fraction of iron ions present as Fe^{2+} | (P) 17/20 |
| (B) The fraction of iron ions present Fe^{3+} | (Q) 3/20 |
| (C) The fraction of octahedral voids occupied by Fe^{2+} | (R) 2/19 |
| (D) The fraction of octahedral voids occupied by Fe^{3+} | (S) 17/19 |

Ans. [(A) → (S); (B) → (R); (C) → (P); (D) → (Q)]

16. A metal 'A' exists in the bcc structure having uncovered length between the atoms along the edge equal to $1.2 \text{ \AA}$. If molar mass of the metal is 307.2 gm/mole then calculate density of the crystal in gm/ml .

[Use : $N_A = 6 \times 10^{23}$ and $\sqrt{3} = 1.7$]

Ans. (2)

17. A solid element (monoatomic) exist as cubic crystal. If its atomic radius is $1.0 \text{ \AA}$ and the ratio of packing fraction and density is $0.1 \text{ cm}^3/\text{gm}$, then the atomic mass of the element is in terms of $x\pi$. What is the value of x ?

($N_A = 6 \times 10^{23}$)

Ans. (8)

18. The density of solid Argon is 1.6 gm/ml at -233°C . If the atomic volume of Argon is assumed to be $\frac{5}{3} \times 10^{-23}$ then what % of solid Argon is apparently empty space.

[Give your answer as sum of digits and fill OMR sheet]

Ans. (6)

19. If B^{-2} ion forms hexagonal close packing and A^{+3} ions occupy appropriate number of 3D-voids then ratio of number of smaller size 3D-voids made by packing of B^{-2} ions and number of A^{+3} ions present in the made ionic solid is

Ans. (3)

Liquid Solution

1. The freezing point of aqueous solution of $0.1 \text{ m CH}_3\text{COOH}$ will be :
(If CH_3COOH is 60% dimerised) :

- (A) $-0.07 K_f$
(B) $-2.6 K_f$
(C) $-4.2 K_f$
(D) $0.07 K_f$

Ans. (A)

2. Identify the option(s) which are not correct w.r.t solubility of a gas in water.
- (A) Value of K_H (Henry's constant) will decrease as the temperature increases.
 - (B) Value of K_H (Henry's constant) at same temperature for O_2 gas will be less as compared to H_2 gas.
 - (C) Soft drinks will have greater "Fizz" if sealed under low pressure.
 - (D) **Scuba** divers when they move underwater will have increased solubility of gases in their blood.

Ans. (A, C)

3. Acetone and carbon disulphide form binary liquid solution showing positive deviation from Raoult's law. The normal boiling point of pure acetone is less than that of pure carbon disulphide. Select the incorrect statement(s) among the following.
- (A) Boiling point of mixture is always less than the boiling point of acetone.
 - (B) Boiling point of the azeotropic mixture is always less than the boiling point of pure carbon disulphide.
 - (C) When a small amount of carbon disulphide is added to an excess of acetone, boiling point of solution increases.
 - (D) A mixture of acetone and carbon disulphide can be completely separated by simple fractional distillation.

Ans. (A, C, D)

4. Select correct statement(s) regarding a solution of non-volatile solid solute in a liquid solvent :
- (A) Heat of vaporisation for a pure solvent and for a solution are similar because similar intermolecular forces between solvent molecules must be overcome in both cases.
 - (B) Entropy change between solution and vapour is similar to the entropy change between pure solvent and vapour.
 - (C) Boiling point of the solution is larger than that of the pure solvent.
 - (D) Freezing point of the solution is larger than that of pure solvent.

Ans. (A, C)

5. Which of the following option(s) is/are correct w.r.t Henry's law ?

[Assuming solvent to be water]

- (A) Value of K_H (Henry's constant in atm) increases with increase in temperature
- (B) The Henry's law constant (K_H in atm) of CH_4 in water will be higher than that in benzene at the same temperature.
- (C) The Henry's law constant (K_H in atm) of CS_2 in water will be higher than that in benzene at the same temperature.
- (D) Henry's law is applicable when solubility of gas is low.

Ans. (A, B, C, D)

6. A non-volatile solute 'X' completely dimerises in water, if the temperature is below -3.72°C and the solute completely dissociates as $\text{X} \longrightarrow \text{Y} + \text{Z}$, if the temperature is above 100.26°C . In between these temperature, (including both temperature) 'X' is neither dissociated nor associated. One mole of 'X' is dissolved in 1 kg water.

Given : K_b of water = $0.52 \text{ K.Kg mol}^{-1}$

K_f of water = $1.86 \text{ K.Kg mol}^{-1}$

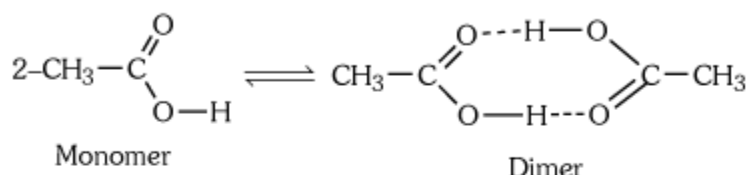
Which of the following statement(s) is / are true regarding the solution?

- (A) The freezing point of solution is -1.86°C .
 (B) The boiling point of solution is 101.04°C .
 (C) When the solution is cooled to -7.44°C , 75% of water present initially will separate as ice.
 (D) When the solution is heated to 102.08°C , 50% of water present initially will escape out as vapour.

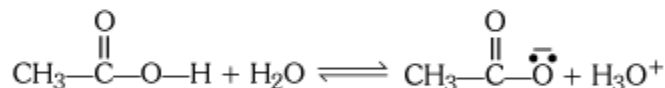
Ans. (A, B, D)

Paragraph-1

Acetic acid tends to form dimer due to the formation of intermolecular hydrogen bonding.



The equilibrium constant for this reaction is $1.5 \times 10^{-2} \text{ M}^{-1}$ in benzene solution and $3.6 \times 10^{-2} \text{ M}^{-1}$ in water. In benzene, monomer does not dissociate but in water, monomer dissociates simultaneously with acid dissociation constant $2.0 \times 10^{-5} \text{ M}$.



Dimer does not dissociate in benzene as well as in water.

[Given : $\log 2 = 0.3$]

7. The molar ratio of dimer to monomer for 0.1 M acetic acid in benzene is equal to
 (A) 150 : 1
 (B) 1 : 150
 (C) 5 : 2
 (D) 2 : 5

Ans. (C)

8. The molar ratio of dimer to monomer for 0.1 M acetic acid in water (neglecting the dissociation of acetic acid in water) is equal to
 (A) 250 : 1
 (B) 1 : 250
 (C) 9 : 2500
 (D) 2500 : 9

Ans. (C)

9. The pH of 0.1 M acetic acid solution in water, considering the simultaneous dimerisation of acid is
 (A) 1
 (B) 2.85
 (C) 5.7
 (D) 3.42

Ans. (B)

Paragraph-2

The solubility of any substance in water may be classified as :

Molecular solubility : $AB(s/1/g) \xrightleftharpoons{aq} AB(aq)$

Ionic solubility : $AB(s/1/g) \xrightleftharpoons{aq} A^+(aq) + B^-(aq)$

For a substance AB(s), the following thermodynamics information are available at 300 K:

Molecular solubility : $\Delta H^\circ = +20 \text{ Kcal/mol}$; $\Delta S^\circ = +40 \text{ Cal/K-mole}$

Ionic solubility : $\Delta H^\circ = -25 \text{ KCal/mol}$; $\Delta S^\circ = -50 \text{ Cal/K-mole}$

10. Which of the following statement is correct for the solubility of AB(s) in water at 300 K ?
 (A) Molecular solubility is favoured
 (B) Ionic solubility is favoured
 (C) Molecular and Ionic solubility is favoured
 (D) It is almost completely insoluble in water.

Ans. (B)

11. At what temperature molecular solubility is spontaneous ?
 (A) 300 K
 (B) 400 K
 (C) 200 K
 (D) 600 K

Ans. (D)

- 12.** When the temperature of aqueous solution of $AB(s)$ is increased from 300 K, then the extent of
- (A) Molecular as well as ionic solubility increases
 - (B) Molecular as well as ionic solubility decreases
 - (C) Molecular solubility increases but ionic solubility decreases.
 - (D) Molecular solubility decreases but ionic solubility increases.

Ans. (C)

Paragraph-3

An element (X) having same chemical properties as that of hydrogen atom. The diatomic molecule X_2 is gaseous in nature. The energy required to remove an electron from the outermost shell of (X) atom is 13.5 eV. It is also found that the time taken for diffusion of equal volume of gaseous X_2 & O_2 at the same pressure is in the ratio $\sqrt{3} : 4$. The normal freezing point of pure X_2O is $0^\circ C$. However on adding 0.02 moles of a non-electrolyte solute to 0.8 moles of X_2O , the freezing point of solution is found to be $-1.25^\circ C$. X-atom also obey Bohr's model.

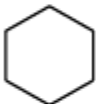
- 13.** When 3 gm of a weak monoprotic acid ($M_w = 60$) is added to 0.8 moles of X_2O , the resulting solution freeze at $-3.75^\circ C$. What is degree of dissociation for weak acid.
- (A) 0.10
 - (B) 0.20
 - (C) 0.30
 - (D) 0.40

Ans. (B)

- 14.** What amount of electricity is required to produce 3.3 gm of X_2 gas by electrolysis of X_2O .
- (A) 0.1 F
 - (B) 1.1 F
 - (C) 0.3 F
 - (D) 0.4 F

Ans. (B)

Answer Q.15, Q.16 & Q.17 by appropriately matching the information given in the three columns of the following table.

	Column-1	Column-2	Column-3
(I)	$C_6H_6 + C_6H_5CH_3$	(i) $\Delta H_{mix.} = 0$	(P) Positive deviation (Minimum boiling azeotrope)
(II)	$H_2O + CS_2$	(ii) $\Delta H_{mix.} = +ve$	(Q) Negative deviation (Maximum boiling azeotrope)
(III)	 + C_2H_5OH	(iii) $\Delta H_{mix.} = -ve$	(R) Follows Raoult's law
(IV)	$CHCl_3 + CH_3-\overset{\overset{O}{\parallel}}{C}-CH_3$	(iv) $\Delta S_{sys.} = +ve$	(S) Immiscible

15. Find the correct match.

- (A) (I) - (i) - (R)
 (B) (I) - (iv) - (S)
 (C) (III) - (ii) - (S)
 (D) (IV) - (iv) - (S)

Ans. (A)

16. Find the correct match.

- (A) (II) - (ii) - (R)
 (B) (II) - (i) - (S)
 (C) (I) - (i) - (Q)
 (D) (I) - (iv) - (P)

Ans. (B)

17. Find the incorrect match.

- (A) (IV) - (iii) - (Q)
 (B) (I) - (iv) - (R)
 (C) (III) - (iii) - (P)
 (D) (II) - (i) - (S)

Ans. (C)

18. List-I

- (A) Propan-2-ol + 2-methyl propanol
 (B) $H_2O + HNO_3$
 (C) $CS_2 + CH_3COCH_3$
 (D) $H_2O + HCOOH$

List-II

- (P) Maximum boiling azeotrope
 (Q) Raoult's law is obeyed
 (R) Positive deviation from Raoult's law
 (S) Fractional distillation column gives azeotrope from residual liquid

Ans. [(A) → (Q); (B) → (P, S); (C) → (R); (D) → (P, S)]

19. Match the List-I with List-II.

List-I

- (A) Acetone + CHCl_3
 (B) Ethanol + Water
 (C) $\text{C}_2\text{H}_5\text{Br} + \text{C}_2\text{H}_5\text{I}$
 (D) $\text{HNO}_3 + \text{H}_2\text{O}$
 azeotrope

List-II

- (P) $\Delta S_{\text{mix}} > 0$
 (Q) $\Delta V_{\text{mix}} > 0$,
 (R) $\Delta H_{\text{mix}} < 0$,
 (S) May have maximum boiling point

Ans. [(A) $\rightarrow$ (P, R, S); (B) $\rightarrow$ (P, Q); (C) $\rightarrow$ (P); (D) $\rightarrow$ (P, R, S)]

20. Find freezing point temperature (in Kelvin) of liquid if vapour pressure of solid & liquid are given by following expression.

$$\ln p_s = 10 - \frac{3000(\text{K})}{T} \quad \text{where } P_s = \text{vapour pressure of solid}$$

$$\ln p_\ell = 5 - \frac{2000(\text{K})}{T} \quad P_\ell = \text{vapour pressure of liquid}$$

Note : All terms are in SI unit

Ans. (200)

21. In an aqueous solution mole fraction of solute is 0.2 and its density is d_1 (g/ml). It is diluted such that mole fraction of solute becomes 0.1 and density becomes d_2 g/ml. If density of new solution is 20% less than initial solution then molarity of new solution is almost half of the original solution. If molar mass of the solute is 'X' gm/mol, then

closest integer value of $\left[\frac{X}{13} \right]$ is

Ans. (6)

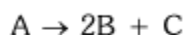
22. The vapour pressure of a saturated solution of sparingly soluble salt (X_3Y_2) was 17.5 mm Hg at 27°C . If the vapour pressure of pure H_2O is 18 mm Hg at 300 K, what is the solubility of sparingly soluble salt X_3Y_2 in mole/Litre.

Give your answer by multiplying 10 and taking greatest integer function.

For example if your answer is 0.51 then $0.51 \times 10 = 5.1$, then your answer is 5, fill OMR sheet.

Ans. (3)

- 23.** Decomposition of non-volatile solute 'A' into another non-volatile solute B and C, when dissolved in water follows as :



When one mole of A is dissolved in 180gm of water and left for decomposition, the vapour pressure of solution was found to be 20 mm Hg. Find moles of A decomposed. Assume constant temperature of 25°C, throughout. The vapour pressure of pure water at 25°C is 24 mmHg.

[Fill your answer by multiplying it with 4]

Ans. (2)

- 24.** x gms of glucose should be dissolved in 500 gms of water in order to produce a solution with 105°C difference between the freezing point and boiling point.

For water $K_f = 1.9 \text{ K-Kg/mol}$ and $K_b = 0.6 \text{ K-Kg/mol}$. Calculate the value of $\left(\frac{x}{36}\right)$

Ans. (5)

- 25.** Depression in freezing point of 0.01 molal aqueous monoacidic solution is 0.020°C. 1.86m urea aqueous solution freezes at -1.86°C. Assuming molality of solution is equal to molarity, pH of solution would be

Ans. (2)

- 26.** Ionic compound AB have NaCl type structure and its density is 3.6 gm/cm³. 4 gm of this compound added in 200 gm of water. Assume 100% ionisation of AB in solution. The freezing point of solution is 269.28 K. If radius of anion is 126.66 pm then calculate the approximate value of Z.

$$\text{where } Z = \frac{\text{radius of anion}}{\text{radius of cation}} - 0.1665$$

Given : $K_f(\text{H}_2\text{O}) = 1.86 \text{ kg mol}^{-1}$, $N_A = 6 \times 10^{23}$, Normal freezing point of water = 273K.

Ans. (3)

Electrochemistry

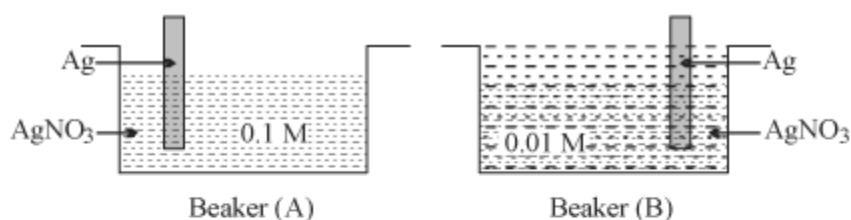
1. Which of the following statements regarding electrolytic conductance is not correct ?
- (A) The value of constant, 'A' in the equation for variation of molar conductance with square root of concentration, $\lambda_m^c = \lambda_m^0 - A\sqrt{C}$ for KCl and NaCl (A_{KCl} & A_{NaCl}) aqueous solution will be same at same temperature.
- (B) The value of limiting molar conductivity (λ_m^0) of CH_3COOH solution can not be obtained on extrapolation of graph of λ_m vs $\sqrt{C}$
- (C) The correct order of limiting molar conductivity of H^+ , Na^+ and Mg^{2+} follow the order $\lambda_{\text{mH}^+}^0 > \lambda_{\text{mMg}^{2+}}^0 > \lambda_{\text{mNa}^+}^0$
- (D) For all weak electrolytes, $K_{\text{eq}} = \frac{C\lambda_{\text{m,C}}^2}{\lambda_m^0(\lambda_m^0 - \lambda_{\text{m,C}}^0)}$, where K_{eq} is the dissociation constant, $\lambda_{\text{m,C}}$ is molar conductivity at any concentration and λ_m^0 is limiting molar conductivity.

Ans. (D)

2. Two beakers A and B contains AgNO_3 solutions of concentrations 0.1 M and 0.01 M respectively as shown in figures. Pure silver electrodes are now lowered in each beaker. Now NH_3 is added in each beaker till conc. of NH_3 in beaker A and B become 0.1 M and 0.2 M respectively. If two beakers are now connected in Galvanic cell arrangement, find the EMF (in Volts) of the resulting cell at 298 K.

[Use: $\log_{10} 2 = 0.3$, $\frac{RT}{F}(2.303) = 0.06$ at 298 K]

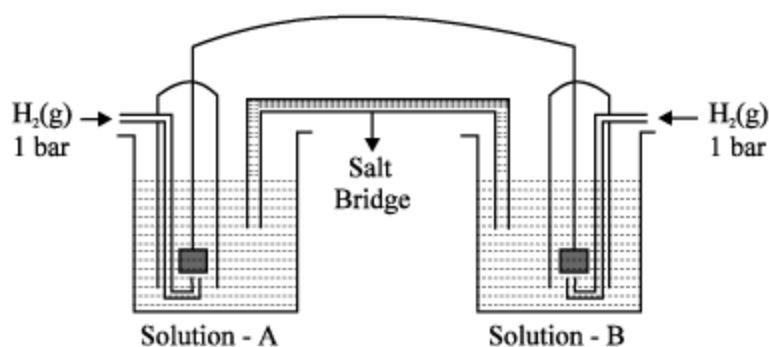
Given: $E_{\text{Ag}^+/\text{Ag}}^\circ = 0.8$ Volt, $K_f[\text{Ag}(\text{NH}_3)_2]^+ = 10^8$



- (A) 9.6×10^{-2}
 (B) 7.8×10^{-2}
 (C) 5.8×10^{-3}
 (D) None of these

Ans. (A)

3. Carefully observe the given figure and using data provided find the EMF of shown Galvanic cell (in volt) :



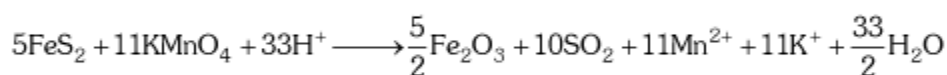
Solution A is 0.1 M each in NH_4OH and NH_4Cl and solution B is 0.1 M CH_3COONa .

[Given : $K_a(\text{CH}_3\text{COOH}) = 10^{-5}$, $K_b(\text{NH}_4\text{OH}) = 10^{-5}$ and $\frac{2.303RT}{F} = 0.06 \text{ volt}$]

- (A) 0.24 V
(B) 0.12 V
(C) 0.06 V
(D) 0 V

Ans. (D)

4. For the reaction,



$E_{\text{Cell}}^\circ = 2 \text{ V}$ and temperature coefficient of EMF = $\frac{1}{298 \times 5.5} \text{ V/K}$. Hence, $\Delta H_{\text{reaction}}^\circ$

for the above reaction at 298 K will be

- (A) 9650 kJ
(B) -9650 J
(C) -96500 kJ
(D) -9650 kJ

Ans. (D)

5. Conductivity of saturated solution of AgCl is given by which of the following options if limiting ionic conductance of Ag^+ and Cl^- ions is $80 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ and $100 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ respectively & K_{sp} of AgCl = 10^{-20}

(A) $6 \times 10^{-7} \Omega^{-1} \text{cm}^{-1}$
(B) $1.8 \times 10^{-13} \Omega^{-1} \text{cm}^{-1}$
(C) $1.8 \times 10^{-11} \Omega^{-1} \text{cm}^{-1}$
(D) $6 \times 10^{-5} \Omega^{-1} \text{cm}^{-1}$

Ans. (C)

6. Current efficiency is defined as the extent of a desired electrochemical reaction divided by the theoretical extent of the reaction times 100%. What is the current efficiency of an electrodeposition of Ag metal in which 9.8 g Ag is deposited by passage of 3A current for 10000 seconds.

(A) 0.9%
(B) 49.9%
(C) 28.9%
(D) 51.1%

Ans. (C)

7. What is the value of $E_{\text{Cd}^{2+}|\text{Cd}}$ at pH = 10

Given : $E_{\text{Cd}|\text{Cd}^{2+}}^0 = 0.40 \text{V}$; $K_{\text{SP}(\text{Cd}(\text{OH})_2)} = 10^{-14}$

(A) -0.5773V
(B) -0.2227V
(C) $+0.2227 \text{V}$
(D) -0.1045V

Ans. (A)

8. Which of the following observations is/are correct regarding electrolysis ?

(A) Electric current is used to drive a non-spontaneous reaction
(B) In electrolysis of molten PbBr_2 oxygen produces at anode.
(C) Cations and anions are discharged at the cathode and anode respectively
(D) Solution become acidic on electrolysis of $\text{CuSO}_4(\text{aq})$

Ans. (A, C, D)

9. A Daniel Cell is made at 25°C by connecting Zn / Zn⁺² (0.1 M, 1 litre) & Cu / Cu⁺² (0.9 M, 1 litre) electrode. The cell is discharged till its emf reaches 1.10 volt, then it is charged (reversing the discharge process) by passing 0.6 F charge.

[Given : $E^\circ_{\text{Zn/Zn}^{+2}} = 0.76\text{ V}$, $E^\circ_{\text{Cu/Cu}^{+2}} = -0.34\text{ V}$, $\frac{2.303RT}{F} = 0.06$, $\log 2 = 0.30$]

Select the correct option(s).

- (A) Concentration of Zn⁺² becomes 1 M when EMF reach 1.10 volt.
- (B) Concentration of Cu⁺² becomes 0.5 M when EMF reach 1.10 volt.
- (C) EMF of the cell just after charging is 1.118 volt
- (D) After charging concentration of Cu⁺² becomes 0.8 volt.

Ans. (B, C, D)

10. The following data are obtained in an experiment

Solution	Conductivity
0.01 M HA	$3.8 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$
100 ml of 0.01 M HA + 1 mmol NaOH	$80.0 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$

Calculate

Given : $\lambda^\infty_{\text{m}}(\text{Na}^+) = 50 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, $\lambda^\infty_{\text{m}}(\text{H}^+) = 350 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

- (A) $\Lambda_{\text{m}}(\text{NaA})$ is $80 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
- (B) $\Lambda_{\text{m}}(\text{A}^-)$ is $30 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
- (C) $\Lambda_{\text{m}}(\text{HA})$ is $3.8 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
- (D) degree of dissociation of 0.01 M HA is 0.01

Ans. (A, B, C, D)

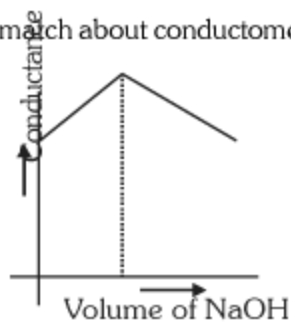
11. Which of the following arrangement will not produce oxygen at anode during electrolysis ?

- (A) Dilute H₂SO₄ solution with Cu electrodes
- (B) Conc. H₂SO₄ solution with inert electrodes
- (C) Fused NaOH with inert electrodes
- (D) Fused NaCl with inert electrodes

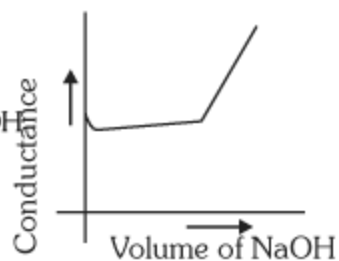
Ans. (A, B, D)

12. Which is/are correct match about conductometric titration.

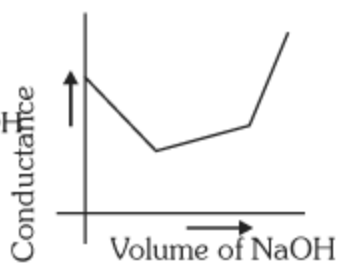
(A) HCl vs NaOH



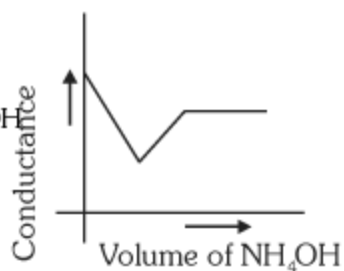
(B) CH_3COOH vs NaOH



(C) $(\text{HCl} + \text{CH}_3\text{COOH})$ vs NaOH

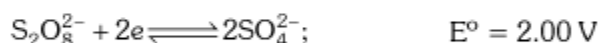
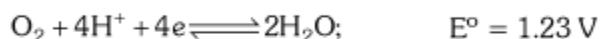


(D) $(\text{CH}_3\text{COOH} + \text{HCl})$ vs NH_4OH



Ans. (B, C, D)

13. An aqueous solution containing 1M-NiSO₄ and 1M-S₂O₈²⁻ is electrolysed using palladium electrode, at 25°C.



pH of solution is 7.

Select the correct statement(s) on the basis of above given information. Ignore over-voltage and the variation of Pd²⁺ | Pd electrode potential with concentration.

- (A) Reaction at anode is $\text{Pd} \rightarrow \text{Pd}^{2+} + 2e$
- (B) Reaction at anode is $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4e$
- (C) Reaction at cathode is $\text{Ni}^{2+} + 2e \rightarrow \text{Ni}$
- (D) Reaction at cathode is $2\text{H}^+ + 2e \rightarrow \text{H}_2$

Ans. (B, C)

14. A Daniel Cell is made at 25°C by connecting Zn / Zn⁺² (0.1 M, 1 litre) & Cu / Cu⁺² (0.9 M, 1 litre) electrode. The cell is discharged till its emf reaches 1.10 volt, then it is charged (reversing the discharge process) by passing 0.6 F charge.

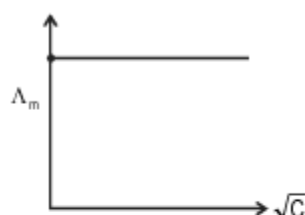
[Given : $E^\circ_{\text{Zn}/\text{Zn}^{+2}} = 0.76 \text{ V}$, $E^\circ_{\text{Cu}/\text{Cu}^{+2}} = -0.34 \text{ V}$, $\frac{2.303RT}{F} = 0.06$, $\log 2 = 0.30$]

Select the correct option(s).

- (A) Concentration of Zn⁺² becomes 1 M when EMF reach 1.10 volt.
- (B) Concentration of Cu⁺² becomes 0.5 M when EMF reach 1.10 volt.
- (C) EMF of the cell just after charging is 1.118 volt
- (D) After charging concentration of Cu⁺² becomes 0.8 volt.

Ans. (B, C, D)

15. Which of the following is/are correct ?



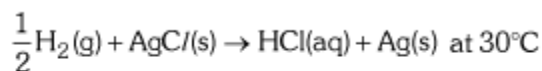
- (A) For a solution of a strong electrolyte, in the absence of any ions-solvent, ions-ions, and solvent-solvent interaction forces, the variation of the molar conductance (Λ_m) with the root of the concentration ($\sqrt{C}$) is as shown in the graph
- (B) The flow of the electrons in the internal circuit (the salt bridge) of an electrochemical cell is from the cathode to the anode
- (C) For a metal X, if $E^0_{X|XI|I^-} < E^0_{X|XCl|Cl^-}$ then : $K_{sp}(XI) < K_{sp}(XCl)$
- (D) Electrolysis of a 1M NaCl(aq) for a very long time will give a mixture of Cl_2 and O_2 gas at the anode.

Ans. (A, D)

16. If E^0_{cell} for the cell

$Pt | H_2(g) | HCl(aq) | AgCl/Ag(s)$ is determined by $E^0_{(volt)} = 0.25 - 5 \times 10^{-3} t(^{\circ}C)$

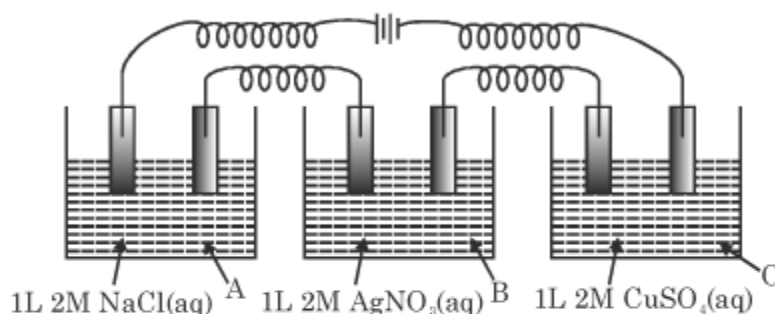
then for cell reaction



- (A) $\Delta_r G^{\circ} \approx -9.6 \text{ KJ/mol}$
- (B) $\Delta_r H^{\circ} \approx -155 \text{ KJ/mol}$
- (C) $\Delta_r S^{\circ} \approx 23 \text{ KJ/K/mole}$
- (D) $\Delta_r C_p^{\circ} = 0$

Ans. (A, B, D)

17. Three electrolytic cell with inert electrodes are connected in series combination as following



If 9.65 amp current is passed from external battery for 100 second then which is / are correct statements ?

(Assume volume of solution in electrolytic cells remain constant during electrolysis)

- (A) pH of electrolytic cell A is 12 after electrolysis.
- (B) 10.8 gm of silver deposit at cathode in electrolytic cell B after electrolysis.
- (C) Blue colour intensity in the solution of electrolyte cell C, decreases during electrolysis.
- (D) Electrolytic cell B and electrolytic cell C have equal pH after electrolysis, assuming no hydrolysis of both salts.

Ans. (A, C, D)

18. $\text{Pb} \mid \text{PbSO}_4 \mid \text{SO}_4^{2-} (0.1\text{M}) \parallel \text{Pb}^{+2} (0.01\text{M}) \mid \text{Pb}$

[Given : $K_{\text{sp}} (\text{PbSO}_4) = 10^{-16} \text{ M}^2$ $E^\circ_{\text{Pb}^{+2}/\text{Pb}} = -0.12 \text{ volt}$

Use $\frac{2.303RT}{F} = 0.06$]

Select the correct option(s).

- (A) $E^\circ_{\text{Pb}/\text{PbSO}_4/\text{SO}_4^{2-}}$ is 0.36 volt
- (B) E_{cell} is 0.39 volt
- (C) E°_{cell} is zero
- (D) $E^\circ_{\text{Pb}/\text{PbSO}_4/\text{SO}_4^{2-}}$ is 0.60 volt

Ans. (B, D)

- 19.** Saturated solution $\text{Co}_2[\text{Fe}(\text{CN})_6]$ is taken in an electrolytic cell of cell constant 0.3 cm^{-1} . If conductivity of saturated solution of $\text{Co}_2[\text{Fe}(\text{CN})_6]$ & water used to make solution is $2.2 \times 10^{-6} \text{ S cm}^{-1}$ and $4 \times 10^{-7} \text{ S cm}^{-1}$ respectively, & molar conductance of Co^{+2} & $[\text{Fe}(\text{CN})_6]^{-4}$ ions at infinite dilution are $80 \text{ S cm}^2 \text{ mol}^{-1}$ & $440 \text{ S cm}^2 \text{ mol}^{-1}$ respectively then
- (A) Solubility of $\text{Co}_2[\text{Fe}(\text{CN})_6]$ is $3 \times 10^{-6} \text{ mole/dm}^3$
 (B) K_{sp} of $\text{Co}_2[\text{Fe}(\text{CN})_6]$ is $1.08 \times 10^{-16} \text{ mole}^3/\text{dm}^9$
 (C) Ionic mobility of Co^{+2} of this solution increases on increasing potential difference across the terminal of the cell.
 (D) On increasing temperature conductivity of this solution increases.

Ans. (A, B, D)

- 20.** Choose the correct statement(s) from the following :
- (i) Van't hof factor for $0.001 \text{ N CH}_3\text{COOH}$ ($\Lambda^0 = 350 \text{ Scm}^2\text{mol}^{-1}$ & $k = 4 \times 10^{-5} \text{ Scm}^{-1}$) is 39/35.
 (ii) If Λ^0 of KCl , LiCl & KNO_3 are 150, 115 & $145 \text{ Scm}^2 \text{ mol}^{-1}$ then that of LiNO_3 will be $110 \text{ Scm}^2 \text{ mol}^{-1}$.
 (iii) In general with increase in concentration specific conductance increases and reaches a maximum and then decreases with further increase in concentration.
- (A) (i), (ii) and (iii)
 (B) (i) and (ii) only
 (C) (i) and (iii) only
 (D) Only (ii)

Ans. (C)

Paragraph-1

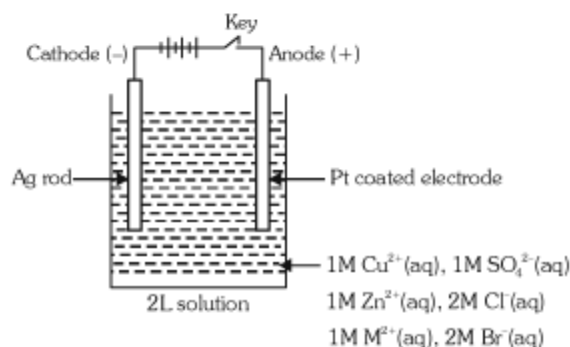
The process of using an electric current to bring about chemical change is called electrolysis.

Electrolysis is a process of oxidation and reduction at the respective electrodes due to external current passed in the electrolyte.

The product obtained during electrolysis depends on following factors.

- * The nature of the electrolyte
- * The concentration of electrolyte
- * The nature of the electrode.

Consider the electrolysis of the following cell containing aq. solution of CuSO_4 , ZnCl_2 and MBr_2 by using pure silver rod as a cathode and Pt electrode as anode.



Assume that M^{2+} does not further oxidise and can not form complex with NH_3 . Assume no hydrolysis of any ion.

$$E_{\text{Cu}^{2+}/\text{Cu}}^0 = 0.34; \quad E_{\text{M}^{2+}/\text{M}}^0 = -0.10\text{V}; \quad E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76; \quad E_{\text{H}_2\text{O}/\text{H}_2}^0 = -0.828\text{V};$$

$$E_{\text{Ag}^+/\text{Ag}}^0 = -0.80\text{V}, \quad \frac{2.303RT}{F} = 0.06$$

21. After passing 20 amp current from battery for 28950 sec. the remaining conc. of ions in solution given in passage would be : (Assume current efficiency to be 100%)

- (A) $[\text{Cu}^{2+}] = 0.5\text{ M}$, $[\text{M}^{2+}] = 0.5\text{ M}$, $[\text{Zn}^{2+}] = 0.1\text{ M}$
- (B) $[\text{Cu}^{2+}] \approx 0\text{ M}$, $[\text{M}^{2+}] = 0.5\text{ M}$, $[\text{Zn}^{2+}] = 0.5\text{ M}$
- (C) $[\text{Cu}^{2+}] = 0.5\text{ M}$, $[\text{M}^{2+}] = 0.5\text{ M}$, $[\text{Zn}^{2+}] = 0.5\text{ M}$
- (D) $[\text{Cu}^{2+}] \approx 0\text{ M}$, $[\text{M}^{2+}] = 0.5\text{ M}$, $[\text{Zn}^{2+}] = 1\text{ M}$

Ans. (D)

- 22.** What will be the volume of gases formed at anode at STP by electrolysis of above solution after passing 20 amp current for 28950 sec. (Assume current efficiency to be 100% and one mole of gas occupies 22.4 L volume at STP).
- (A) 44.8 L Br₂
 (B) 22.4 L Br₂
 (C) 22.4 L Cl₂
 (D) 44.8 L Cl₂

Ans. (A, C)

- 23.** If 36 mol of NH₃(g) is passed in electrolytic solution given in comprehension (assume no volume change by addition of NH₃), then what would be decreasing order of reduction potential of following :

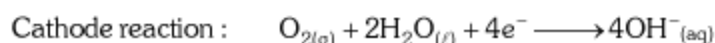
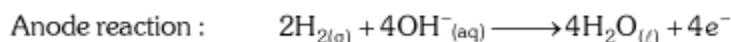
$$\{K_f [\text{Cu}(\text{NH}_3)_4]^{2+} = 1 \times 10^{12} \text{ and } K_f [\text{Zn}(\text{NH}_3)_4]^{2+} = 1 \times 10^9\}$$

- (A) $\text{M}^{2+} > \text{Cu}^{2+} > \text{Zn}^{2+}$
 (B) $\text{Cu}^{2+} > \text{M}^{2+} > \text{Zn}^{2+}$
 (C) $\text{M}^{2+} > \text{Zn}^{2+} > \text{Cu}^{2+}$
 (D) $\text{Cu}^{2+} > \text{M}^{2+} > \text{Zn}^{2+}$

Ans. (A)

Paragraph-2

Combustion of H₂(g) is a highly exothermic reaction. The energy of a hydrogen gas fuel can be converted to electrical energy either through heat engines or through fuel cells. Fuel cells are important since indefinite amount of electrical energy can be obtained from a fuel cell as long as the outside supply of fuel is maintained. In a H₂/O₂ fuel cell, following reaction occurs.



- 24.** What will be the value of $\Delta G^{\circ}_{\text{formation}}$ of H₂O_(l) at 298 K?
- (A) -231.6 kJ/mole
 (B) -463.2 kJ/mole
 (C) -231.6 J/mole
 (D) -463.2 J/mole

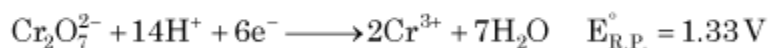
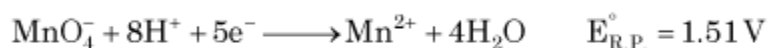
Ans. (A)

25. If the temperature coefficient of the E.M.F. is -10^3 V/Kelvin then $\Delta S_{\text{formation}}^\circ$ of $\text{H}_2\text{O}_{(l)}$ will be :
- (A) -193 J/K-mole
 (B) 193 J/K-mole
 (C) 386 J/K-mole
 (D) -386 J/K-mole

Ans. (A)

Paragraph-3

In quantitative estimation of any element using an oxidizing agent it is essential to predict which substance(s) gets oxidized by the oxidizing agent. In case more than one substance are getting oxidized then the oxidizing agents gets distributed in all the reaction taking place. From this information and the data given below answer the questions that follow.



26. Which of the following cannot be used for quantitative estimation of FeC_2O_4 ?
- (A) $\text{K}_2\text{Cr}_2\text{O}_7$ in presence of dil H_2SO_4
 (B) $\text{K}_2\text{Cr}_2\text{O}_7$ in presence of dil. HCl
 (C) KMnO_4 in presence of dil. HCl
 (D) KMnO_4 in presence of dil. H_2SO_4

Ans. (C)

27. Millimoles of FeC_2O_4 in the solution if 50 ml of 0.1 M KMnO_4 is used for its oxidation in presence of dil. HCl if 3.5 millimoles of Cl_2 is obtained along with other products.
- (A) $25/3$ millimoles
 (B) 6 millimoles
 (C) 12.5 millimoles
 (D) quantitative estimation can't be made from given data.

Ans. (B)

Paragraph-4

Aluminium is produced from bauxite by electro-reduction of alumina in Hall-Herault's process. Alumina, dissolved in a conducting medium like cryolite (Na_3AlF_6) and NaF, is added to decrease the temperature of melting of cryolite. In the electrolytic bath of molten mixture of Al_2O_3 , Na_3AlF_6 and NaF, aluminium gets deposited at cathode and liberated $\text{O}_2(\text{g})$ at anode attacks the graphite anode and forms $\text{CO}_2(\text{g})$.

(1 F = 96500 coulomb)

28. If the electrolysis is carried out with a potential difference of 9 V by passing 180kAmp of current with current efficiency (η) of 96.5% then electrical energy required to produce 10 kg of aluminium will be

(A) $3 \times 10^9 \text{ J}$
(B) 10^9 J
(C) $4.5 \times 10^6 \text{ J}$
(D) $3 \times 10^6 \text{ J}$

Ans. (B)

29. Find the rate of formation of CO_2 in g/sec using data of above question.

(A) 79.2
(B) 39.6
(C) 19.8
(D) 59.4

Ans. (C)

Paragraph-5

The ability of a material to conduct electricity varies a great deal and depends on the nature of the material. Conductivity defined as reciprocal of resistivity is the parameter used to compare conducting ability of a material. Conductivity varies a great deal in magnitude and depends on the nature of a material. Metals & alloys have high conductivity and are called conductors. Aqueous solutions of strong electrolytes are also good conductors. Glass, ceramics have very low conductivity and are called insulators. Substances like silicon, doped silicon and gallium arsenide having conductivity between conductors and insulators are called semi-conductors. Now answer the following questions -

30. Arrange the following in decreasing order of conductivity

(I) 0.1M CH_3COOH (aq.) (II) Ag(s)
(III) Teflon (s) (IV) Germanium (s)
(A) I > II > IV > III
(B) II > IV > III > I
(C) II > I > IV > III
(D) II > IV > I > III

Ans. (C)

- 31.** Identify the incorrect statement :
- (A) Conductivity of metals decreases with increasing temperature
 - (B) Conductivity of electrolytic conductors increases with increasing temperature
 - (C) Conductivity of semi- conductors increases with increasing temperature
 - (D) Conductivity of electrolytic conductors decreases with increase in temperature.

Ans. (D)

Paragraph-6

The conductivity of saturated solution of sparingly soluble salt, $\text{Ba}_3(\text{PO}_4)_2$ is $1.2 \times 10^{-5} \text{ ohm}^{-1} \text{ cm}^{-1}$. The limiting equivalent conductance of BaCl_2 , $\text{K}_3(\text{PO}_4)$ and KCl are 160, 140 and $100 \text{ ohm}^{-1} \text{ cm}^2/\text{eq.}$ respectively

- 32.** Calculate solubility product (K_{sp}) of $\text{Ba}_3(\text{PO}_4)_2$
- (A) 52×10^{-15}
 - (B) 52×10^{-20}
 - (C) 108×10^{-20}
 - (D) 108×10^{-25}

Ans. (D)

Paragraph-6

The conductivity of saturated solution of sparingly soluble salt, $\text{Ba}_3(\text{PO}_4)_2$ is $1.2 \times 10^{-5} \text{ ohm}^{-1} \text{ cm}^{-1}$. The limiting equivalent conductance of BaCl_2 , $\text{K}_3(\text{PO}_4)$ and KCl are 160, 140 and $100 \text{ ohm}^{-1} \text{ cm}^2/\text{eq.}$ respectively

- 33.** Calculate solubility of $\text{Ba}_3(\text{PO}_4)_2$ in $0.1 \text{ M Na}_3(\text{PO}_4)$ solution,
- (A) $1.2 \times 10^{-6} \text{ M}$
 - (B) $3.42 \times 10^{-8} \text{ M}$
 - (C) $2 \times 10^{-10} \text{ M}$
 - (D) $1.52 \times 10^{-12} \text{ M}$

Ans. (B)

- 34.** Match the Column.

Column-I
(Quantities)

- (A) Molar conductance
- (B) emf of a cell in operation
- (C) Electrode potential
- (D) Standard reduction potential

Column-II
(Factors on which dependency exist)

- (P) Temperature
- (Q) Concentration of species involved
- (R) Nature of substance involved
- (S) No. of electrons lost or gained in the reaction

Ans. [(A) → (P, Q, R); (B) → (P, Q, R); (C) → (P, Q, R); (D) → (R)]

35.

List-I

- (A) Conductivity (k) increases
- (B) Molar conductance (λ_m) increases
- (C) Conductance (G)
- (D) Molar conductance (λ_m^∞) at infinite dilution.

List-II

- (P) Depends on cell constant as well as concentration
- (Q) Independent of concentration as well as cell constant
- (R) With decreases in concentration but independent of cell constant
- (S) With increase in concentration but independent of cell constant

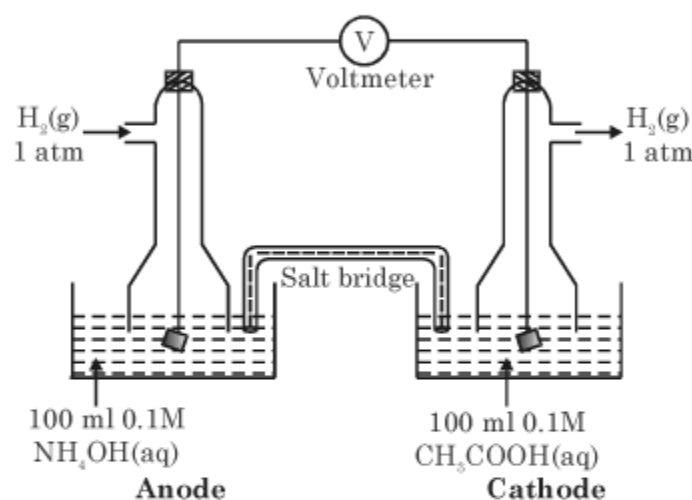
Ans. [(A) $\rightarrow$ (S); (B) $\rightarrow$ (R); (C) $\rightarrow$ (P); (D) $\rightarrow$ (Q)]

36. Given :

$$K_b(\text{CH}_3\text{COO}^-) = 10^{-9} \quad K_h(\text{NH}_4^+) = 10^{-9}$$

$$K_w = 10^{-14} \quad \log 2 = 0.30$$

$$\frac{2.303RT}{F} = 0.06$$



Column-I

Column-II

- | | |
|--|--|
| (A) Addition of 50 ml 0.1 M HCl in anode compartment | (P) pOH of anode compartment is 5 |
| (B) Addition of 100 ml 0.1 M HCl in anode compartment | (Q) pH of cathode of compartment is 8.85 |
| (C) Addition of 50 ml 0.1 M NaOH in cathode compartment | (R) Emf of cell = 0.36 V |
| (D) Addition of 100 ml 0.1 M NaOH in cathode compartment | (S) Emf of cell = 0.129 V |

$$(T) \Delta G_{\text{cell}}^0 = 0$$

Ans. [(A) → (P, R); (B) → (S); (C) → (R); (D) → (Q, S)]

37. A saturated solution in AgA ($K_{\text{sp}} = 3 \times 10^{-14}$) and AgB ($K_{\text{sp}} = 1 \times 10^{-14}$) has conductivity of $375 \times 10^{-10} \text{ Scm}^{-1}$ and limiting molar conductivity of Ag^+ and A^- are $60 \text{ Scm}^2 \text{ mol}^{-1}$ and $80 \text{ Scm}^2 \text{ mol}^{-1}$ respectively then what will be the limiting molar conductivity of B^- (in $\text{Scm}^2 \text{ mol}^{-1}$)

Ans. (270)

38. Calculate total moles of the gases liberated during Kolbe electrolysis of aq. CH_3COONa solution for 400sec. using 965 amp. Current.

Ans. (8)

39. For a saturated solution of AgCl at 25°C , specific conductance is $3.4 \times 10^{-6} \text{ ohm}^{-1} \text{ cm}^{-1}$ and that of H_2O used for preparing the solution was $1.6 \times 10^{-6} \text{ ohm}^{-1} \text{ cm}^{-1}$. What is the solubility of AgCl ?

[Given $\Lambda_{\text{eq}}^\circ(\text{AgCl}) = 150 \text{ ohm}^{-1} \text{ cm}^{-1} \text{ eq}^{-1}$]

If answer of solution is $x \times 10^{-y} \text{ M}$, then find y.

Ans. (5)

Chemical Kinetics

1. For the following reaction : $\text{A}_{(\text{g})} + \text{B}_{(\text{g})} \longrightarrow 2\text{C}_{(\text{g})} + \text{D}_{(\text{g})}$, the rate law is given as rate of disappearance of 'A' = $K[\text{A}]^{2/3} [\text{B}]^{1/3}$. If initial concentration of A and B are 2M each and no C and D were present initially then the time at which total concentration (sum of concentrations of A, B, C and D) will become 5.5 M, is :

[Given : $K = 1.386 \times 10^{-1} \text{ min}^{-1}$]

- (A) 5 min
(B) 10 sec
(C) 600 sec
(D) $\frac{1}{2}$ min

Ans. (C)

2. A substance 'A' undergoes decomposition in an elementary reaction as shown below.

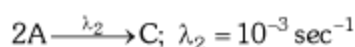
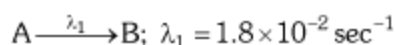


Starting only with A, after 13.86 minutes, concentration of both A & B was found to be 0.5M in a 2 L rigid vessel. What was the rate of heat release (in cal/min) at the start of the reaction?

- (A) 1500
(B) 3000
(C) 1000
(D) 15000

Ans. (B)

3. The average (mean) life of a radio nuclide which decays by parallel path is



- (A) 52.63 sec
(B) 500 sec
(C) 50 sec
(D) None

Ans. (C)

4. In the Lindemann theory of unimolecular reactions, it is shown that the apparent rate

constant for such a reaction is $k_{\text{app}} = \frac{k_1 C}{1 + k_2 C}$, where C is the concentration of the

reactant, k_1 and k_2 are constants ($k_2 = 9 \times 10^5$). The value of C for which k_{app} has 90% of its limiting value at C tending to infinitely large is :

- (A) 10^{-6} M
(B) 10^{-4} M
(C) 10^{-5} M
(D) 5×10^{-5} M

Ans. (C)

5. The dependence of rate constant of a reaction on temperature may be given as

$$\frac{d(\ln K)}{dT} = \frac{E_a}{RT^2}$$

where E_a is activation energy for the reaction. For a reaction, the dependence of rate constant on temperature is given as

$$\ln K = 10 - \frac{2500}{T} + 3 \ln T$$

The activation energy of reaction is

- (A) $2500 + 3T$
(B) $2500 R$
(C) $(2500 + 3T)R$
(D) $(2500 - 3 \ln T)R$

Ans. (D)

6. The order of reaction $A \rightarrow \text{product}$ can be given by the expression
[where r = rate of reaction ; $[A]_1$ = concentration at time t_1 ; $[A]_2$ = concentration at time t_2]

(A) $\frac{\ell n r_2 - \ell n r_1}{\ell n [A]_2 - \ell n [A]_1}$

(B) $\frac{\ell n [A_0]_2 - \ell n [A_0]_1}{\ell n [t_{1/2}]_2 - \ell n [t_{1/2}]_1}$

(C) $\ell n \left(\frac{-d[A]}{k \cdot dt} \right) / \ell n [A]$

(D) $\frac{\ell n (r/k)}{\ell n [A]}$

Ans. (A, C, D)

7. For a reaction the Rate Law is

$$\text{Rate} = k[A]^a[C]^b$$

Where 'C' act as catalyst. At an initial concentration $[A] = 0.1 \text{ M}$, $[C] = 0.01 \text{ M}$, $t_{1/3}$ and $t_{5/9}$ was found to be 30 min and 75 min respectively. But $t_{1/3}$ was found to be 7.5 min if initial concentration are $[A] = 0.2 \text{ M}$ and $[C] = 0.02 \text{ M}$.

Select the correct option(s).

(A) $a = 2$

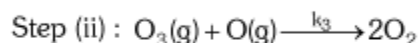
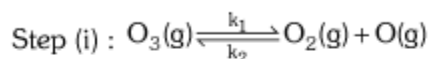
(B) $b = 2$

(C) When initial concentration are $[A] = 0.2 \text{ M}$, $[C] = 0.02 \text{ M}$, $t_{5/9}$ is 18.75 min.

(D) When initial concentration are $[A] = 0.2 \text{ M}$, $[C] = 0.02 \text{ M}$, $t_{5/9}$ is 22.5 min.

Ans. (A, C)

8. The ozone decomposes according to the equation $2\text{O}_3(\text{g}) \rightarrow 3\text{O}_2(\text{g})$. The mechanism of the reaction is as follows :



If $k_3 \ll k_1$ and $k_3 \ll k_2$ then select the correct option(s) :

(A) Step (i) is the RDS

(B) Rate law for the reaction is $-\frac{1}{2} \frac{d[\text{O}_3]}{dt} = \frac{k_1 k_3}{k_2} [\text{O}_3]^2 [\text{O}_2]^{-1}$

(C) For step (ii), Molecularity = 2

(D) O(g) is an intermediate

Ans. (C, D)

9. Identify the incorrect options regarding increase in rate of reaction.

(I) $E_a = 40 \text{ kJ/mole}$ Temperature changes from 300 to 310 K

(II) $E_a = 80 \text{ kJ/mole}$ Temperature changes from 200 to 210 K

(III) $E_a = 60 \text{ kJ/mole}$ Temperature changes from 400 to 410 K

(A) Since temperature changes by same amount therefore all four experiences same change.

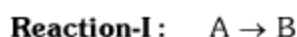
(B) I experiences highest increase in rate of reaction.

(C) III experiences lowest increase in rate of reaction.

(D) II experiences lowest increase in rate of reaction.

Ans. (A, B, D)

10. For the two reactions



following curves are plotted

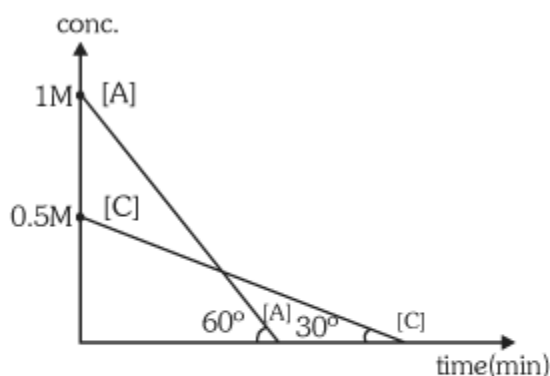
Which of the following is/are true :

(A) $(t_{100\%})_{\text{reaction I}} = \frac{2}{3}(t_{100\%})_{\text{reaction II}}$

(B) $[A] = [C]$ at $t = \frac{\sqrt{3}}{4}$ minutes

(C) When $[B] = [A]$ at that time $[C] = [D]$

(D) If both $[A]_0$ & $[C]_0$ are 1 M then $(t_{50\%})_I = \frac{1}{3} (t_{50\%})_{II}$



Ans. (A, B, D)

11. Which of the following statement is/are correct with respect to radioactive decay ?

(A) Fraction of radioactive atoms decayed in time 't' $= 1 - e^{-\lambda t}$.

(B) ${}^{24}_{11}\text{Na}$ emits $-\beta$ radiation.

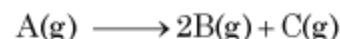
(C) The number of β -particles emitted during the change ${}_a^c\text{X} \rightarrow {}_d^b\text{Y}$ is equal to

$$d + \left[\frac{a-b}{2} \right] - c$$

(D) The total number of α -particles emitted during thorium series is equal to six.

Ans. (A, B, D)

12. In the following gas phase first order reaction :



initial pressure was found to be 400 mm of Hg and it changed to 1000 mm Hg after 20 min. Then :

(A) Half life of A is 10 min.

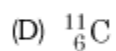
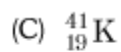
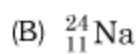
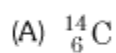
(B) Rate constant is 0.0693 min^{-1}

(C) Partial pressure of C at 30 min is 350 mm Hg

(D) Total pressure after 30 min is 1150 mm Hg

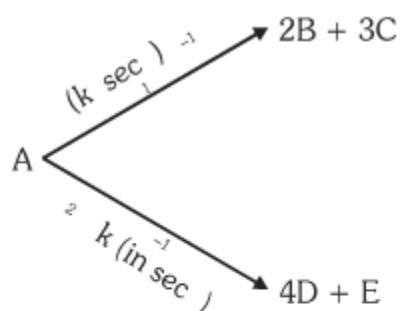
Ans. (A, B, C)

13. Which of the following is the β^- emitter



Ans. (A, B, C)

14. For given parallel reaction



Which of the following is/are correct for % yield of mole in product mixture ?

(A) $\% \text{ B} = \frac{40k_1}{k_1 + k_2}$

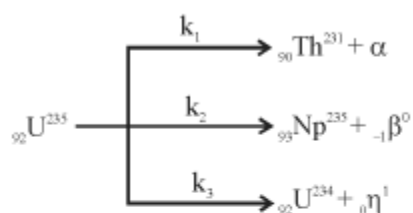
(B) $\% \text{ D} = \frac{80k_2}{k_1 + k_2}$

(C) $\% \text{ C} = \frac{60k_2}{k_1 + k_2}$

(D) $\% \text{ E} = \frac{10k_2}{k_1 + k_2}$

Ans. (A, B, C)

15. For the radioactive parallel decay



After 30 min. from the start of decay the mole ratio of products obtained are :

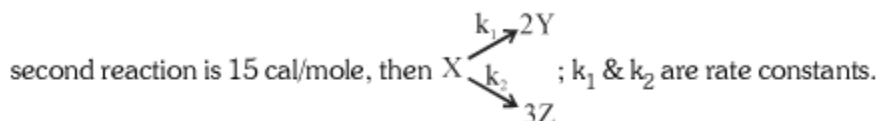
$$\text{Th}^{231} : \text{Np}^{235} : \text{U}^{234} :: 4 : 2 : 1.$$

If $k_2 = 2 \times 10^{-2} \text{ sec}^{-1}$ then the overall decay constant of ${}_{92}\text{U}^{235}$ is -

- (A) $3 \times 10^{-2} \text{ sec}^{-1}$
 (B) $8 \times 10^{-2} \text{ sec}^{-1}$
 (C) $2 \times 10^{-2} \text{ sec}^{-1}$
 (D) $7 \times 10^{-2} \text{ sec}^{-1}$

Ans. (D)

16. Y and Z are formed in 5 : 3 molar ratio if substance X undergoes 1st order parallel reaction as shown. If activation energy of first reaction is 20 cal/mole and that of



(A) Net activation energy of the reaction is 18.6 cal/mole (approx)

(B) $\frac{k_1}{k_2}$ is $\frac{5}{2}$

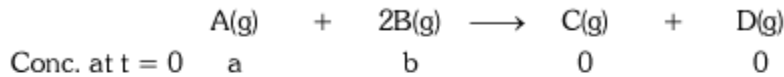
(C) Ratio of rate of formation of Y & Z is $\frac{2}{3}$

(D) $k_2 > k_1$

Ans. (A, B)

Paragraph-1

Consider the following reaction occurring at constant volume.



Given : Rate of reaction = kC_B

where C_B = concentration of B at any time 't' and k is specific rate of reaction.

- 17.** Concentration of 'A' at time 't' is given as :

(A) $C_A = ae^{-kt}$

(B) $C_A = a - \frac{b}{2}(1 - e^{-2kt})$

(C) $C_A = a - kt$

(D) $C_A = a - b(1 - e^{-kt})$

Ans. (B)

- 18.** Time at which concentration of A reduces to half is given as :

(A) $(t_{1/2})_A = \frac{a}{2k}$

(B) $(t_{1/2})_A = \frac{a}{k}$

(C) $(t_{1/2})_A = \frac{1}{2k} \ln \frac{b}{(b-a)}$

(D) $(t_{1/2})_A = \frac{\ln 2}{k}$

Ans. (C)

- 19.** For what condition, $(t_{1/2})_A = (t_{1/2})_B$:

(A) $\frac{a}{b} = \frac{1}{2}$

(B) $\frac{a}{2k} = \frac{\ln 2}{k}$

(C) $\frac{a}{b} = 1$

(D) None of these

Ans. (A)

Paragraph-2

The reaction $A(g) \longrightarrow B(g) + 2C(g)$ is a first order reaction with rate constant $2.772 \times 10^{-3} \text{ sec}^{-1}$. Reaction is started with only 0.1 mol of A in a container (piston type) with volume 2 litre and is allowed to take place at constant pressure and at constant temperature 300 K. $[R = 0.082 \text{ litre atm mol}^{-1}\text{K}^{-1}]$

20. Concentration of A after 250 sec will be

(A) 0.125 M
(B) 0.0125 M
(C) 0.05 M
(D) 0.025 M

Ans. (B)

21. Concentration of C after 250 sec will be

(A) 0.05 M
(B) 0.025 M
(C) 0.1 M
(D) 0.25 M

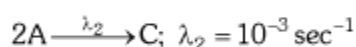
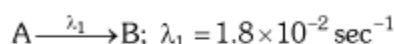
Ans. (B)

22. Partial pressure of C after 250 sec will be

(A) 1.23 atm
(B) 0.3075 atm
(C) 2.46 atm
(D) 0.615 atm

Ans. (D)

23. The average (mean) life of a radio nuclide which decays by parallel path is



[Divide your answer by 10 and fill OMR sheet]

Ans. (5)

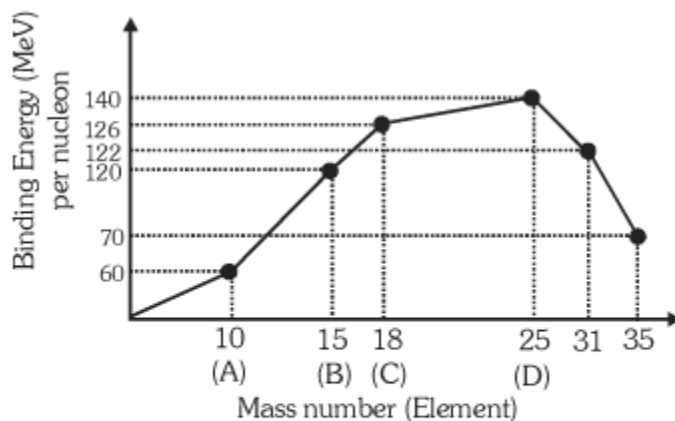
24. For a two step reaction



(Where, R is a reactive intermediate concentration is maintained at some low steady state throughout the reaction). If the concentration of C is very high then the order of reaction for formation of "P" is

Ans. (1)

25.



(All the graphs are straight line)

Energy released when most stable gets converted to fourth most stable ?

[Give your answer as sum of digits and fill OMR sheet]

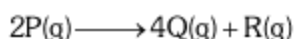
Ans. (8)

26. Two substance A and B react as per the reaction : $A + B \rightarrow C$. If a term $t_{1/2}$ is defined as the time in which half of the reactant present in limited amount gets consumed, then calculate order of the reaction from the following data.

	[A](M)	[B](M)	$t_{1/2}(\text{min})$
1.	10^{-3}M	10^2M	10min
2.	10^{-3}M	10^{-1}M	15min
3.	$\frac{10^{-3}}{2}\text{M}$	10^2M	40min
4.	10^{-3}M	0.2M	7.5min

Ans. (5)

27. For a gaseous reaction obeying first order kinetics



Total pressure at various instants was calculated as shown

Time	t = 8 min	t = 28 min	t = ∞
Total pressure	1375 mm of Hg	1562.5	1750 mm of Hg

Calculate the time for $\frac{7}{8}$ th of the reactant to get consumed.

[Express your answer in minutes and Give your answer as sum of digits and fill OMR sheet]

Ans. (6)

28. Consider the following reaction $2(A)(g) \longrightarrow 3B(g) + C(g)$, inclosed rigid container.

Starting with pure A having pressure 2 atm initially, the total pressure is exactly doubled in 2 hrs. The possible order of reaction is

Ans. (0)

29. For the following reaction : $A_{(g)} + B_{(g)} \longrightarrow 2C_{(g)} + D_{(g)}$, the rate law is given as rate of disappearance of 'A' = $K[A]^{2/3} [B]^{1/3}$. If initial concentration of A and B are 2M each and no C and D were present initially then the time at which total concentration (sum of concentrations of A, B, C and D) will become 5.5 M, is :

[Given : $K = 1.386 \times 10^{-1} \text{ min}^{-1}$]

[Give your answer as sum of digits and fill OMR sheet]

Ans. (6)

30. A first order reaction occur 20% in 15 min. In what time (hrs), it will complete 80%, at the same temperature ? [Given : $\ln 10 = 2.3$, $\ln 2 = 0.7$].

Ans. (2)

31. For the following reaction : $A_{(g)} + B_{(g)} \longrightarrow 2C_{(g)} + D_{(g)}$, the rate law is given as rate of disappearance of 'A' = $K[A]^{2/3} [B]^{1/3}$. If initial concentration of A and B are 2M each and no C and D were present initially then the time at which total concentration (sum of concentrations of A, B, C and D) will become 5.5 M, is :

[Given : $K = 1.386 \times 10^{-1} \text{ min}^{-1}$]

[Give your answer as sum of digits and Fill OMR sheet]

Ans. (6)

Surface Chemistry

1. Which of the following options correctly represent set of True (T) or False (F) statements.
- (I) Colloidal solution of gelatin gets stabilised by solvation of colloidal particles by the dispersion medium.
 - (II) An aqueous solution of KI will show Tyndall effect, however when added with excess AgNO_3 solution it will not show Tyndall effect.
 - (III) At isoelectric point all the colloids will have maximum stability.
 - (IV) In an autocatalysis reaction, rate of reaction will decrease with time.
 - (V) Physisorption is an enthalpy driven process.
- (A) T T F T T
 (B) F T T T F
 (C) T F T F T
 (D) F T F F F

Ans. (C)

2. At 70 K, the adsorption of N_2 gas at iron surface obeys Freundlich adsorption isotherm. The experimental data collected is

P(bar)	4	25	64
$\frac{x}{m}$	0.2	0.5	0.8

where $\frac{x}{m}$ is the mass (**in gm**) of N_2 gas adsorbed per gm of iron at P bar pressure. The moles of N_2 gas adsorbed per gm of iron at 36 bar and 70 K, is

- (A) $\frac{3}{70}$
 (B) $\frac{3}{140}$
 (C) $\frac{3}{280}$
 (D) 0.3

Ans. (B)

3. Which of the following options is correct for Langmuir isotherm.

- (A) At very low pressure $\frac{x}{m}$ will become constant.
- (B) At moderate pressures $\frac{x}{m}$ will be linearly dependent on pressure.
- (C) At high pressures $\frac{x}{m}$ should be constant independent of pressure.
- (D) At high pressures $\frac{x}{m}$ should be varying linearly with pressure.

Ans. (C)

4. Choose the **INCORRECT** statement(s) in the following

- (A) Chemical adsorption can not take place between any two phases
- (B) Chemical adsorption may be multilayer in nature.
- (C) Physical adsorption increases with increase in pressure
- (D) Chemisorption increase with increase in temperature

Ans. (B)

5. Which of the following statements is/are correct with respect to surface phenomenon ?

- (A) Potassium ferrocyanide cause less coagulation in a basic dye as compared to Na_2HPO_3 .
- (B) A starch aqua-sol can act as a protective colloid for $\text{Fe}(\text{OH})_3$ sol.
- (C) The slope of the Freundlich Isotherm ($\log \frac{x}{m}$ vs $\log P$) keeps on changing for a long range of pressure and is constant over a limited range of pressure.
- (D) On mixing AgNO_3 with large amount of KI and subjecting the colloidal state to electrophoresis, coagulation is obtained at anode.

Ans. (B, C, D)

6. Identify the statements which are not correct.

- (A) Zeolites are selective type of catalyst which can catalyse only specific reaction.
- (B) Micelle formation shows increases in entropy.
- (C) An emulsifier destabilises a liq-liq. colloidal system.
- (D) Coagulation value of PO_4^{3-} ions is less as compared to SO_4^{2-} ions for coagulation of positive sols.

Ans. (B, C)

7. Which of the following statements regarding sols is/are **correct**.

- (A) Brownian movement stabilises sols.
- (B) In a reaction catalyst may change physically.
- (C) An emulsion can be broken both by heating as well as freezing.
- (D) On prolong dialysis colloids become stable.

Ans. (A, B, C)

8. Which of the following statements is/are correct with respect to surface phenomenon.

- (A) As per Freundlich theory; slope of the graph between $\log x/m$ vs $\log P$ remains same for a large range of pressure.
- (B) Potassium ferrocyanide can cause greater coagulation in a basic dye as compared to what Na_2HPO_4 can cause.
- (C) Arsenious sulphide is stabilised by brownian motion & hydration both.
- (D) Catalyst can change the final major product obtained in certain reactions.

Ans. (B, D)

9. Which of the option(s) regarding true/false nature of the statements is/are correct ?

S-1 : Out of all colligative properties, osmotic pressure is easiest to determine experimentally.

S-2 : Chemisorption is a multilayer process.

S-3 : Solubility of gases in water increases with increase in temperature.

S-4 : Addition of NaCl can cause coagulation in both +vely charged and -vely charged sols.

S-5 : Enzyme catalysed reactions are elementary reactions.

S-6 : NaCl crystal shows Frenkel defects.

- (A) There are more false statements than true statements.
- (B) S-3 and S-5 are false statements.
- (C) S-1 and S-4 are the only true statements.
- (D) Only S-1 is a true statement.

Ans. (A, B, C)

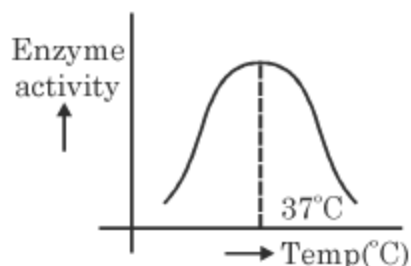
10. Identify the statements which are not correct.

- (A) Zeolites are non selective type of catalyst which can catalyse any reaction.
- (B) Micelle formation shows increases in entropy.
- (C) An emulsifier stabilises a liq-liq. colloidal system.
- (D) Coagulation value of PO_4^{3-} ions is less as compared to SO_4^{2-} ions for coagulation of positive sols.

Ans. (A, B)

11. Which is / are correct options ?

(A) For Enzyme catalyst



(B) Surfactants decreases the surface tension of liquid on dissolving in it.

(C) Conductance of oil in water emulsion increase on addition of common salt into it.

(D) At CMC, molecules associate to form particles of colloidal dimension.

Ans. (A, B, C, D)

12. Which of the following is/are correct about the adsorption theory of catalysis ?

(A) Catalysis occur through physisorption

(B) The increase in concentration of reactants on the surface increases the rate of reaction

(C) The heat of adsorption is utilised in enhancing the rate of reaction

(D) The adsorbate molecules become immobile at the catalyst surface

Ans. (B, C)

13. Which of the following are correct for physical adsorption ?

(A) Adsorption increases with increases in temperature.

(B) Adsorption is spontaneous.

(C) Both change in enthalpy and entropy of adsorption are negative.

(D) Adsorption on solid is reversible.

Ans. (B, C, D)

Paragraph-1

1 gm of activated charcoal has a surface area of 10^3 m^2 . Complete coverage of the surface of charcoal by NH_3 molecules is assumed by monolayer formation. Diameter of each NH_3 molecule is $4\text{\AA}$. Ammonia adsorbed on the charcoal by two dimensional closed packed arrangements with maximum efficiency. In another experiment $10\sqrt{3}$ gm of the given charcoal is treated with 100 mL of (5 M) ammonia solution. After adsorption the excess ammonia solution is removed.

Given : $N_A = 6 \times 10^{23}$

14. What is the effective area of charcoal occupied by per ammonia molecule :

- (A) $\frac{1}{2}\sqrt{3} \times 10^{-20} \text{ m}^2$
- (B) $4\sqrt{3} \times 10^{-20} \text{ m}^2$
- (C) $2\sqrt{3} \times 10^{-20} \text{ m}^2$
- (D) $8\sqrt{3} \times 10^{-20} \text{ m}^2$

Ans. (D)

15. In 1st experiment (1 gram sample) number of NH_3 molecules adsorbs is :

- (A) 0.07×10^{23}
- (B) 0.14×10^{23}
- (C) 0.21×10^{23}
- (D) 0.28×10^{23}

Ans. (A)

16. As per the second experiment what is the molarity of excess ammonia solution removed (nearly) ?

- (A) 4 M
- (B) 3 M
- (C) 2 M
- (D) 1 M

Ans. (B)

17. 526.3 ml of 0.5 M HCl is shaken with 0.5 g of activated charcoal and filtered. The concentration of the filtrate is reduced to 0.4 M. What is the amount of adsorption (x/m) ?

(Round off your answer to nearest integer)

Ans. (4)

18. 526.3 ml of 0.5 M HCl is shaken with 0.5 g of activated charcoal and filtered. The concentration of the filtrate is reduced to 0.4 M. What is the amount of adsorption (x/m) ?

Ans. (4)

19. Molecular formula of starch can be represented as $(C_6H_{10}O_5)_n$. If the gold number of one such starch molecule is 6.48 and 0.01 millimoles of the above starch are required to be added to 10 ml of gold sol to prevent coagulation by 1 ml of 10% sodium chloride solution, then calculate the value of 'n'.

Ans. (4)

20. How many of the following statements are correct :

- (i) Butter is an example of a Lyophobic colloid
- (ii) Soaps that form micelles at concentrations above CMC, form true solutions at the lower concentrations
- (iii) A colloid obtained by using 'Bredig's arc method' can not have negatively charged dispersed phase in general
- (iv) $FeCl_3$ is a better coagulant for the human blood (a negative colloid) as compared to Na_3PO_4
- (v) Delta is formed at the mouth of the river, where it meets the ocean, due to peptisation.
- (vi) Enzymes are globular proteins which work over a wide range of pH and temperatures.

Ans. (3)

Transition Element

1. When dilute H_2SO_4 and H_2O_2 are added to a solution of chromate ions, an intense blue color is produced which is stable in ether. This is due to the formation of

- (A) $Cr_2O_7^{2-}$
- (B) Cr_2O_3
- (C) CrO_5
- (D) CrO_3

Ans. (C)

2. $Mn^{2+} + \text{oxidising agent} \rightarrow MnO_4^{-}$; The oxidising agent could be :

- (A) $S_2O_8^{2-}$
- (B) $NaBiO_3$
- (C) PbO_2
- (D) All

Ans. (D)

3. When silver nitrate is reacted with excess of I_2 in aqueous solution then which of the following compounds are formed ?

(A) $AgIO_3$
(B) AgI
(C) HIO_3
(D) HNO_3

Ans. (B, C, D)

4. Which of the following is correct trend in the melting point order ?

(A) $V > Mn > Ni$
(B) $V > Ni > Mn$
(C) $V > Cr > Ti > Sc > Fe$
(D) $Co > Ni > Cu > Zn$

Ans. (B, C, D)

5. Which of the following statement(s) is/are correct with respect to bromide ions ?

(A) KBr on heating with MnO_2 and concentrated H_2SO_4 liberates Br_2 and SO_2 gases.
(B) KBr on heating with concentrated H_2SO_4 liberates Br_2 and SO_2 gases.
(C) KBr forms HBr with concentrated H_3PO_4 .
(D) $KBr(s)$ liberates Br_2 on gentle warming with concentrated H_2SO_4 and $K_2Cr_2O_7(s)$.

Ans. (B, C, D)

6. When MnO_2 is fused with KOH , a purple green coloured compound is formed. Choose correct statements about purple green coloured compound

(A) It disproportionates in acidic medium
(B) It is paramagnetic in nature
(C) Geometry is tetrahedral
(D) It uses non axial d-orbital in hybridisation

Ans. (A, B, C, D)

7. Which of the following statements is/are correct.

(A) $AgCl$ is not soluble in H_2O but soluble in KCN .
(B) Zn when reacts with cold very-very dil HNO_3 gives NH_4NO_3 .
(C) $CuSO_4$ reacts with KI to form only $Cu_2I_2 + K_2SO_4$.
(D) $CuSO_4 \cdot 5H_2O \xrightarrow{1000^\circ C}$ The gases liberated are SO_2 and O_2 .

Ans. (A, B, D)

8. Identify the correct statements
 (A) Only CuF_2 is ionic amongst cupric halides
 (B) Fe does not form amalgam
 (C) Cr^{2+} is reducing but Mn^{3+} is oxidising
 (D) KMnO_4 on reaction with conc. NaOH followed by dilution gives brown residue

Ans. (A, B, C, D)

9. $\text{FeSO}_4(\text{s}) \xrightarrow{\Delta} \text{Solid X} + \text{gas A} + \text{gas B}$
 $\text{X} + \text{NaOH} + \text{Cl}_2 \rightarrow \text{Z} + \text{NaCl} + \text{H}_2\text{O} + \text{H}_2$
 $\text{Z} \xrightarrow{\text{readily decomposes}} \text{Solid X} + \text{O}_2(\text{g})$

gas A + $\text{K}_2\text{Cr}_2\text{O}_7 \longrightarrow$ green solution

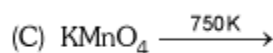
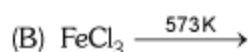
Choose the correct statements

- (A) The trimeric form of gas B has no S-S bond
 (B) Z has Fe in +6 oxidation state
 (C) gas A gives white turbidity with H_2S gas
 (D) None of these

Ans. (A, B, C)

10. Match the following columns :

Column-I



Column-II

(P) One of the products is bright orange coloured but diamagnetic

(Q) One of the products is green coloured and paramagnetic

(R) One of the products is violet and paramagnetic

(S) One of the products exists as dimer.

Ans. [(A) $\rightarrow$ (Q, R); (B) $\rightarrow$ (S); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (P)]

11. Match the column

Column-I

(Elements)

- (A) Rhombic sulphur
 (B) Tin
 (C) White phosphorus
 (D) Copper

Column-II

(Correct Characteristics)

- (P) Insoluble in water at room temperature
 (Q) Oxidised by concentrated nitric acid
 (R) Disproportionates in caustic soda solution
 (S) Oxidised by hot and concentrated sulphuric acid
 (T) Soluble in carbon disulphide

Ans. [(A) $\rightarrow$ (P,Q,R,S,T); (B) $\rightarrow$ (P,Q,S); (C) $\rightarrow$ (P,Q,R,S,T); (D) $\rightarrow$ (P,Q,S)]

12. Find the number of yellow coloured compound from the following compound.
 Ag_2CrO_4 , BaCrO_4 , $\text{Ag}_2\text{S}_2\text{O}_3$, PbSO_3 , PbI_2 , PbCrO_4 , CdS , $[\text{Ni}(\text{dmg})_2]$, $\text{Ni}(\text{OH})_2$

Ans. (4)

13. How many moles of a colourless gas which burns with blue flame are produced when one mole of $\text{K}_4[\text{Fe}(\text{CN})_6]$ is treated with excess of hot and conc. sulphuric acid.

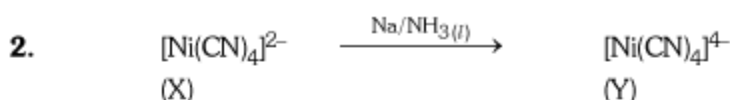
Ans. (6)

Co-Ordination Compounds

1. In $[\text{Ni}(\text{CO})_4]$, metal act as :

- (A) π -donar σ -acceptor
- (B) σ -donar π -acceptor
- (C) π -donar π -acceptor
- (D) σ -donar σ -acceptor

Ans. (A)



Which of the following properties changed between X and Y.

- (A) Hybridisation of central atom
- (B) Geometry
- (C) Magnetic nature
- (D) Both (A) and (B)

Ans. (D)

3. Which of the following is not correctly matched ?

- (A) Sodium (ethylenediaminetetraacetato)chromate(II) – $\text{Na}_2[\text{Cr}(\text{CH}_3\text{COO})_4(\text{en})]$
- (B) Dichlorido bis (ethane-1, 2-diamine)cobalt(III) ion – $[\text{Co}(\text{Cl})_2(\text{en})_2]^+$
- (C) Tris(bipyridyl) iron(II)ion – $[\text{Fe}(\text{NH}_4\text{C}_5\text{H}_4\text{N})_3]^{2+}$
- (D) Ammineaquadibromidocopper(II) – $[\text{Cu}(\text{NH}_3)(\text{H}_2\text{O})\text{Br}_2]$

Ans. (A)

4. Wilkinson's catalyst $[\text{RhCl}(\text{PPh}_3)_3]$ react with H_2 to form an octahedral complex in which Rh ($Z = 45$) has the following electronic configuration in the ligand field $t_{2g}^{2,2,2}, e_g^{0,0}$.
Then which of the following is correct about this new complex.
- (A) It is paramagnetic.
(B) Its IUPAC name is chloridohydridotris(triphenylphosphine) rhodium (III).
(C) Hybridisation of Rh(I) is d^2sp^3 .
(D) Hybridisation of Rh(III) is sp^3d^2 .

Ans. (B)

5. Which of the following are square planar complexes ?
- (A) $[\text{Ni}(\text{CO})_4]$
(B) $[\text{PtCl}_2(\text{NH}_3)_2]$
(C) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
(D) $[\text{FeCl}_4]^-$

Ans. (B, C)

6. In which of the following cases, the synergic bonding takes place at the π^* -orbital of the ligand.
- (A) $[\text{PtCl}_3(\text{C}_2\text{H}_4)]^-$
(B) $[\text{Ni}(\text{PF}_3)_4]$
(C) $[\text{Cr}(\text{C}_6\text{H}_6)_2]$
(D) $[\text{Fe}(\pi\text{-C}_5\text{H}_5)_2]$

Ans. (A, C, D)

7. In which of the following cases, the synergic bonding takes place at the π^* -orbital of the ligand.
- (A) $[\text{PtCl}_3(\text{C}_2\text{H}_4)]^-$
(B) $[\text{Ni}(\text{PF}_3)_4]$
(C) $[\text{Cr}(\text{C}_6\text{H}_6)_2]$
(D) $[\text{Fe}(\pi\text{-C}_5\text{H}_5)_2]$

Ans. (A, C, D)

8. $\text{CoCl}_2 + 4\text{NH}_4\text{SCN} \longrightarrow (\text{NH}_4)_2[\text{Co}(\text{SCN})_4]$
Select the **CORRECT** options regarding the test :
- (A) Excess of NH_4SCN is added in order to suppress the dissociation of the complex.
(B) The complex forms blue solution with amyl alcohol.
(C) This test is interfered by Fe^{3+} ion.
(D) The complex formed is paramagnetic with $\mu(\text{spin only}) = \sqrt{15}$ B.M.

Ans. (A, B, C, D)

9. When an alkaline solution of potassium chromate is treated with 30% H_2O_2 , the oxidation state of chromium changes by a unit and a brown complex $\text{K}_3[\text{CrO}_8]$ is formed.

Select **CORRECT** statement(s) about the complex :

- (A) It contains diamagnetic ligand.
- (B) It is paramagnetic compound.
- (C) It is a tetraperoxo species.
- (D) O–O bond order is an integer.

Ans. (A, B, C, D)

10. Which of the following salts can decolourise the blood red colouration of $\text{Fe}(\text{SCN})_3$?

- (A) NaF
- (B) $\text{Na}_2\text{C}_2\text{O}_4$
- (C) HgCl_2
- (D) KF

Ans. (A, B, C, D)

11. Which of the following statement(s) is/are **CORRECT** ?

- (A) In $[\text{CoFCl}(\text{en})_2]^+$ geometrical isomerism exists, while optical isomerism does not exist.
- (B) In $[\text{Fe}(\text{CN})_6]^{4-}$ ion EAN (effective atomic number) of Fe is 36
- (C) 'en' is a bidentate ligand
- (D) $[\text{Co}(\text{SCN})_2(\text{H}_2\text{O})_4]\text{Br}$ can show linkage isomerism

Ans. (B, C, D)

Paragraph-1

$\text{Ni}^{II}(\text{NH}_3)_4(\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$ molecule may have two unpaired electrons or zero unpaired electron and measurement of magnetic moment helps us to predict the geometry.

12. If Magnetic moment value is zero, then the formula of the complex will be :

- (A) $[\text{Ni}(\text{NH}_3)_4](\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$
- (B) $[\text{Ni}(\text{NH}_3)_2(\text{H}_2\text{O})_2](\text{NO}_3)_2 \cdot 2\text{NH}_3$
- (C) $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2](\text{NO}_3)_2$
- (D) $[\text{Ni}(\text{NO}_3)_2(\text{H}_2\text{O})_2] \cdot 4\text{NH}_3$

Ans. (A)

13. If the magnetic moment value is $2\sqrt{2}$ and conducts electricity, then the formula of the complex is :

- (A) $[\text{Ni}(\text{NH}_3)_4](\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$
- (B) $[\text{Ni}(\text{NH}_3)_2(\text{H}_2\text{O})_2](\text{NO}_3)_2 \cdot 2\text{NH}_3$
- (C) $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2](\text{NO}_3)_2$
- (D) $[\text{Ni}(\text{NH}_3)_4](\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$

Ans. (C)

14. The higher and lower values of magnetic moment of the given complex corresponds to the following geometries, respectively:

- (A) Octahedron and Tetrahedron
- (B) Octahedron and square planar
- (C) Square planar and Octahedron
- (D) Octahedron and Octahedron

Ans. (B)

Paragraph-2 (One or More than One Correct)

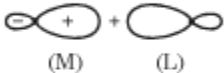
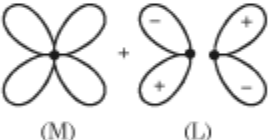
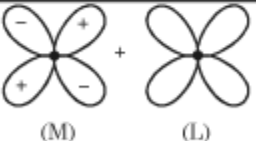
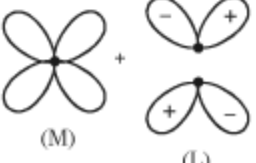
The following column I, II, III represent the different type of overlapping involve during the bonding of metal and π -acceptor ligands.

Answer the questions that follow

Column-I - Overlapping between metal(M) & ligand(L)

Column-II - Involve electron transfer

Column-III - Given overlapping is responsible for

Column-I (Overlapping between metal(M) & ligand(L))	Column-II (Involve electron transfer)	Column-III (Given overlapping is responsible for)
(1) 	(P) Full orbital on the metal with the empty antibonding molecular orbital (π^*) of the carbon monoxide	(i) π -bond between metal and atom of σ -donor ligand
(2) 	(Q) π back bond formed by the flow of electron density from a filled d-orbital of the metal into the vacant π antibonding molecular orbital of the alkene ligand	(ii) π -bond between metal and π -donor ligand
(3) 	(R) Donation of an electron pair from carbon to vacant orbital of metal	(iii) Interaction between olefinic ligand and metal
(4) 	(S) Fully filled orbital of metal with vacant d-orbital of ligand	(iv) σ -bond between metal and ligand

15. Select **CORRECT** option(s) for $[\text{Ni}(\text{CO})_3(\text{PMe}_3)]$:

- (A) (3), (S), (i)
- (B) (4), (R), (iii)
- (C) (1), (R), (iv)
- (D) (2), (P), (i)

Ans. (A, C, D)

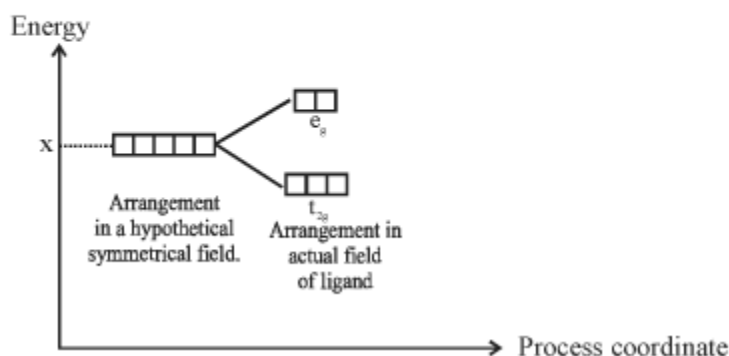
16. Select the **INCORRECT** option(s) for Zeises salt :

- (A) (3) (S) (iii)
- (B) (2) (Q) (i)
- (C) (4) (Q) (ii)
- (D) (4) (S) (ii)

Ans. (A, B, D)

Paragraph-3

According to crystal field theory, splitting of d orbitals in an octahedral complex occurs as follows :



If the energy of the arrangement in a hypothetical symmetrical field = x
and the crystal field splitting energy = Δ_o

17. Energies of e_g & t_{2g} orbitals are respectively :

- (A) $0.6\Delta_o$, $-0.4\Delta_o$
- (B) $0.6\Delta_o + x$, $-0.4\Delta_o - x$
- (C) $0.6\Delta_o + x$, $-0.4\Delta_o + x$
- (D) $0.6\Delta_o + \frac{x}{5}$, $-0.4\Delta_o + \frac{x}{5}$

Ans. (C)

18. Crystal field stabilisation energy for $\text{Fe}(\text{CN})_6^{-4}$ is (consider no change in energy on pairing)

- (A) $-2.4 \Delta_o + 6x$
- (B) $-2.4 \Delta_o - 6x$
- (C) $-2.4 \Delta_o$
- (D) $-2.4 \Delta_o - 1.2x$

Ans. (C)

Answer Q.19, Q.20 and Q.21 by appropriately matching the information given in the three columns of the following table.

	Column-1	Column-2	Column-3
(I)	$[\text{NiF}_6]^{2-}$	(i) d^2sp^3	(P) Outer orbital complex
(II)	$[\text{Cu}(\text{NH}_3)_4]^{2+}$	(ii) dsp^2	(Q) Inner orbital complex
(III)	$[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$	(iii) sp^3d^2	(R) Paramagnetic
(IV)	$[\text{Ni}(\text{NH}_3)_6]^{2+}$	(iv) sp^3	(S) Diamagnetic

19. Find the correct match.

- (A) (I) - (iii) - (R)
- (B) (I) - (iii) - (S)
- (C) (II) - (ii) - (R)
- (D) (II) - (iv) - (Q)

Ans. (C)

20. Find the correct match.

- (A) (III) - (i) - (P)
- (B) (III) - (i) - (R)
- (C) (IV) - (iii) - (R)
- (D) (IV) - (i) - (R)

Ans. (C)

21. If in complex (III) all existing ligands are replaced by NH_3 , then find the correct match.

- (A) (iii) - (S) - (Q)
- (B) (iii) - (R) - (P)
- (C) (i) - (P) - (R)
- (D) (i) - (Q) - (P)

Ans. (B)

Answer Q.22, Q.23 & Q.24 by appropriately matching the information given in the three columns of the following table.

Column - 1 Aq. Solution of Complex / Ion	Column - 2 Reagent which react with given complex	Column - 3 Observation after reaction
(I) $[\text{NiCl}_4]^{2-}$	(P) Excess conc. HCl	(i) Precipitation of metal ion
(II) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_2$	(Q) Excess KCN_{Aq}	(ii) Change in coordination number of central metal ion
(III) $[\text{Cu}(\text{H}_2\text{O})_6]\text{SO}_4$	(R) Excess NH_4OH in the presence of Air	(iii) Change in oxidation number of central metal ion
(IV) $[\text{Ag}(\text{NH}_3)_2]\text{NO}_3$	(S) $\text{H}^+/\text{H}_2\text{S}$	(iv) Change in magnetic nature

22. Select the **CORRECT** for the complex in column-1 which have spin magnetic moment

$$\mu = \sqrt{3} \text{ B.M.}$$

- (A) (I), (Q), (iv)
- (B) (III), (Q), (iv)
- (C) (III), (S), (iii)
- (D) (I), (R), (iii)

Ans. (B)

23. Select the **INCORRECT** for reaction of complex in column-1 which produces dark blue solution as product

- (A) I, (R), (ii)
- (B) II, (P), (ii)
- (C) III, (P), (ii)
- (D) III, (R), (iv)

Ans. (D)

24. Select the **INCORRECT** option

- (A) II, (R), (iii)
- (B) III, (S), (i)
- (C) IV, (P), (i)
- (D) II, (S), (i)

Ans. (D)

25. There are some coordination compounds given in Column I which may exist in different isomeric forms as given in Column II. Select the correct option(s) for the coordination compounds and their respective isomeric forms.

Column-I

- (A) $[\text{Co}(\text{en})_2\text{NH}_3\text{Cl}]\text{SO}_4$
 (B) $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2](\text{NO}_3)$
 (C) $[\text{Co}(\text{en})(\text{pn})(\text{NO}_2)_2]\text{Cl}$
 (D) $[\text{Co}(\text{gly})_3]$

Column-II

- (P) Optical isomer
 (Q) Geometrical isomer
 (R) Ionization isomer
 (S) Linkage isomer

Ans. [(A) → (P, Q, R); (B) → (Q, R, S); (C) → (P, Q, R, S); (D) → (P, Q)]

26. Match the column.

Column-I

- (A) $[\text{M}(\text{AA})\text{b}_2\text{cd}]^{n\pm}$
 (B) $[\text{Ma}_3\text{bcd}]^{n\pm}$
 (C) $[\text{M}(\text{AB})\text{b}_2\text{c}_2]^{n\pm}$
 (D) $[\text{M}(\text{AB})_3]^{n\pm}$

Column-II

- (P) Four optically active isomer
 (Q) Four geometrical isomer
 (R) All geometrical isomers are optically active.
 (S) Two pair of enantiomer

Note : $\widehat{\text{AA}}$, $\widehat{\text{AB}}$, **a, b, c, d** are not having chiral centre.

Ans. [(A) → (P, Q, S); (B) → (Q); (C) → (P, Q, S); (D) → (P, R, S)]

- 27.

**Column-I
Complexes**

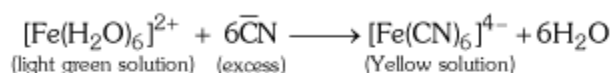
- (A) $[\text{Cr}(\text{H}_2\text{O})_5\text{Br}]^{2+}$
 (B) $[\text{Cu}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)\text{Cl}_4]^{2-}$
 (C) $[\text{Pt}(\text{OX})_2]^{2-}$
 (D) $[\text{Fe}(\text{OH})_4]^-$

**Column-II
Correct Characteristics**

- (P) Paramagnetic
 (Q) Geometrical isomerism is exhibited
 (R) Optical isomerism is exhibited
 (S) Do not follow sidgwick EAN rule
 (T) Complex having symmetrical bidentate ligand

Ans. [(A) → (P, S); (B) → (P, S, T); (C) → (S, T); (D) → (P, S)]

28. Consider given reaction for matching list type question :



Column-I

(Characteristic of complex)

- (P) No π -bond
 (Q) Crystal field splitting energy is higher as compared to $[\text{Fe}(\text{SCN})_6]^{3-}$ (thiocyanato-S)
 (R) Synergic bonding between metal and ligand
 (S) $\mu_{\text{spin only}} = \sqrt{35}$ B.M.

Column-II

(Possible in)

- (1) Reactant complex
 (2) Product complex
 (3) Both reactant and product complex
 (4) Neither reactant nor product complex

Codes :	P	Q	R	S
(A)	4	3	2	1
(B)	3	2	4	1
(C)	1	3	2	4
(D)	1	2	3	4

Ans. (C)

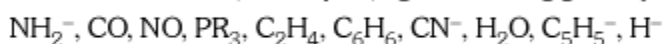
29. The octahedral complex $[\text{Rh}(\text{NO}_2)(\text{SCN})(\text{en})_2]^+$ can exist in a total number of isomeric forms including stereoisomers :

Ans. (12)

30. E.A.N. of $\text{K}[\text{PtCl}_3(\eta^2-\text{C}_2\text{H}_4)]$ is

Ans. (84)

31. Total number of π -acid (π -acceptor) ligands among given species is/are



Ans. (7)

32. In the following reaction 'A' is the number of reactions that are redox and 'B' is the number of reactions which involve in complex formation, (A + B) is

- (i) $\text{CuSO}_4 + \text{Excess KI} \longrightarrow$ (ii) $\text{CuSO}_4 + \text{Excess NaSCN} \longrightarrow$
 (iii) $\text{CuSO}_4 + \text{Excess NaCN} \longrightarrow$ (iv) $\text{FeCl}_3 + \text{Excess Na}_2\text{S}_2\text{O}_3 \longrightarrow$
 (v) $\text{CuSO}_4 + \text{Excess Na}_2\text{S}_2\text{O}_3 \longrightarrow$

Ans. (8)

33. Consider following reactions :

- (i) $\text{CuSO}_4 + \text{KCN} \longrightarrow \text{products}$
- (ii) $\text{NO}_3^{-1} + \text{cons. H}_2\text{SO}_4 + \text{FeSO}_4 \longrightarrow \text{products}$
- (iii) $\text{Fe}^{2+} + [\text{Fe}(\text{CN})_6]^{4-} \xrightarrow[\text{of air}]{\text{absence}} \text{products}$
- (iv) $\text{Fe}(\text{SCN})_3 + \text{C}_2\text{O}_4^{2-} \longrightarrow \text{products}$
- (v) $\text{Co}^{3+} + \text{aq. NH}_3 \longrightarrow \text{products}$

In how many of the reactions, the product contains paramagnetic central metal ion ?

Ans. (2)

34. Find the number of Ligand(s) which is/are chelating ?

$(\text{C}_6\text{H}_5)_3\text{P}$, en, $\text{C}_2\text{O}_4^{2-}$, acac⁻, DMG⁻, gly⁻, Py

Ans. (5)

35. How many of the following complexes are correctly matched with given properties ?

- (a) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ – octahedral outer orbital complex and two unpaired electrons.
- (b) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ – octahedral inner orbital complex and one unpaired electron.
- (c) $[\text{Zn}(\text{NH}_3)_6]^{2+}$ – octahedral outer orbital complex and diamagnetic.
- (d) $[\text{Pt}(\text{NH}_3)_4]^{2+}$ – square planar and diamagnetic
- (e) $[\text{Co}(\text{SCN})_4]^{2-}$ – tetrahedral and three unpaired electrons.
- (f) $[\text{NiBr}_2(\text{PPh}_3)_2]$ – square planar and diamagnetic.

Ans. (4)

36. How many of the following species would have C–O bond length longer than that of free CO molecules.

$[\text{V}(\text{CO})_6]^{-}$, $[\text{Ti}(\text{CO})_6]^{2-}$, $[\text{Cr}(\text{CO})_6]$, $[\text{Sc}(\text{CO})_6]^{3+}$, $\text{Ni}(\text{CO})_4$, $[\text{Fe}(\text{CN})_5(\text{CO})]^{3-}$, CO_2 , CO_3^{2-} .

Ans. (7)

37. (a) From Meridional and facial isomer of $[\text{Ma}_3\text{b}_3]^{n\pm}$ on replacement of only one 'a' by 'b', the number of isomer of the product obtained are respectively x and y.

(b) The number of corner shared per tetrahedron for 2D-silicate is z.

Then $x + y + z$ is :

Ans. (6)

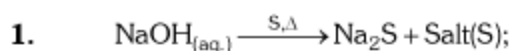
38. Consider the following species assuming no distortion due to lone pair :

XeF_4 , XeO_2F_2 , TeCl_4 , SbCl_5 , BrF_5 , $[\text{PtCl}_4]^{-2}$, $[\text{Ni}(\text{CN})_4]^{-2}$, $[\text{Cu}(\text{NH}_3)_4]^{2+}$,
 $[\text{CoCl}_4]^{-2}$

How many of these have atleast 5 atoms present in one plane.

Ans. (5)

Qualitative Analysis



Product (P) is :

- (A) white ppt.
- (B) black ppt.
- (C) white turbidity
- (D) clear solution

Ans. (B)

2. Identify the **CORRECT** statements.

- (I) NH_4Cl is added along with NH_4OH so that only IIIrd group cations can be precipitated as their hydroxides.
 - (II) In IIIrd group in place of NH_4Cl , $(\text{NH}_4)_2\text{SO}_4$ can not be used as Vth group radical(s) will also be precipitated as MSO_4 along with Al^{3+} , Fe^{3+} & Cr^{3+} .
 - (III) Aqueous solution of ammonium sulphate produces white ppt. with BaCl_2 solution which is insoluble in dil. HCl .
- (A) I and II only
 - (B) II and III only
 - (C) I and III only
 - (D) I, II and III

Ans. (D)

3. Select the incorrect statement(s).

- (A) Ammonium ions produce yellow colour solution with sodium hexanitrito-N-cobaltate (III).
- (B) Ammonia gas develops a brown colour on filter paper moistened with a solution of MnCl_2 and H_2O_2 .
- (C) Ammonium ions produce white precipitate with saturated sodium hydrogen tartrate solution.
- (D) Ammonium salts in presence of sodium hydroxide solution produces red precipitate with 4-nitrobenzene diazonium chloride reagent.

Ans. (A, D)

4. The reaction which yield a black precipitate is/are :

- (A) $\text{Hg}_2\text{Cl}_2 + \text{NH}_3 \rightarrow$
- (B) $\text{Bi}(\text{OH})_3 \downarrow + \text{Na}_2[\text{Sn}(\text{OH})_4] \rightarrow$
- (C) $\text{Fe}^{3+} + (\text{NH}_4)_2\text{S}(\text{neutral}) \rightarrow$
- (D) None of these

Ans. (A, B, C)

5. Match List-I with List-II.

List-I

- (I) A blue solution in water
- (II) A blue precipitate
- (III) Red colour compound/ppt
- (IV) Tetra coordinated compound

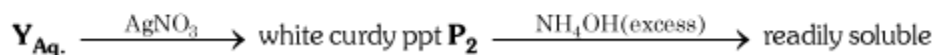
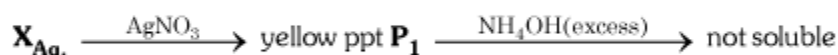
List-II

- (i) $\text{Cu}(\text{OH})_2$
- (ii) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- (iii) $[\text{Co}(\text{SCN})_4]^{2-}$
- (iv) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- (v) $[\text{Ni}(\text{NH}_3)_6](\text{OH})_2$
- (vi) Pb_3O_4
- (vii) HgI_2

- (A) (I) – (ii, iii, v)
- (B) (II) – (i, iv)
- (C) (III) – (vi, vii)
- (D) (IV) – (iii)

Ans. (A, B, C, D)

6. When I_2 is added in hot and conc. caustic soda solution, two iodine containing products 'X' and 'Y' are obtained.



Select **CORRECT** option(s).

- (A) ppt P_1 is AgIO_3
- (B) ppt P_2 is AgIO_3
- (C) Acidic solution of **Y** produced blue colour with H_2S and starch.
- (D) Alakaline solution of **X** produced **Y** with KMnO_4

Ans. (B, C, D)

7. Which of the following produce(s) H_2S gas when treated with water.

- (A) Cr_2S_3
- (B) Al_2S_3
- (C) Na_2S
- (D) Ag_2S

Ans. (A, B)

8. An alloy contains copper, silver, cadmium and zinc. It is dissolved in con. HNO_3 and then dil. HCl is added in the solution. Precipitate formed in the reaction is filtered and the filtrate is treated with excess KCN followed by H_2S gas that again gives yellow precipitate alongwith the solution that contains,

(A) $\text{K}[\text{Ag}(\text{CN})_2]$
(B) $\text{K}_2[\text{Cu}(\text{CN})_4]$
(C) $\text{K}_3[\text{Cu}(\text{CN})_4]$
(D) $\text{K}_2[\text{Zn}(\text{CN})_4]$

Ans. (C, D)

9. $\text{Co}^{2+} \xrightarrow{\text{KCN}} \text{X} \xrightarrow{\text{xs KCN}} \text{Y} \xrightarrow[\Delta]{\text{air}} \text{Z}$

Select the correct statement regarding the above experiment

(A) X is yellowish brown ppt.
(B) Y is brown solution of $[\text{Co}(\text{CN})_6]^{4-}$
(C) Y is paramagnetic
(D) Z is diamagnetic

Ans. (B, C, D)

Paragraph-1

For the purpose of systematic qualitative analysis cations are classified into various groups on the basis of their behaviour against some reagents. The group reagents used for the classification of most common cations are hydrochloric acid, hydrogen sulphide, ammonium hydroxide and ammonium carbonate. Classification is based on whether a cation reacts with these reagents by the formation of precipitates or not.

10. Which one among the following pairs of ions cannot be separated by H_2S in the presence of dilute hydrochloric acid ?

(A) Bi^{3+} , Cd^{2+}
(B) Al^{3+} , Hg^{2+}
(C) Zn^{2+} , Cu^{2+}
(D) Ni^{2+} , Cu^{2+}

Ans. (A)

11. An aqueous solution contains Hg^{2+} , Hg_2^{2+} , Pb^{2+} and Cd^{2+} . The addition of 2 M HCl will precipitate :

(A) HgCl_2 only
(B) PbCl_2 only
(C) PbCl_2 and Hg_2Cl_2
(D) PbCl_2 and CdCl_2

Ans. (C)

12. An aqueous solution which is slightly acidic contain cations Fe^{3+} , Zn^{2+} and Cu^{2+} . The reagent that when added in excess to this solution would identify the separate Fe^{3+} ion in one step is :
- (A) 2M HCl
 - (B) 6M NH_3
 - (C) 6M NaOH
 - (D) H_2S gas

Ans. (B)

Paragraph-3

Two students were given mixture of Al^{3+} , Cr^{3+} , Fe^{3+} , Zn^{2+} , Ni^{2+} , Co^{2+} and Mn^{2+} ions as their soluble salt which were not interfering radicals. They performed the experiment as follows :

Student I : In the aqueous solution of ions he added first NH_4Cl then NH_4OH and then passed H_2S gas. He got the precipitate (X).

Student II : In the aqueous solution of ions he first added NH_4Cl and then NH_4OH . He got the precipitate (Y) and a filtrate. In the filtrate, he passed H_2S gas and again got a precipitate (Z).

Based on above information answer the following three questions :

13. Which of the following is not the precipitate (X) ?
- (A) ZnS
 - (B) MnS
 - (C) $\text{Cr}(\text{OH})_3$
 - (D) Al_2S_3

Ans. (D)

14. When precipitate (Z) is dissolved in acetic acid and then treated with NaBiO_3 in HNO_3 , a purple colour (P) solution is obtained. Which of the following is incorrect about (P) ?
- (A) (P) is a peroxy acid.
 - (B) Potassium salt of (P) is used in redox titrations.
 - (C) Potassium salt of (P) can not be used to titrate Mohr's salt in presence of HCl or HNO_3 .
 - (D) Potassium salt of (P) is a self indicator in redox titrations.

Ans. (A)

Paragraph-4

An aqueous solution of salt **(P)** gives white precipitate with Barium chloride solution which is insoluble in dilute HCl & dilute HNO_3 . When ammonia (aq) is added to the aqueous solution of salt **(P)** blue precipitate is obtained which dissolves in excess ammonia producing deep blue colouration due to formation complex **(Q)**. With KCN, **(P)** gives first a yellow precipitate which quickly decomposes to white precipitate along with liberating highly poisonous gas **(R)**. **(R)** on hydrogenation produces **(S)** which gives yellowish orange coloured complex **(T)** with CoCl_3 .

15. Which of the following options is correct with respect to P, Q, R, S and T.

- (A) 'P' is Ag_2SO_4 , 'Q' is $[\text{Ag}(\text{NH}_3)_2]\text{SO}_4$
- (B) 'S' is a bidentate ligand & 'R' is poisonous gas
- (C) 'R' is CH_2N_2 & 'P' is CuSO_4
- (D) None of the above options is correct.

Ans. (B)

16. Which of the following options regarding complex Q & T is not correct.

- (A) Q is paramagnetic where as T is diamagnetic.
- (B) Q has a square planar geometry where as T has octahedral geometry.
- (C) When dissolved in water, Van't Hoff coefficient for Q is 2 & that of T is 4 assuming 100% dissociation.
- (D) Only one of the above statements is correct.

Ans. (D)

17. **Column-I**

Column-II

- | | |
|---|--|
| (A) NH_4^+ (aq) + $[\text{Co}(\text{NO}_2)_6]^{3-}$ (aq) $\rightarrow$ complex | (P) Colourless water soluble complex. |
| (B) $\text{Cr}(\text{OH})_3 \downarrow + \text{NH}_3$ (aq) $\rightarrow$ complex | (Q) Yellow coloured water insoluble complex. |
| (C) $\text{Fe}(\text{SCN})_3$ (aq) + F^- (aq) $\rightarrow$ complex | (R) Value of oxidation state of metal ion and number of unpaired electrons are same. |
| (D) $\text{Ni}(\text{OH})_2 \downarrow + \text{NH}_3$ (aq) $\rightarrow$ complex | (S) d^2sp^3 hybridisation. |

Ans. [(A) $\rightarrow$ (Q, S); (B) $\rightarrow$ (R, S); (C) $\rightarrow$ (P); (D) $\rightarrow$ (R)]

18. Match the matrix.

Column-I

Column-II

(Aq. soln. of metal ions)

- | | |
|----------------------|--|
| (A) Ag^{2+} | (P) Precipitated by Na_2S |
| (B) Pb^{2+} | (Q) Yellow precipitate with KI |
| (C) Bi^{3+} | (R) Produce precipitate on dilution by water |
| (D) Zn^{2+} | (S) Produce amphoteric hydroxide |

Ans. [(A) $\rightarrow$ (P, Q); (B) $\rightarrow$ (P, Q, S); (C) $\rightarrow$ (P, R); (D) $\rightarrow$ (P, S)]

19. Match the chemicals used to differentiate pairs of compounds.

Column-I

- (A) $\text{Fe}(\text{OH})_3, \text{Cr}(\text{OH})_3$
 (B) $\text{PbCl}_2, \text{AgCl}$
 (C) HgS, PbS
 (D) $\text{CuSO}_{4(\text{anhy.})}, \text{Bi}_2(\text{SO}_4)_3$

Column-II

- (P) Hot water
 (Q) 50% HNO_3
 (R) Na_2O_2
 (S) NH_4OH (excess)
 (T) NaOH (excess) solution

Ans. [(A) $\rightarrow$ (RT); (B) $\rightarrow$ (P, S); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (S)]

20. Match the following column

List-I

- (P) $\text{Zn}[\text{Fe}(\text{CN})_5(\text{NOS})]$
 (Q) $\text{K}_2[\text{HgI}_4] + \text{KOH}$
 (R) $\text{NaHC}_4\text{H}_4\text{O}_6$
 (S) NO_3^-

List-II

- (1) Gives brown colour ppt with NH_3
 (2) Gives pungent odour with Devarda Alloy in presence of alkaline medium
 (3) Gives unknown compound of red colour with SO_2
 (4) Gives white ppt. with K^+

Codes :	P	Q	R	S
(A)	3	2	4	1
(B)	2	1	3	4
(C)	3	1	2	4
(D)	3	1	4	2

Ans. (D)

21. Find total number of conditions in which oxidation of Fe^{+2} (aq) ions to iron(III) takes place

- (i) On exposure to air
 (ii) On addition of conc. HNO_3
 (iii) On reaction with SnCl_2
 (iv) On reaction with H_2O_2
 (v) On reaction with $\text{MnO}_4^-/\text{H}^+$
 (vi) On reaction with KI
 (vii) On reaction with $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$

Ans. (5)

22. How many of the following precipitates is/are soluble in acetic acid?

$\text{ZnS}, \text{MnS}, \text{CuS}, \text{HgS}, \text{BaC}_2\text{O}_4, \text{SrC}_2\text{O}_4, \text{CaC}_2\text{O}_4, \text{BaCrO}_4, \text{SrCrO}_4, \text{CaCrO}_4$

Ans. (4)

23. How many of given sulphides are soluble in $(\text{NH}_4)_2\text{S}_2$ solution ?

$\text{PbS}, \text{SnS}_2, \text{As}_2\text{S}_3, \text{SnS}, \text{As}_2\text{S}_5, \text{ZnS}, \text{Sb}_2\text{S}_3, \text{HgS}$

Ans. (5)

- 24.** The most acidic oxide of chromium on treatment with calculated amount of ammonia, followed by acidification with dilute HCl gives a compound 'A'. 1 mole of 'A' on heating produces 1 mole of a colourless diatomic gas 'B' and residue 'C'. Find bond order (x) in 'B' and oxidation number (y) of Cr in residue 'C'.

Your answer in OMR is (x + y)

Ans. (6)

- 25.** Solution containing phosphate $\xrightarrow[\text{H}^+]{\text{Ammonium molybdate}}$ Canary yellow ppt. of (A).

Find the number of atoms present in one molecule of 'A'

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

Ans. (5)

- 26.** Find correct number of match.

- (i) Ag_3PO_4 yellow ppt.
- (ii) $\text{BaHPO}_4 \downarrow$ white ppt.
- (iii) $(\text{NH}_4)_3[\text{P}(\text{Mo}_3\text{O}_{10})_4]$ Canary yellow solution.
- (iv) $\text{Ag}_2\text{Cr}_2\text{O}_7$ reddish brown ppt.
- (v) $[\text{HgS}_2]^{2-}$ colorless complex.
- (vi) $[\text{Co}(\text{NH}_3)_6]^{3+}$ pink solution.
- (vii) $[\text{BiI}_4]^-$ orange solution.
- (viii) $\text{K}_3[\text{Fe}(\text{CN})_6]$ yellow solution.
- (ix) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ prussian blue solution.
- (x) $\text{Hg}(\text{SCN})_2$ colorless.

Ans. (8)

27. Find correct number of reactions/statements.

(i) KO_2 produces observable change with $\text{Mg}^{2+}(\text{aq.})$ and $\text{Cr}^{3+}(\text{aq.})$

(ii) Heating effect $\text{NH}_4\text{IO}_3 \longrightarrow \text{N}_2 + \text{I}_2 + \text{O}_2$

(iii) Heating effect $\underset{\text{Pink}}{\text{CoCl}_2 \cdot 6\text{H}_2\text{O}} \xrightarrow{140^\circ\text{C}} \underset{\text{Blue}}{\text{CoCl}_2}$

(iv) Oxides of all alkaline earth metal are basic in nature

(v) $\text{Na}_3\text{PO}_4 \xrightarrow{\Delta} \text{P}_4$

(vi) $\underset{\text{Yellow}}{\text{HgO}} \xrightarrow{\Delta} \underset{\text{Red}}{\text{HgO}} \xrightarrow{400^\circ\text{C}} \text{Hg} + \frac{1}{2}\text{O}_2$

(vii) $\text{NH}_3 > \text{SbH}_3 > \text{AsH}_3 < \text{PH}_3$: Thermal stability

(viii) Halogen molecules are diamagnetic

(ix) $\text{SnC}_2\text{O}_4 \xrightarrow{\Delta} \text{SnO} + \text{CO} + \text{CO}_2$

(x) $\text{Ag}_2\text{C}_2\text{O}_4 \xrightarrow{\Delta} 2\text{Ag} + 2\text{CO}_2$

Ans. (7)

28. In how many of the following reactions one of the products is obtained as a black precipitate ?

(i) $\text{Bi}(\text{OH})_3 \downarrow + [\text{Sn}(\text{OH})_4]^{2-}(\text{aq}) \longrightarrow \text{Products}$

(ii) $\text{Bi}^{3+}(\text{aq}) + \text{I}^-(\text{aq})$ (not in excess) $\longrightarrow \text{Products}$

(iii) $\text{Bi}^{3+}(\text{aq}) + \text{H}_2\text{S}(\text{g}) \xrightarrow{\text{H}^+} \text{Products}$

(iv) $[\text{BiI}_4]^- (\text{aq}) + \text{H}_2\text{O}(\ell) \xrightarrow{\text{dilution}} \text{Products}$

(v) $\text{Bi}^{3+}(\text{aq}) + \text{C}_6\text{H}_3(\text{OH})_3$ (10%) $\longrightarrow \text{Products}$

(vi) $\text{BiO.OH} \downarrow + \text{H}_2\text{O}_2(\text{aq}) \longrightarrow \text{Products}$

Ans. (3)

29. Find the number of reactions in which SO_2 gas is produced :

- Heating of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$
- Roasting of Fool's gold
- Heating Cu_2O with chalcocite
- Reaction of Cu with dil. H_2SO_4
- Reaction of Cu with conc. H_2SO_4
- Treatment of washing soda with dil. H_2SO_4
- Hypo solution reacts with dil. HCl
- White phosphorus react with SO_2Cl_2

Ans. (6)

- 30.** The total number of species which give black precipitate with H_2S
 $\text{Hg}(\text{NO}_3)_2$, MgCl_2 , CuSO_4 , AgNO_3 , $\text{Pb}(\text{NO}_3)_2$, $\text{Sn}(\text{NO}_3)_2$

Ans. (4)

- 31.** How many species are coloured ?

AgCl , CaC_2O_4 , $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$, $[\text{Cu}(\text{NH}_3)_4](\text{NO}_3)_2$, $\text{Zn}_2[\text{Fe}(\text{CN})_6]$, $[\text{Ni}(\text{DMG})_2]$, NO_2 , CrO_4^{2-}

Ans. (5)

- 32.** How many of the following reactions give $\text{NH}_4\text{OH}/\text{NH}_3$?

- | | |
|---|---|
| (i) $\text{NH}_4\text{NO}_3 + \text{NaOH} \longrightarrow$ | (ii) $\text{NH}_4\text{Cl} + \text{NaNO}_2 \xrightarrow{\Delta}$ |
| (iii) $(\text{NH}_4)_2\text{SO}_4 + \text{CaO} \rightarrow$ | (iv) $(\text{NH}_4)_2\text{CO}_3 + \text{HCl} \rightarrow$ |
| (v) $(\text{NH}_4)_2\text{SO}_4 + \text{KNO}_3 \rightarrow$ | (vi) $\text{AlN} + \text{H}_2\text{O} \rightarrow$ |
| (vii) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \xrightarrow{\Delta}$ | (viii) $(\text{NH}_4)_2\text{C}_2\text{O}_4 \xrightarrow{\Delta}$ |
| (ix) $(\text{NH}_4)_2\text{SO}_4 \xrightarrow{\Delta}$ | (x) $\text{Ca} + \text{C} + \text{N}_2 \xrightarrow{\Delta} \xrightarrow{\text{H}_2\text{O}}$ |
| (xi) $(\text{NH}_4)_2\text{S} + \text{KOH} \longrightarrow$ | |

Ans. (7)

- 33.** How many of the following salts give precipitate with aq. NaOH solution which dissolve in excess of the reagent ?

CdSO_4 , $\text{Pb}(\text{NO}_3)_2$, CrCl_3 , MnCl_2 , ZnSO_4 , CuSO_4 , $\text{Bi}(\text{NO}_3)_3$, SnCl_4

Ans. (4)

